A hitch-hiker's guide to the metaplectic trace formula for SL(2): basic theory *

David Joyner

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Abstract

This is an account of the Arthur trace formula for the n-fold metaplectic cover of $SL(2)$, following closely Arthur's derivation of the invariant trace formula in this case. We also discuss the stabilized trace formula of the covers of $SL(2)$ by defining stable conjugacy on a cover of $SL(2)$ with conjugacy in a corresponding cover of $GL(2)$, then sketch a $\overline{GL(2,\mathbb{A}_S)}$-invariant trace formula as in [LL].

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1 Introduction

We will write down (with some proofs) the invariant trace formula following [A2]–[A15] in the case of the n-fold metaplectic cover of $G = SL(2)$. This is essentially an explicit formula for the trace of the right regular representation of $\overline{G(\mathbb{A})}$ on the space of square-integrable genuine cusp forms. Then we shall sketch a partial stabilization following [LL]. Our formula resembles that of [LL] except that there is no term in our formula corresponding to the contribution to the residual spectrum [LL, (5.5), p. 754]. Moreover, we do not stabilize the spectral side of the trace formula.

We have left out almost entirely the details of the theory of Eisenstein series (there are several excellent presentations available - [L1, L2], [K2]). We have also left out most of the local representation theory needed, for which we refer to the papers [J1, J3].

Any comments on this paper may be sent to (wdj@sma.usna.navy.mil or Math Dept, USNA, Annapolis MD 21402).

2 Background on metaplectic extensions

Notation: Let k denote a number field containing the group of nth roots of unity $\mu_n \subset \mathbb{C}^\times$. In particular, if $n > 2$ then k is totally imaginary. Let Σ_k denote the set of places of k, Σ_k^∞ denote the set of archimedian places of k (which we may identify with the set of inequivalent complex embeddings $k \hookrightarrow \mathbb{C}$), and let Σ_k^f denote the set of non-archimedian places of k (which we may identify with the set of non-zero prime ideals of $\mathcal{O}_k$). Let $\mathbb{A}$ denote the adele ring of $\mathbb{Q}$ and let $\mathbb{A}_k$ denote the adele ring of k and, for $v \in \Sigma_k$, let k_v denote the completion of k with respect to v. For $F = k_v$ p-adic, let $\mathcal{O} = \mathcal{O}_F = \mathcal{O}_v$ denote the ring of integers of F, $\pi = \pi_F = \pi_v$ a local

uniformizer, q the cardinality of the residue field, and $\nu = \nu_F = \nu_v : F^\times \to \mathbb{Z}$ the normalized valuation, so that $\nu(\pi) = q^{-1}$.

Let G denote the reductive group over $\mathbb{Q}$ defined by $SL(2)$ via restriction of scalars $k/\mathbb{Q}$, so for example $G(\mathbb{Q}) = SL(2,k)$.

2.1 Local aspects

Let $F = k_v$, where $v \in \Sigma_k$. For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(F)$, let

$$x(g) = \begin{cases} c, & c \neq 0, \\ d, & c = 0, \end{cases}$$

and let $\beta = \beta_v : SL_2(F) \times SL_2(F) \to \mu_n$ be defined by

$$\begin{aligned} \beta(g_1, g_2) &= (x(g_1), x(g_2))_n \cdot (-x(g_1)^{-1} x(g_2), x(g_1 g_2))_n \\ &= (\frac{x(g_1 g_2)}{x(g_1)}, \frac{x(g_1 g_2)}{x(g_2)})_n, \end{aligned}$$

where $(...,...)_n = (...,...)_{n,v} : F^\times \times F^\times \to \mu_n$ denotes the Hilbert symbol [We1]. Note that this is trivial if $F = \mathbb{C}$.

Lemma 2.1. *(Kubota [K1], Moore [M]) The map β defines a continuous Borel 2-cocycle (giving μ_n the discrete topology of course). Its cohomology class in the continuous cohomology group $H^2(SL_2(F), \mu_n)$ corresponds to an n-fold topological central extension $\overline{SL_2(F)}$ which fits into the short exact sequence*

$$1 \to \mu_n \to \overline{SL_2(F)} \xrightarrow{p} SL_2(F) \to 1.$$

If $H < SL_2(F)$ is a subgroup then we define $\overline{H} = p^{-1}(H) \subset \overline{SL_2(F)}$.

Elements of $\overline{SL_2(F)}$ will be denoted by (g, ς), where $g \in SL_2(F)$, $\varsigma \in \mu_n$. Multiplication on $\overline{SL_2(F)}$ is given by

$$(g_1, \varsigma_1) \cdot (g_2, \varsigma_2) = (g_1 g_2, \beta(g_1, g_2)\varsigma_1\varsigma_2). \tag{2.1}$$

so inverses are given by

$$(g, \varsigma)^{-1} = (g^{-1}, \beta(g, g^{-1})^{-1}\varsigma^{-1}) \tag{2.2}$$

and conjugation is given by

$$(g, \varsigma) \cdot (h, \varsigma') \cdot (g, \varsigma)^{-1} = (ghg^{-1}, \beta(g,h)\beta(gh, g^{-1})\beta(g, g^{-1})^{-1}\varsigma'). \tag{2.3}$$

We temporarily let $B = \{\begin{pmatrix} * & * \\ 0 & * \end{pmatrix}\} \subset SL_2$ denote the standard Borel, N its unipotent radical, and A its Levi component. For $\theta \in F^\times - F^{\times 2}$, let

$$T = T_\theta = \{\begin{pmatrix} a & b\theta \\ b & a \end{pmatrix} \mid a^2 + b^2\theta = 1,\ a, b \in F\}.$$

Lemma 2.2. *(a)* $\overline{N(F)} = N(F) \times \mu_n$ *(direct product),*

(b) if p *is relatively prime to* n *then* $\overline{SL_2(\mathcal{O})} = SL_2(\mathcal{O}) \times \mu_n$ *(direct product),*

(c) if $K_0 = SL_2(\mathcal{O})$ *and*

$$K_m = \{\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathcal{O}) \mid a - 1, d - 1, b, c \in \pi^m\mathcal{O}\},\ m > 0,$$

then $\overline{K_m} = K_m \times \mu_n$ *(direct product) for all* $m \geq n$,

(d) if $F = k_v = \mathbb{R}$ *then* $\overline{A(k_v)^0} = A(k_v)^0 \times \mu_n$,

(e) The center of $\overline{SL_2(F)}$ *is*

$$Z(\overline{SL_2(F)}) = \{\pm 1_2\} \times \mu_n,$$

(f) For $a_i = \begin{pmatrix} t_i & 0 \\ 0 & t_i^{-1} \end{pmatrix} \in A(F)$, $i = 1, 2$, *and* $\varsigma_i \in \mu_n$, *we have* $(a_1, \varsigma_1)^{-1} \cdot (a_2, \varsigma_2) \cdot (a_1, \varsigma_1) = (a_2, (t_1, t_2)_n^2 \varsigma_2)$. *(In particular,* $\overline{A(F)}$ *is abelian if* $n = 1, 2$.*)*

(g) $\overline{T}$ *is abelian.*

(h) If $g \in G(F)$ *and* $n = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ *then* $(g, \zeta)(n, \zeta')(g, \zeta)^{-1} = (gng^{-1}, \zeta')$.

proof: Except for (g), (h) these are all immediate conseqences of the well-known properties of the Hilbert symbol [We1]. Part (g) is equivalent to the statement $\beta(\gamma_1, \gamma_2) = \beta(\gamma_2, \gamma_1)$, for all $\gamma_1, \gamma_2 \in T$. This is a special case of the calculation in [Fl, p. 128]. Part (h) is a direct calculation which we omit. □

The covers of the maximal comapct subgroups do not always split at the archimedian places either. We recall (see for example [Ge2]) that the 2-fold cover of $SO(2, \mathbb{R})$ does not split, though for $n > 2$ the n-fold covers of $SU(2, \mathbb{C})$ do split.

Note that $G(\mathbb{Q}_p) = \prod_{v|p} SL_2(k_v)$. Define $\beta_p : G(\mathbb{Q}_p) \times G(\mathbb{Q}_p) \to \mu_n$ by

$$\beta_p(g_1, g_2) = \prod_{v|p} \beta_v(g_{1,v}, g_{2,v}).$$

This is a 2-cocycle by Lemma 2.1 above. Define $\overline{G(\mathbb{Q}_p)}$ to be the n-fold covering of $G(\mathbb{Q}_p)$ associated to this 2-cocycle. We may identify $\overline{G(\mathbb{Q}_p)}$ with a subgroup of $\prod_{v|p}\overline{SL_2(k_v)}$.

Let $B = \{\begin{pmatrix} * & * \\ 0 & * \end{pmatrix}\} \subset G$ denote the standard minimal parabolic defined over $\mathbb{Q}$, N its unipotent radical, and A its Levi component. Let $T \subset G(\mathbb{Q}_p)$ denote a Cartan subgroup of the form $T = \prod_{v|p} T_v$, where T_v is a non-split Cartan of $SL_2(k_v)$.

Lemma 2.3. *(a)* $\overline{N(\mathbb{Q}_p)} = N(\mathbb{Q}_p) \times \mu_n$ *(direct product),*
(b) if p *is relatively prime to* n *then* $\overline{G(\mathbb{Z}_p)} = G(\mathbb{Z}_p) \times \mu_n$ *(direct product),*
(c) if $K_0 = G(\mathbb{Z}_p)$ *and*

$$K_m = \prod_{v|p}\{\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(k_v) \mid a-1, d-1, b, c \in \pi_v^m \mathcal{O}_v\},\ m > 0,$$

then $\overline{K_m} = K_m \times \mu_n$ *(direct product) for all* $m \geq n$,
(d) if $n = 2$, $k = \mathbb{Q}$, $p = \infty$ *then* $\overline{A(\mathbb{R})^0} = A(\mathbb{R})^0 \times \mu_n$,
(e) The center of $\overline{G(\mathbb{Q}_p)}$ *is*

$$Z(\overline{G(\mathbb{Q}_p)}) = \{\pm 1_2\} \times \mu_n,$$

(f) For $a_i = \begin{pmatrix} t_i & 0 \\ 0 & t_i^{-1} \end{pmatrix} \in A(\mathbb{Q}_p)$, $i = 1, 2$, *and* $\varsigma_i \in \mu_n$, *we have* $(a_1, \varsigma_1)^{-1} \cdot (a_2, \varsigma_2) \cdot (a_1, \varsigma_1) = (a_2, \prod_{v|p}(t_1, t_2)_{n,v}^2 \varsigma_2)$. *(In particular,* $\overline{A(\mathbb{Q}_p)}$ *is abelian if* $n = 1, 2$.*)*
(g) $\overline{T}$ *is abelian.*

proof: These are all immediate conseqences of the previous lemma. □

2.1.1 NAK decompositions

By the above lemma, we may identify $N(\mathbb{Q}_p)$ with a subgroup of $\overline{G(\mathbb{Q}_p)}$ via $n \longmapsto (n, 1)$. We may identify $K_0 = K_{0,p}$ with a subset of $\overline{G(\mathbb{Q}_p)}$ via $k \longmapsto (k, 1)$. If $p|n$ then this image is a subgroup.

Lemma 2.4. *We have*

$$\overline{G(\mathbb{Q}_p)} = N(\mathbb{Q}_p) \cdot \overline{A(\mathbb{Q}_p)} \cdot K_p.$$

Furthermore, if $(g,\varsigma) = (n,1)\cdot(a,\varsigma')\cdot(k,1)$ *then the map*

$$(g,\varsigma) \longmapsto a \pmod{Z(G(\mathbb{Q}_p))}$$

is well-defined.

2.1.2 Heights

For $x \in \prod_{v|p} x_v$, let $|x|_p = \max_{v|p}\{|x_v|_v\}$. Let $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G(\mathbb{Q}_p)$ and define

$$||g||_p = \begin{cases} \max(|a|_p, |b|_p, |c|_p, |d|_p), & p < \infty, \\ \sqrt{|a|_\infty^2 + |b|_\infty^2 + |c|_\infty^2 + |d|_\infty^2}, & p = \infty. \end{cases}$$

This is called the height or norm of g. We extend this to $\overline{G(\mathbb{Q}_p)}$ by $||(g,\zeta)||_p = ||g||_p$.

2.1.3 Characteristic functions of Weyl chambers

Let $P_0 = B$ denote the minimal parabolic defined above Lemma 2.3. This will be our fixed minimal parabolic in the set-up of [A2, A3]. The set of all standard parabolics is simply the set $\{B, G\}$. Let $P = MN$ denote a standard parabolic and let A_P denote the center of M. Thus $A_B = A$ and $A_G = \{\pm 1\}$.

Define $H = H_B : \overline{G(\mathbb{Q}_p)} \to \mathbb{R}$ by $H(g) = \log|t|_{\mathbb{Q}_p}$, where $g = nak$ and $a = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \in A(\mathbb{Q}_p)$, $t \in \mathbb{Q}_p$.

Lemma 2.5. *(a)* $H(aa') = H(a) + H(a')$, $\forall a, a' \in \overline{A(\mathbb{A})}$.
(b) $H(kxk') = H(x)$, $\forall k, k' \in K$.
(c) $|H(x)| << 1 + \log||x||$, $\forall x \in \overline{G(\mathbb{A})}$.
(d) For all $x \in \overline{G(\mathbb{A})}$, $\gamma \in B(\mathbb{Q})$, $H(\gamma x) = H(x)$.

Let $X(A) = Hom_{\mathbb{Q}}(A, \mathbb{G}_m) \cong \mathbb{Z}$ (send the character χ_n defined by $\chi_n(a) = t^n$ to $n \in \mathbb{Z}$) and identify $\mathfrak{a} = Hom(X(A), \mathbb{R})$ with $\mathbb{R}$ (send $X \in \mathfrak{a}$ to $X(\chi_1)$). Let $X_\alpha \in \mathfrak{a}$ denote the map such that $X_\alpha(\chi_1) = 1$. Let $\mathfrak{a}^* = Hom(\mathfrak{a}, \mathbb{R})$ and let $\alpha \in \mathfrak{a}^*$ be defined by $\alpha(X) = 2X(\chi_1)$. We may identify $\mathfrak{a}$ and $\mathfrak{a}^*$ by means of the linear map which sends $X_\alpha \in \mathfrak{a}$ to $\alpha \in \mathfrak{a}^*$. Note $\alpha(X_\alpha) = 2$. We also let $\alpha \in X(A)$ be the element such that $\alpha(a) = t^2$, where $a = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \in A$, so α the simple positive root of (G, A). By duality,

there is an $\varpi \in (\mathfrak{a}^*)^* \cong \mathfrak{a}$ such that $\varpi(\alpha) = 1$, so ϖ is the dual root of α. We may identify ϖ with $\frac{1}{2}X_alpha$.

Let $\mathfrak{a}^+ = \{X \in \mathfrak{a} \mid \alpha(X) > 0\}$ denote the positive Weyl chamber and let $(\mathfrak{a}^*)^+ = \{X \in \mathfrak{a}^* \mid \varpi(X) > 0\}$ denote the dual positive Weyl chamber. The elements of $\mathfrak{a}^+$ are called **regular**. Let τ_B denote the characteristic function of $\mathfrak{a}^+$ and let $\tau_B^\wedge$ denote the characteristic function of $(\mathfrak{a}^*)^+$. Define $\tau_G = \tau_G^\wedge = 1$.

More generally, if $P_1 \subset P_2$ denote two standard parabolics. let $\Delta_1^2 \subset \Delta_1$, τ_1^2, and $\tau_1^{\wedge,2} = (\tau^\wedge)_1^2$ be as in [A2]. If we identify $\mathfrak{a} = \mathfrak{a}_B$ with R as above then we have

$$\begin{aligned} \tau_0 &= \tau_B = \tau_0^G = \tau_B^G = \text{char. fcn. of } \{u \in \mathbb{R} \mid u > 0\} \\ \tau_0^\wedge &= \tau_B^\wedge = \tau_0^{\wedge,G} = \tau_B^{\wedge,G} = \text{char. fcn. of } \{u \in \mathbb{R} \mid u > 0\} \\ \tau_1^2 &= \begin{cases} \tau_1, & \text{if } P_2 = G, \\ 1, & \text{if } P_2 \neq G, \end{cases} \end{aligned} \tag{2.4}$$

Finally, if σ_1^2 is as in [A2], then

$$\sigma_1^2 = \begin{cases} \tau_1, & \text{if } P_1 = B,\ P_2 = G, \\ 0, & \text{if } P_2 = P_1 = B, \\ 1, & \text{if } P_2 = P_1 = G. \end{cases} \tag{2.5}$$

2.1.4 Algebras of functions

If H is the group of F-rational points of a reductive algebraic group over a real or p-adic field F, let $C_c^\infty(H)$ denote the usual algebra of compactly supported smooth functions under convolution and $C^\infty(H//K)$ the space of smooth functions which are bi-K-ivariant. A function $f : \overline{G(\mathbb{Q}_p)} \to \mathbb{C}$ such that

$$f(g, \varsigma) = \varsigma \cdot f(g, 1) \tag{2.6}$$

will be called **genuine**. Let

$$\begin{aligned} C_c^\infty(\overline{G(\mathbb{Q}_p)})_{gen} &= \{f : \overline{G(\mathbb{Q}_p)} \to \mathbb{C} \mid (i)\ f|_{G(\mathbb{Q}_p)} \in C_c^\infty(G(\mathbb{Q}_p)), \\ &\qquad (ii)\ f \text{ is genuine}\} \end{aligned}$$

denote the Schwartz-Bruhat algebra of smooth genuine functions. If $(p, 2n) = 1$ then let

$$\begin{aligned} \mathcal{H}(\overline{G(\mathbb{Q}_p)}, K_p) &= \{f : \overline{G(\mathbb{Q}_p)} \to \mathbb{C} \mid (i)\ f|_{G(\mathbb{Q}_p)} \in C^\infty(G(\mathbb{Q}_p)//K_p), \\ &\qquad (ii)\ f \text{ is genuine}\} \end{aligned}$$

denote the Hecke algebra of genuine spherical functions.

2.1.5 Measures

Let dk denote the Haar measure on K_p normalized so that K_p has volume 1. Fix an additive non-trivial character ψ_v of k_v, for each $v \in \Sigma_k$, and let dn denote a Haar measure on $N(\mathbb{Q}_p)$ which is induced by a self-dual measure on the additive group

$$\bigoplus_{v\in\Sigma_k, v|p} k_v$$

(with respect to the ψ_v, $v|p$). Let da be a Haar measure on $A(\mathbb{Q}_p)$. If T is a Cartan subgroup of G defined over $\mathbb{Q}_p$ and $\psi : T \to A$ is a $\overline{\mathbb{Q}_p}$-isomorphism given by $h \longmapsto t^h$, some $h \in G(\mathbb{Q}_p)$, then we define dt to be the pull-back of da from A to T as in [Rog]. If T' is another Cartan subgroup of G defined over $\mathbb{Q}_p$ which is stably conjugate to T then we assume that its Haar measure dt' is compatible with dt as in [Rog]. Let dg be a Haar measure on $G(\mathbb{Q}_p)$ such that $dg = \delta_B(a)^{1/2} dk da dn$, $g = kan$, where

$$\delta_B(a) = |t|^2, \quad a = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}. \tag{2.7}$$

We extend these measures to the n-fold cover by imposing the discrete measure $d\varsigma$ on μ_n which gives μ_n total volume 1. For example, the measure on $\overline{G(\mathbb{Q}_p)}$ will be denoted $d\overline{g} = dg \cdot d\varsigma$.

2.2 Global aspects

Define a map $\beta = \beta_{\mathbb{A}} : G(\mathbb{A}) \times G(\mathbb{A}) \to \mu_n$ by

$$\beta(g_1, g_2) = \prod_{p\leq\infty} \beta_p(g_{1,v}, g_{2,v}).$$

This product converges by Lemma 2.3(b) and the definition of the restricted direct product. As a product of local 2-cocycles it inherits the cocycle property and therefore defines an element of $H^2(G(\mathbb{A}), \mu_n)$ associated to an n-fold central topological covering group $\overline{G(\mathbb{A})}$ fitting into the following exact sequence

$$1 \to \mu_n \to \overline{G(\mathbb{A})} \xrightarrow{p} G(\mathbb{A}) \to 1. \tag{2.8}$$

The rules of multiplication, inverses, and conjugation are still as in equations (2.1)–(2.3) above.

Note that we may identify $\overline{G(\mathbb{A})}$ with a subgroup of the restricted direct product $\prod_{v\in\Sigma_k}\overline{SL_2(k_v)}$. Again, if $H < G(\mathbb{A})$ is a subgroup then we define $\overline{H} = p^{-1}(H) < \overline{G(\mathbb{A})}$.

Lemma 2.6. *(a)* $\overline{G(\mathbb{Q})} \cong G(\mathbb{Q})\times\mu_n$ *(direct product), where the isomorphism in* $\cong$ *will be described below,*

(b) $\overline{N(\mathbb{Q})} = N(\mathbb{Q}) \times \mu_n$ *(direct product),*

(c) if $K(n)^f = \prod_{p|n} K_{n,p}$, $K(n)_f = \prod_{(p,n)=1} K_{0,p}$, *and* $K(n) = K(n)^f \times K(n)_f$ *then*

$$\overline{K(n)} = K(n) \times \mu_n, \quad \text{(directproduct)}.$$

(d) The center of $\overline{G(\mathbb{A})}$ *is* $Z(\overline{G(\mathbb{A})}) = Z(G(\mathbb{A})) \times \mu_n$,

(e) For $a_i = \begin{pmatrix} t_i & 0 \\ 0 & t_i^{-1} \end{pmatrix} \in A(\mathbb{A})$, $i = 1, 2$, *and* $\varsigma_i \in \mu_n$, *we have* $(a_1, \varsigma_1)^{-1}\cdot (a_2, \varsigma_2) \cdot (a_1, \varsigma_1) = (a_2, (t_1, t_2)_n^2\varsigma_2)$. *Here* $(...,...)_n$ *denotes the global Hilbert symbol.*

proof: This is an immediate consequnce of the above lemmas or well-known properties of the global Hilbert symbol [We1]. □

From [K2], we see that the splitting of the above exact sequence indicated in part (a) of the above lemma is not given by $\gamma \longmapsto (\gamma, 1)$ but by $\gamma \longmapsto (\gamma, s_{\mathbb{A}}(\gamma)^{-1})$, where

$$s_{\mathbb{A}}(\gamma) = \prod_{v\in\Sigma_k} s_v(\gamma),$$
$$s_v(\gamma) = \begin{cases} (c,d)_{v,n}^{-1}, & cd \neq 0 \text{ and } n \text{ does not divide } v(c) \text{ and } v < \infty, \\ 1, & \text{otherwise.} \end{cases} \tag{2.9}$$

Here $\gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SL_2(k)$.

From now on, we identify $G(\mathbb{Q})$ *with its image in* $\overline{G(\mathbb{A})}$ *under this splitting (and the diagonal embedding).* We identify $N(\mathbb{A})$ with the subgroup $N(\mathbb{A}) \times 1 \subset \overline{G(\mathbb{A})}$.

Let $K_0 = \Pi_{p\leq infty} K_{0,p}$ denote the "usual" maximal compact subgroup of $G(\mathbb{A})$. Here, if $p < \infty$ then $K_{0,p} = G(\mathbb{Z}_p)$ and otherwise

$$K_{0,\infty} = \prod_{v|\infty, v\in\Sigma_k} K_v,$$

where $K_v = SO(2,\mathbb{R})$ if v is real and $K_v = SU(2,\mathbb{C})$ if v is complex.

The following result is essentially due to Kubota [K2, Proposition 5] (see also [Ge2, Lemma 3.2])

Lemma 2.7. *For each compact open subgroup $K = \Pi_{p\leq\infty} K_p \subset K_0 \subset G(\mathbb{A})$, there are discrete subgroups of cofinite volume $\Gamma_i \subset \overline{G(\mathbb{R})}$ such that*

$$G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}/\overline{K} \cong \coprod_i \Gamma_i\backslash\overline{G(\mathbb{R})}/\overline{K_\infty}.$$

2.2.1 NAK decompositions

By the above lemma, we may identify $N(\mathbb{A})$ with a subgroup of $\overline{G(\mathbb{A})}$ via $n \longmapsto (n,1)$. We may identify $K_0 = \prod_{p\leq\infty} K_{0,p}$ with a *subset* of $\overline{G(\mathbb{A})}$ via $k \longmapsto (k,1)$.

Lemma 2.8. *We have*

$$\overline{G(\mathbb{A})} = N(\mathbb{A}) \cdot \overline{A(\mathbb{A})} \cdot K_0.$$

Furthermore, if $(g,\varsigma) = (n,1)\cdot(a,\varsigma')\cdot(k,1)$ then the map $(g,\varsigma) \longmapsto a \pmod{Z(G(\mathbb{A}))}$ is well-defined.

2.2.2 Heights

For $g \in \prod_{p\leq\infty} g_p \in G(\mathbb{A})$, $g_p = \begin{pmatrix} a_p & b_p \\ c_p & d_p \end{pmatrix} \in G(\mathbb{Q}_p)$, define

$$||g|| = \prod_{p\leq\infty} ||g_p||_p \tag{2.10}$$

This is called the height or norm of g. Extend this to $\overline{G(\mathbb{A})}$ by $||(g,\zeta)|| = ||g||$.

Lemma 2.9. *(a) $||x|| \geq 1$, $\forall x \in \overline{G(\mathbb{A})}$.*
(b) $||kxk'|| = ||x||$, $\forall k,k' \in K$.
(c) $||xy|| \leq ||x||\cdot||y||$, $\forall x,y \in \overline{G(\mathbb{A})}$.
(d) There is a constant $N > 0$ such that for all $x \in \overline{G(\mathbb{A})}$, $||x^{-1}|| \leq ||x||^N$.

2.2.3 Algebras of functions

A function $f : \overline{G(\mathbb{A})} \to \mathbb{C}$ such that (2.6) holds will be called **genuine**. Let $\mathcal{H}(\overline{G(\mathbb{A})})$ denote the algebra of functions generated by the elements of the restricted tensor product $\bigotimes_{p\leq\infty} C_c^\infty(\overline{G(\mathbb{Q}_p)})_{gen}$, where almost all f_p in the product $f = \otimes_{p\leq\infty} f_p \in \bigotimes_{p\leq\infty} C_c^\infty(\overline{G(\mathbb{Q}_p)})_{gen}$ are spherical in the sense of §2.1.4.

2.2.4 Measures

Let dk denote the Haar measure on $K = \prod_{p\leq\infty} K_p$ defined by $dk = \prod_{p\leq\infty} dk_p$. Let dg be the Tamagawa measure on $G(\mathbb{A})$ [We2]. For any Cartan subgroup T of G defined over $\mathbb{Q}$, let dt denote the global Tamagawa measure on $T(\mathbb{A})$. If $T = A$ then this will be denoted by da. Let dn denote the product measure on $N(\mathbb{A})$.

We extend these measures to the n-fold cover by imposing the discrete measure $d\varsigma$ on μ_n which gives μ_n total volume 1. For example, the measure on $\overline{G(\mathbb{A})}$ will be denoted $d\overline{g} = dg \cdot d\varsigma$.

3 The rank 0 case

It might be appropriate here to discuss the rank 0 case. The metaplectic covers of $A(\mathbb{A})$ are not in general abelian but nonetheless, the trace formula takes an especially simple form in this case. In this section, we shall prove the trace formula for metaplectic covers of $A(\mathbb{A})$ and apply it to prove a "matching" between automorphic forms on $A(\mathbb{A})$ and on metaplectic covers of $A(\mathbb{A})$. At the same time, we construct invariant distributions $I_{\mathcal{O}}^A$ and I_χ^A which will be needed in an inductive definition of certain invariant distributions on $\overline{G(\mathbb{A})}$ in Section 6.

3.1 Metaplectic lifting over the split torus

Suppose that $k/\mathbb{Q}$ is a number field containing the $n-th$ roots of unity. (If $n > 2$ then this implies k has no real places, i.e, $k_v = \mathbb{C}$ for all $v \in \Sigma_k^\infty$.) Let A and $A(\mathbb{A})^1$ be as in the previous section, so $A(\mathbb{Q}) = k^\times$. *We shall often identify* $A(\mathbb{A})$ *with* $\mathbb{A}_k^\times$ *in this section.* Let $(.,.)_n : A(\mathbb{A}) \times A(\mathbb{A}) \cong \mathbb{A}_k^\times \times \mathbb{A}_k^\times \to \mu_n$ denote the global Hilbert symbol defined by [We1, section XIII.5,

proposition 7] as a product of local Hilbert symbols $(...,...)_{n,v}$, where μ_n is the group of n-th roots of unity. Let μ_n act on $G(\mathbb{A}) = \mathbb{A}_k^\times$ trivially. A 2-cocycle on $A(\mathbb{A})$ with values in μ_n is a continuous mapping $\beta : A(\mathbb{A}) \times A(\mathbb{A}) \to \mu_n$ satisfying

$$\beta(g_2, g_3)\beta(g_1, g_2 g_3) = \beta(g_1, g_2)\beta(g_1 g_2, g_3), \tag{3.1}$$

$g_1, g_2, g_3 \in \mathbb{A}_k^\times$. We also denote by β the restriction $\beta : A(\mathbb{A})^1 \times A(\mathbb{A})^1 \to \mu_n$. This is a product of local 2-cocycles, denoted β_v. Let $\overline{A(\mathbb{A})}^1$ denote the n-fold metaplectic cover of $A(\mathbb{A})^1$ defined by the 2-cocycle β given by

$$\beta(g_1, g_2) = (g_1^{-1}, g_2^{-1})_n \cdot (-g_1 g_2^{-1}, g_1^{-1} g_2^{-1})_n. \tag{3.2}$$

The fact that this satisfies the condition above may be verified by multiplying each term out and using the properties of the Hilbert symbol listed in [Sha, Proposition 60, p. 176] (this is long but straightforward and in any case well-known). By definition of group extension, multiplication on $\overline{A(\mathbb{A})^1} = A(\mathbb{A})^1 \times \mu_n$ (as *sets*) is given by

$$(g, \varsigma) \cdot (g', \varsigma') = (gg', \beta(g_1, g_2)\varsigma\varsigma'), \quad g, g' \in A(\mathbb{A}), \ \ \varsigma, \varsigma' \in \mu_n. \tag{3.3}$$

This is a central extension of $A(\mathbb{A})^1$ (i.e., μ_n belongs to the center of the metaplectic cover) which fits into the following exact sequence

$$1 \to \mu_n \to \overline{A(\mathbb{A})^1} \to A(\mathbb{A})^1 \to 1,$$

where $\mu_n \to \overline{A(\mathbb{A})^1}$ is the inclusion of μ_n into the second component of $\overline{A(\mathbb{A})}^1$, and the map $\overline{A(\mathbb{A})^1} \to A(\mathbb{A})^1$ is the projection map.

Note that $\overline{A(\mathbb{A})}^1 = \overline{A(\mathbb{A})^1}$.

Using the local 2-cocycles β_v, we may in turn define local n-fold covers $\overline{k_v^\times}$ of $k_v^\times$, $v \in \Sigma_k$. The rule for multiplication on the local covers

$$\overline{A(\mathbb{Q}_p)} = \overline{\prod_{v|p} k_v^\times} \subset \prod_{v|p} \overline{k_v^\times}$$

of $\prod_{v|p} k_v^\times$ is the same, except one must replace β by $\prod_{v|p} \beta_v$. The topology on $\overline{k_v^\times}$ is defined by stipulating

$$\{U_v \times 1 \mid \beta_v(U_v, U_v) = 1 \text{ and } 1 \in U_v \subset k_v^\times \text{ is open}\}$$

to be a basis of open neighborhoods of the identity. If $(p,n)=1$ and if $v|p$ then the n-fold covers $\overline{A(\mathbb{Z}_p)}$ and $\overline{\mathcal{O}_v^\times}$ split by the properties of the Hilbert symbol listed in [We1, section XIII.5].

The global cover $\overline{A(\mathbb{A})}$ is not the restricted direct product of the local covers $\overline{k_v^\times}$. However, there is an embedding

$$\overline{A(\mathbb{A})} \hookrightarrow \prod_{v\in\Sigma_k}{}' \overline{k_v^\times}.$$

We call a representation π of $\overline{A(\mathbb{A})^1}$ **genuine** if it satisfies

$$\pi(g,\varsigma) = \varsigma^{-1}\cdot\pi(g,1). \tag{3.4}$$

A *function* satisfying this property will sometimes be called **anti-genuine**. Similarly, we define a representation π of $\overline{A(\mathbb{A})} = \overline{\mathbb{A}_k^\times}$ or of $\overline{k_v^\times}$ to be **genuine** if it satisfies (3.4).

If $H < \overline{A(\mathbb{A})}$ a locally compact totally disconnected group of type 1, let $\Pi(H)$ denote the set of equivalence classes of admissible genuine representations of H and let $\Pi_u(H)$ denote the subset of unitary ones. Let $\Pi_f(\overline{A(\mathbb{A})})$ denote the subset (equivalance classes of) representations in $\Pi(\overline{A(\mathbb{A})})$ which can be written as the restriction to $\overline{A(\mathbb{A})}$ of a *factorizable* representation $\overline{\pi}$ of the restricted direct product

$$\Pi(\prod_v \overline{k_v^\times}) = \prod_v \Pi(\overline{k_v^\times}).$$

Let $\Pi_f(\overline{A(\mathbb{A})}^1)$ denote the set of equivalence classes of representations of $\overline{A(\mathbb{A})^1}$ which can be written as the restriction to $\overline{A(\mathbb{A})^1}$ of an element of $\Pi_f(\overline{A(\mathbb{A})})$. Let $\Pi_{u,f}(\overline{A(\mathbb{A})})$, $\Pi_{u,f}(\overline{A(\mathbb{A})^1})$ denote the corresponding subset of factorizable unitary representations.

Lemma 3.1. *([Ge2, §3])*

$$\begin{aligned}
\Pi_f(\overline{A(\mathbb{A})}) &= \Pi(\overline{A(\mathbb{A})}),\\
\Pi_{u,f}(\overline{A(\mathbb{A})}) &= \Pi_u(\overline{A(\mathbb{A})}),\\
\Pi_f(\overline{A(\mathbb{A})^1}) &= \Pi(\overline{A(\mathbb{A})^1}),\\
\Pi_{u,f}(\overline{A(\mathbb{A})^1}) &= \Pi_u(\overline{A(\mathbb{A})^1})
\end{aligned}$$

See also [Ge1], [Fla] for the non-metaplectic version.

We shall sketch a proof of the following

Theorem 3.2. *There is a natural 1-1 correspondence between the subset of all genuine irreducible representations $\overline{\chi} \in \Pi(\overline{A(\mathbb{A})}^1)$ of $\overline{A(\mathbb{A})^1}$ and characters of the form $\chi^{n/2}$ (n even), χ^n (n odd), for $\chi \in \Pi(A(\mathbb{A})^1)$.*

First we prove a trace formula for $\overline{A(\mathbb{A})}^1$.

3.2 Irreducible representations of $\overline{A(\mathbb{A})}^1$

For any group H, let Let

$$H^m = \{x \in H \mid x = y^m, \text{ some } y \in H\}. \tag{3.5}$$

Let

$$N = \begin{cases} n, & n \text{ odd}, \\ n/2, & n \text{ even}. \end{cases} \tag{3.6}$$

We have, by the properties of the Hilbert symbol, $\overline{A(\mathbb{A})^{1,n}} = A(\mathbb{A})^{1,n} \times \mu_n$, that is to say, the metaplectic cover splits over $A(\mathbb{A})^{1,n}$. By Lemma 2.6(e) above, the cover $\overline{A(\mathbb{A})^{1,N}}$ is abelian (though it does not split).

For $\chi \in \Pi(\overline{A(\mathbb{A})^1})$, $h \in \overline{A(\mathbb{A})}$, define $\chi^h(g) = \chi(h^{-1}gh)$, for $g \in \overline{A(\mathbb{A})}^1$. We call such a χ **self-conjugate** if there is an h as above such that $\chi = \chi^h$ in $\Pi(\overline{A(\mathbb{A})^1})$. If $h = (x, \varsigma)$, $\varsigma \in \mu_n$, then

$$\chi^h(g) = \chi(h^{-1}gh) = \chi(x', (x, x')_n^2 \varsigma'), \qquad g = (x', \varsigma') \in \overline{A(\mathbb{A})^1},$$

If $p < \infty$, let $C_p \subset A(\mathbb{Q}_p)$ denote a subgroup of finite index such that $\overline{C_p} \subset \overline{A(\mathbb{Q}_p)}$ is an abelian subgroup. (In particular, we may take $C_p = A(\mathbb{R})^N$.) If $p = \infty$ let $C_p = A(\mathbb{R})$. Let

$$C_{\mathbb{A}} = \prod_{p \le \infty} C_p \tag{3.7}$$

denote the restricted direct product.

We have the following result (due to Clifford, Mackey, ...)

Lemma 3.3. *If $\chi \in \Pi(\overline{C_{\mathbb{A}}})$ is not self-conjugate then*

$$Ind_{\overline{C_{\mathbb{A}}}}^{\overline{A(\mathbb{A})}}(\chi) \cong Ind_{\overline{C_{\mathbb{A}}}}^{\overline{A(\mathbb{A})}}(\chi^h), \quad h \in \overline{A(\mathbb{A})},$$

is irreducible and belongs to $\Pi(\overline{A(\mathbb{A})})$.

Remark 3.4. *By analogy with Mackey's irreduciblity criterion in [Se, §7.4], the converse to the avove result also holds. The detailed proof of the local result can be found in [J3].*

proof: We must first reduce this down to a statement about local representations and then prove the local statement.

Pick a $p \leq \infty$. By Clifford-Mackey theory ([J3], [Se, §7.4]), if $\chi \in \Pi(\overline{A(\mathbb{Q}_p)})$ is not self-conjugate then

$$Ind_{\overline{C_p}}^{\overline{A(\mathbb{Q}_p)}}(\chi) \cong Ind_{\overline{C_p}}^{\overline{A(\mathbb{Q}_p)}}(\chi^h), \quad h \in \overline{A(\mathbb{Q}_p)}, \tag{3.8}$$

is irreducible and belongs to $\Pi(\overline{A(\mathbb{Q}_p)})$. We may take the restricted direct product of these local representations to obtain an irreducible (by Schur's lemma) factorizable representation on the restricted direct product of of these local covers. $\square$

In our case, it is easy to see that for any $\chi \in \Pi(\overline{C_p})$, the restriction of $Ind_{\overline{C_p}}^{\overline{A(\mathbb{Q}_p)}}(\chi)$ to the abelian subgroup $\overline{C_p}$ must decompose into a direct sum of characters:

$$Ind_{\overline{C_p}}^{\overline{A(\mathbb{Q}_p)}}(\chi)|_{\overline{A_p}} \cong \bigoplus_{i=1}^{d} m_i \chi_i, \quad \chi_i \in \Pi(\overline{C_p}), \tag{3.9}$$

where m_i, d is are some integers and one of these χ_i must be χ itself. By the isomorphism (for any $h \in \overline{A(\mathbb{Q}_p)}$)

$$Ind_{\overline{C_p}}^{\overline{A(\mathbb{Q}_p)}}(\chi) \cong Ind_{\overline{C_p}}^{\overline{A(\mathbb{Q}_p)}}(\chi^h), \quad h \in \overline{A(\mathbb{Q}_p)}, \tag{3.10}$$

one of these χ_i must be equal to χ^h. In fact, we have

Lemma 3.5. *([GeKn, Lemma 2.1]) If $\chi \in \Pi(\overline{C_p})$ then all the $m_i's$ in (3.9) are equal (to m, say) and*

$$Ind_{\overline{C_p}}^{\overline{A(\mathbb{Q}_p)}}(\chi)|_{\overline{C_p}} \cong \bigoplus_{h \in \overline{A(\mathbb{Q}_p)}/\overline{A(\mathbb{Q}_p)}_\chi} m\chi^h,$$

where

$$\overline{A(\mathbb{Q}_p)}_\chi = \{a \in \overline{A(\mathbb{Q}_p)} \mid \chi^a = \chi\}.$$

Remark 3.6. *By analogy with [Se, ch 7], one might expect there to be a more general form of this result, closely related to Frobenius reciprocity.*

Remark 3.7. *An aside: Even using the explicit local recipocity law of (Sen, Wiles, ...) it does not appear to be easy to explicitly determine C_p such that $A(\mathbb{Q}_p)/C_p$ is minimal.*

3.3 The trace formula for $\overline{A(\mathbb{A})^1}$

There is natural projection $pr : \overline{A(\mathbb{A})} \to A(\mathbb{A})$, $(x, \varsigma) \longmapsto x$. Let $\mathcal{S}(\overline{A(\mathbb{A})}^1)$ denote the collection of all genuine functions f on $\overline{A(\mathbb{A})^1}$ such that $f \circ pr \in \mathcal{S}(A(\mathbb{A})^1)$. Let

$$H = L^2(\overline{A(\mathbb{Q})} \backslash \overline{A(\mathbb{A})^1})_{gen},$$

where $\overline{A(\mathbb{Q})}$ is regarded as a subgroup of $\overline{A(\mathbb{A})^1}$ via the diagonal embedding. Let R denote the right regular representation:

$$R : \overline{A(\mathbb{A})^1} \to End(H),$$

defined by $(R(y)\varphi)(x) = \varphi(xy)$, $x, y \in \overline{A(\mathbb{A})}^1$, $\varphi \in H$. For $\overline{f} \in \mathcal{S}(\overline{A(\mathbb{A})}^1)$, let

$$R(\overline{f}) = \int_{\overline{A(\mathbb{A})}^1} \overline{f}(x) R(x) dx.$$

Since $R(x) : H \to H$ is a unitary operator (in particular bounded), this integral converges. This operator $R(\overline{f}) : H \to H$ may be written

$$\begin{aligned} (R(\overline{f})\varphi)(x) &= \textstyle\int_{\overline{A(\mathbb{A})}^1} \overline{f}(x)(R(y)\varphi)(x) dy, \quad \varphi \in H \\ &= \textstyle\int_{\overline{A(\mathbb{A})}^1} \overline{f}(x^{-1}y)\varphi(y) dy \\ &= \textstyle\int_{A(\mathbb{Q})\backslash\overline{A(\mathbb{A})}^1} \sum_{\delta \in A(\mathbb{Q})} \overline{f}(x^{-1}\delta y)\varphi(\delta y) dy \\ &= \textstyle\int_{A(\mathbb{Q})\backslash\overline{A(\mathbb{A})^1}} \varphi(y) \sum_{\delta \in A(\mathbb{Q})} \overline{f}(x^{-1}\delta y) dy. \end{aligned} \tag{3.11}$$

Let

$$K(x, y) = K_{\overline{f}}(x, y) = \sum_{\delta \in A(\mathbb{Q})} \overline{f}(x^{-1}\delta y).$$

For each $\overline{\chi} \in \Pi(\overline{A(\mathbb{A})}^1)$, let

$$\begin{aligned} \overline{\chi}(\overline{f}) &= \textstyle\int_{\overline{A(\mathbb{A})}^1} \overline{f}(y)\overline{\chi}(y) dy, \\ tr\ \overline{\chi}(\overline{f}) &= \textstyle\int_{\overline{A(\mathbb{A})^1}} \overline{f}(y) tr\ (\overline{\chi}(y)) dy = \overline{f}^{\wedge}(\overline{\chi}). \end{aligned} \tag{3.12}$$

We call the latter integral the **Fourier transform** of $\overline{f}$ at $\overline{\chi}$, and the former integral the **operator-valued Fourier transform** of $\overline{f}$ at $\overline{\chi}$. Note that these are 0 if $\overline{f}$ is not genuine.

Finally, for each $h \in \overline{A(\mathbb{A})^1}$, let $\overline{A(\mathbb{A})}_h^1$ denote the **centralizer** of h in $\overline{A(\mathbb{A})^1}$:

$$\overline{A(\mathbb{A})}_h^1 = \{A \in \overline{A(\mathbb{A})}^1 \mid ghg^{-1} = h\}. \tag{3.13}$$

Theorem 3.8. *(a)* $tr\ R(\overline{f}) = \int_{A(\mathbb{Q})\backslash\overline{A(\mathbb{A})}^1} K_{\overline{f}}(x,x)dx$.

(b) For each $\overline{\chi} \in \Pi(\overline{A(\mathbb{A})}^1)$*, there is an integer* $m(\overline{\chi}) \geq 0$ *such that*

$$R(\overline{f}) = \bigoplus_{\overline{\chi}\in\Pi_u(\overline{A(\mathbb{A})}^1)} m(\overline{\chi})\overline{\chi}(\overline{f}).$$

Thus we have

$$\begin{aligned} tr\ R(\overline{f}) &= \textstyle\sum_{\overline{\chi}\in\Pi(\overline{A(\mathbb{A})}^1)} m(\overline{\chi})tr(\overline{\chi}(\overline{f})) \\ &= \textstyle\int_{A(\mathbb{Q})\backslash\overline{A(\mathbb{A})}^1} \sum_{\delta\in A(\mathbb{Q})} \overline{f}(x^{-1}\delta x)dx \\ &= \textstyle\int_{A(\mathbb{Q})\backslash\overline{A(\mathbb{A})}^1} \sum_{\{\delta\}} \sum_{\gamma\in A(\mathbb{Q})_\delta\backslash A(\mathbb{Q})} \overline{f}(x^{-1}\gamma^{-1}\delta\gamma x)dx \\ &= \textstyle\sum_{\{\delta\}} \int_{A(\mathbb{Q})\backslash\overline{A(\mathbb{A})}^1} \sum_{\gamma\in A(\mathbb{Q})_\delta\backslash A(\mathbb{Q})} \overline{f}(x^{-1}\gamma^{-1}\delta\gamma x)dx \\ &= \textstyle\sum_{\{\delta\}} \int_{A(\mathbb{Q})_\delta\backslash\overline{A(\mathbb{A})}^1} \overline{f}(x^{-1}\delta x)dx \\ &= \textstyle\sum_{\{\delta\}} vol(A(\mathbb{Q})_\delta\backslash\overline{A(\mathbb{A})}_\delta^1) \sum_{x\in\overline{A(\mathbb{A})}_\delta^1\backslash\overline{A(\mathbb{A})}^1} \overline{f}(x^{-1}\delta x), \end{aligned} \tag{3.14}$$

where $\{\delta\}$ runs over a complete set of representatives of $\overline{A(\mathbb{A})}^1$-conjugacy classes of $A(\mathbb{Q})$.

This proves the

Theorem 3.9. *(trace formula on* $\overline{A(\mathbb{A})}^1$*): If* $\overline{f} \in \mathcal{S}(\overline{A(\mathbb{A})}^1)$ *then*

$$\sum_{\overline{\chi}\in\Pi_u(\overline{A(\mathbb{A})}^1)} m(\overline{\chi})tr(\overline{\chi}(\overline{f})) = \sum_{\{\delta\}} vol(A(\mathbb{Q})_\delta\backslash\overline{A(\mathbb{A})}_\delta^1) \sum_{x\in\overline{A(\mathbb{A})}_\delta^1\backslash\overline{A(\mathbb{A})^1}} \overline{f}(x^{-1}\delta x),$$

where $\{\delta\}$ *runs over a complete set of representatives of* $\overline{A(\mathbb{A})}^1$*-conjugacy classes of* $A(\mathbb{Q})$.

3.4 proof of the correspondence

Let $\delta \in A(\mathbb{Q})$.

We now prove the correspondence theorem mentioned at the beginning of this subsection. To do this we need to relate the above twisted trace formula on $\overline{A(\mathbb{A})}^1$ to the ordinary trace formula (Poisson summation) on $A(\mathbb{A})^N$. This boils down to relating the metaplectic orbital integrals

$$\int_{\overline{A(\mathbb{A})}^1_\delta \backslash \overline{A(\mathbb{A})^1}} \overline{f}(x^{-1}\delta x)dx$$

on $\overline{A(\mathbb{A})}^1$ to ordinary orbital integrals on $A(\mathbb{A})^N$ (which are "trivial" since $\overline{A(\mathbb{A})}$ is abelian).

Let $\overline{f}$ be factorizable:

$$\overline{f}(x) = \prod_{p\leq\infty} \overline{f}_p(x_p), \qquad \overline{f}_p \in \mathcal{S}(\overline{A(\mathbb{Q}_p)}),$$

so

$$\int_{\overline{A(\mathbb{A})}^1_\delta \backslash \overline{A(\mathbb{A})^1}} \overline{f}(x^{-1}\delta x)dx = \prod_{p\leq\infty} \sum_{x_p \in \mathbb{Q}^\times_{p,\delta}\backslash\mathbb{Q}^\times_p} \overline{f}_p(\delta, (\delta, x_p)_n^2 s_{\mathbb{A}}(\delta)^{-1}), \tag{3.15}$$

where

$$\mathbb{Q}^\times_{p,\delta} = \{x_p \in \mathbb{Q}^\times_p \mid (x_p, \delta)_n = 1\},$$
$$(x_p, \delta)_n = \textstyle\prod_{v|p}(x_v, \delta_v)_{v,n},$$

for $x_p = \prod_{v|p} x_v,\ \delta = \prod_{v|p} \delta_v$.

Let

$$f_p(y) = \sum_{x \in \mathbb{Q}^\times_{p,y}\backslash\mathbb{Q}^\times_p} \overline{f}_p(y, (y, x)_n^2 s_{\mathbb{A}}(y)^{-1}), \tag{3.16}$$

for $y \in \prod_{v|p} k_v^\times \cong A(\mathbb{Q}_p)$.

Remark 3.10. *Let $(h(t), 1) \in \overline{A(\mathbb{Q}_p}$ be the element such that $h(t)$ is diagonal with diagonal entries t, t^{-1}. For $y \in \mathbb{Q}_p^\times$, we have $(h(y), 1)^{(h(x),1)} = (h(y), (y, x)_n^2)$. Therefore, if $\overline{f}_p$ is genuine then the sum in (3.16) is zero unless $y \in (\prod_{v|p} k_v^\times)^N \cong A(\mathbb{Q}_p)^N$.*

Lemma 3.11. *One may choose Haar measures so that*

(a) If $p < \infty$ then the map $\overline{f}_p \longmapsto f_p$ defines a map $\mathcal{S}(\overline{A(\mathbb{Q}_p)}) \to \mathcal{S}(A(\mathbb{Q}_p))$.

(b) If $p = \infty$ then the map $\overline{f}_p \longmapsto f_p$ defines a map $\mathcal{S}(\overline{A(\mathbb{R})}) \to \mathcal{S}(A(\mathbb{R}))$.

(c) If $p < \infty$ and if $\overline{f}_p$ is a unit in the convolution algebra $\mathcal{S}(\overline{A(\mathbb{Q}_p)})$ and if $\overline{f}_p \longmapsto f_p$ then f_p is the unit element of the convolution algebra $\mathcal{S}(A(\mathbb{Q}_p)^N)$.

For $\overline{f} \in \mathcal{S}(\overline{A(\mathbb{A})^1})$ factorizable, let $\overline{f} \longmapsto f$ denote the map obtained by taking the product of the maps $\overline{f}_p \longmapsto f_p$. By the above lemma, this defines a map

$$\mathcal{S}(\overline{A(\mathbb{A})}^1) \to \mathcal{S}(C_{\mathbb{A}}). \tag{3.17}$$

As a corollary to the above lemma, we obtain

Lemma 3.12. *Let $\overline{f} \in \mathcal{S}(\overline{A(\mathbb{A})}^1)$ be a factorizable function and define $f \in \mathcal{S}(C_{\mathbb{A}})$ as above via $\overline{f} \longmapsto f$. Then*

$$\int_{\overline{A(\mathbb{A})}^1_\delta \backslash \overline{A(\mathbb{A})^1}} \overline{f}(x^{-1}\delta x)dx = f(\delta),$$

if $\delta \in A(\mathbb{A})^N$ and

$$\int_{\overline{A(\mathbb{A})}^1_\delta \backslash \overline{A(\mathbb{A})^1}} \overline{f}(x^{-1}\delta x)dx = 0,$$

if $\delta \in A(\mathbb{A}) - A(\mathbb{A})^N$.

Note that if $\delta \in A(\mathbb{Q})^N$ then $\overline{A(\mathbb{A})^1}_\delta = \overline{A(\mathbb{A})^1}$. We choose global measures so that

$$vol(A(\mathbb{Q})\backslash\overline{A(\mathbb{A})^1}) = vol(A(\mathbb{Q})^N \backslash A(\mathbb{A})^{1,N}).$$

This and the above lemma imply that

$$\begin{array}{c} \sum_{\{\delta\}} vol(A(\mathbb{Q})_\delta \backslash \overline{A(\mathbb{A})^1}_\delta) \int_{\overline{A(\mathbb{A})^1}_\delta \backslash \overline{A(\mathbb{A})^1}} \overline{f}(x^{-1}\delta x)dx \\ = vol(A(\mathbb{Q})^N \backslash A(\mathbb{A})^{1,N}) \sum_{\gamma \in A(\mathbb{Q})^N} f(\gamma), \end{array} \tag{3.18}$$

by equation (3.16) and the remark above.

In this case, we have, by the Poisson summation formula for $A(\mathbb{Q})^N \backslash A(\mathbb{A})^{1,N}$, we have

$$\sum_{\overline{\chi} \in \Pi_u(\overline{A(\mathbb{A})^1})} m(\overline{\chi}) tr(\overline{\chi}(\overline{f})) = \sum_{\chi \in \Pi(A(\mathbb{A})^{1,N})} f^\wedge(\chi), \tag{3.19}$$

for each factorizable $\overline{f} \in \mathcal{S}(\overline{A(\mathbb{A})}^1)$ which is genuine and $f \in \mathcal{S}(A(\mathbb{A})^{1,N})$ is defined as above via $\overline{f} \longmapsto f$.

Our correspondence theorem follows immediately from the two claims below:

Claim 1: For each fixed $\overline{\chi} \in \Pi(\overline{A(\mathbb{A})}^1)$ there is a sequence of functions $\overline{f}_m = \overline{f}_{\overline{\chi},m}$ such that

$$\lim_{n\to\infty} tr(\overline{\chi}'(\overline{f}_m)) = \begin{cases} 1, & \overline{\chi}' = \overline{\chi}, \\ 0, & \text{otherwise.} \end{cases}$$

Claim 2: For each fixed $\overline{\chi} \in \Pi(\overline{A(\mathbb{A})}^1)$ there is a unique $\chi \in \Pi(A(\mathbb{A})^{1,N})$ such that $\overline{f}_m \longmapsto f_n = f_{\chi,m}$ and

$$\lim_{m\to\infty} f_m^\wedge(\chi') = \begin{cases} 1, & \chi' = \chi, \\ 0, & \text{otherwise.} \end{cases}$$

Claim 1 is an exercise in harmonic analysis on $A(\mathbb{A})^1$ since $tr(\overline{\chi}'(\overline{f}_m)) = \overline{f}_m^\wedge(\overline{\chi}')$.

To prove claim 2 we fix a genuine factorizable function f on $A(\mathbb{A})$ and work locally. If

$$\overline{\chi} = Ind_{\overline{A(\mathbb{Q}_p)^1}}^{\overline{A(\mathbb{Q}_p)}}(\chi) \in \Pi_u(\overline{A(\mathbb{A})}^1)$$

then we have

$$\begin{aligned} \overline{f}^\wedge(\overline{\chi}) &= \int_{\overline{A(\mathbb{Q}_p)}} \overline{f}(y) tr(\overline{\chi}(y)) dy \\ &= \sum_{h \in \overline{A(\mathbb{Q}_p)^N} \backslash \overline{A(\mathbb{Q}_p)}} \int_{\overline{A(\mathbb{Q}_p)^N}} \overline{f}(a) \chi^h(a) da \\ &= \sum_{h \in \overline{A(\mathbb{Q}_p)^N} \backslash \overline{A(\mathbb{Q}_p)}} \int_{\overline{A(\mathbb{Q}_p)^N}} \overline{f}(h^{-1}ah) \chi(a) da \\ &= \int_{A(\mathbb{Q}_p)^N \times 1} f(a) \chi(a) da \\ &= f^\wedge(\chi). \end{aligned} \tag{3.20}$$

Claim 2 follows.

The 1-1 correspondence between $\Pi(\overline{A(\mathbb{A})^1})$ and $\Pi(A(\mathbb{A})^{1,N}) = \Pi(\overline{A(\mathbb{A})^{1,N}})_{gen}$ is given as follows. The map

$$\Pi(\overline{A(\mathbb{A})^{1,N}}) \to \Pi(\overline{A(\mathbb{A})}^1)$$

is the induction map. In fact, the map $\Pi(\overline{A(\mathbb{A})^1}) \to \Pi(A(\mathbb{A})^{1,N})$ is given locally by the map

$$\overline{\chi} = Ind_{\overline{A(\mathbb{Q}_p)^N}}^{\overline{A(\mathbb{Q}_p)}}(\chi) \longmapsto \chi$$

This proves the desired correspondence. □

Let $\mathcal{O}$ denote a (singleton) conjugacy class in $A(\mathbb{Q})$ and let $\chi \in \Pi_u(\overline{A(\mathbb{A})})$ denote a genuine irreducible representation. Define $I^A_{\mathcal{O}}$ by

$$I^A_{\mathcal{O}}(f) = \sum_{\delta \in \mathcal{O}} f(\delta) \tag{3.21}$$

and I^A_χ by

$$I^A_\chi(f) = \mathrm{tr}(\chi(f)) \tag{3.22}$$

for $f \in \mathcal{H}(\overline{A(\mathbb{A})})$, where $\mathcal{H}(\overline{A(\mathbb{A})})$ is as in §4.3.1 below.

4 The geometric side of the metaplectic trace formula

We follow [A2, A3].

4.1 The kernel function

Let $\mathcal{C}$ denote the set of equivalence classes of $\gamma \in G(\mathbb{Q})$ as in [A2, section 2]. Recall that $\mathcal{C}$ is in bijection with the set of equivalence classes of pairs $(M, \mathcal{O}_M)$, where $P = MN$ is a standard parabolic and $\mathcal{O}_M \subset M(\mathbb{Q})$ is a conjugacy class of elliptic elements.

For $\mathcal{O} \in \mathcal{C}$, let

$$K_{\mathcal{O}}(x, y) = \sum_{\gamma \in \mathcal{O}} f(x\gamma y^{-1}),$$

and more generally, let

$$K_{P,\mathcal{O}}(x, y) = \sum_{\gamma \in M(\mathbb{Q}) \cap \mathcal{O}} \int_{N(\mathbb{A})} f(x\gamma y^{-1}) dn, \tag{4.1}$$

where P is a parabolic as above.

Lemma 4.1. *For $\overline{K_0}$-finite $f \in \mathcal{H}(\overline{G(\mathbb{A})})$, the sum $K_P(x, y) = \sum_{\mathcal{O} \in \mathcal{C}} K_{\mathcal{O}}(x, y)$ converges to the kernel function of R_P, where R_P denotes the right regular representation of $\mathcal{H}(\overline{G(\mathbb{A})})_{gen}$ on $L^2(N(\mathbb{A})M(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{gen}$.*

proof: For $\phi \in C_c^\infty(N(\mathbb{A})M(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{gen}$, we have

$$\begin{aligned}
(R_P(f)\phi)(x) &= \int_{\overline{G(\mathbb{A})}} \phi(g) f(xg^{-1}) dg \\
&= \int_{\overline{G(\mathbb{A})}} \phi(n^{-1}\gamma^{-1}g) f(xg^{-1}) dg \\
&= \int_{\overline{G(\mathbb{A})}} \phi(g) f(xn\gamma g^{-1}) dg \\
&= \int_{M(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} \phi(g) \sum_{\gamma \in M(\mathbb{Q})} f(xn\gamma g^{-1}) dg \\
&= \int_{N(\mathbb{A})M(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} \phi(g) \sum_{\gamma \in M(\mathbb{Q})} \int_{N(\mathbb{A})} f(xn\gamma g^{-1}) dn dg.
\end{aligned}$$

The convergence of the inner sum is established, for example, by passing to the place at infinity and carrying out the estimates in the usual manner. For the description of $G(\mathbb{Q}) \cap (\overline{G(\mathbb{R}) \times K^f})$, see [K2] for example.□

4.1.1 Some analytic reduction theory

Let

$$N(\mathbb{A})_u = \{ \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \mid |n|_{\mathbb{A}} \leq u \}.$$

For $\omega \subset N(\mathbb{A})_u \cdot A(\mathbb{A})^1$ relatively compact, $T_0 \in \mathfrak{a}^+$, let

$$S(T_0, \omega) = \{ nak \in \overline{\omega \cdot A(\mathbb{R})^0 \cdot K_0} \mid \alpha(H(a) - T_0) > 0 \}.$$

This is a Siegel domain of $\overline{G(\mathbb{A})}$. It has finite volume with respect to the measure $\delta(a)dndadk$. For $T_0 \in \mathfrak{a}^+$ and $\omega \subset N(\mathbb{A})_u \cdot A(\mathbb{A})^1$ relatively compact both "sufficiently large" ($u \geq 1/2$, $T_0 > \sqrt{3}/2$), we have

$$\overline{G(\mathbb{A})} = G(\mathbb{Q}) \cdot S(T_0, \omega).$$

In particular, $G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$ is finite volume.

Lemma 4.2. *There exists an $N > 0$ such that for all $g \in \overline{G(\mathbb{A})}$, $T_0 \in \mathfrak{a}^+$, we have*

$$\sum_{\delta \in B(\mathbb{Q})\backslash G(\mathbb{Q})} \tau_B^\wedge(H(\delta x) - T_0) \leq 1.$$

This is an immediate consequence of the corresponding result in [A2].

For $T_0 > T_1$ in $\mathfrak{a}^+$, let

$$S(T_0, T_1, \omega) = \{nak \in S(T_0, \omega) \mid \omega(H(a) - T_0) \leq 0\}.$$

Let $F^G(x, T_1, T_0)$ denote the characteristic function of the image $\overline{S(T_0, T_1, \omega)} \subset G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$ of $S(T_0, T_1, \omega)$ under the quotient map $\overline{G(\mathbb{A})} \to G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$. Let $F^B(x, T_1, T_0)$ denote the characteristic function of the set of all $xB(\mathbb{Q}) \in B(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$ such that $\delta x \in \overline{\omega \cdot A(\mathbb{R})^o \cdot K_0}$ for some $\delta \in B(\mathbb{Q})$.

Lemma 4.3. *For $\omega \subset N(\mathbb{A})_u \cdot A(\mathbb{A})^1$ relatively compact and $T_0 \in \mathfrak{a}^+$ sufficiently large, $x \in \overline{G(\mathbb{A})}$, we have*

$$\sum_{P \in \{B,G\}} \sum_{\delta \in P(\mathbb{Q})\backslash G(\mathbb{Q})} F^P(\delta x, T_1, T_0)\tau_P(H(\delta x) - T_1) = 1.$$

This is an immediate consequence of the corresponding result in [A2].

4.1.2 The truncated kernel

The auxiliary kernel function is by definition

$$k^T(x, f) = \sum_{P \in \{B,G\}} (-1)^{\dim(A_P)} \sum_{\delta \in P(\mathbb{Q})\backslash G(\mathbb{Q})} K_P(\delta x, \delta x)\tau_P^\wedge(H(\delta x) - T).$$

This sum is finite by Lemma 4.2 above. Using the previous Lemma 4.3, this may be rewritten as

$$\begin{aligned} k^T(x, f) &= \sum_{P,Q \in \{B,G\}} (-1)^{\dim(A_P)} \times \\ &\quad \times \sum_{\delta \in Q(\mathbb{Q})\backslash G(\mathbb{Q})} K_Q(\delta x, \delta x)\tau_Q^\wedge(H(\delta x) - T)F^P(\delta x, T, T_0)\tau_P^Q(H(\delta x) - T), \end{aligned}$$

for $T > T_0$ regular. This may be rewritten as

$$\begin{aligned} k^T(x, f) &= K_G(x, x)F^G(x, T, T_0) + \\ &\quad \sum_{\delta \in B(\mathbb{Q})\backslash G(\mathbb{Q})} \tau_B(H(\delta x) - T)F^B(\delta x, T, T_0)(K_G(\delta x, \delta x) - K_B(\delta x, \delta x)). \end{aligned} \tag{4.2}$$

We want to prove

Lemma 4.4. $\int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} |k^T(x,f)|dx < \infty$.

We majorize this by the sum of the integrals of the absolute values of the two terms on the right hand side of (4.2). The integral over the first term on the right hand side of (4.2),

$$\int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} |K_G(x,x)F^G(x,T,T_0)|dx$$

clearly converges since $F^G(x,T,T_0)$ is compactly supported. To estimate the second term, we first write that

$$\int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} |\sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})} \tau_B(H(\delta x)-T)\times \tag{4.3}$$

$$\times F^B(\delta x,T,T_0)(K_G(\delta x,\delta x)-K_B(\delta x,\delta x))|dx \tag{4.4}$$

$$< < \int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} \tau_B(H(\delta x)-T)F^B(\delta x,T,T_0)\times$$

$$\times|\sum_{\delta\in G(\mathbb{Q})} f(x^{-1}\delta x) - \int_{N(\mathbb{A})} f(x^{-1}\mu n' x)dn'|dx. \tag{4.5}$$

Next, we again pass to the archimedian place. Let

$$\Gamma_0 = N(\mathbb{Q})\cap(\overline{G(\mathbb{R})\times K^f}),\ \Gamma_\infty = N(\mathbb{R})\cap(\overline{G(\mathbb{R})\times K_0^f}), \tag{4.6}$$

$$A(\mathbb{R})_T = \{\prod_{v|\infty}\begin{pmatrix} t_v & 0\\ 0 & t_v^{-1}\end{pmatrix}\in A(\mathbb{R})\mid |t_v|>T^{1/2}\},\ G(\mathbb{R})_T = N(\mathbb{R})\cdot A(\mathbb{R})_T\cdot K_\infty,$$

so $\Gamma_0\subset\Gamma_\infty$ is a co-compact discrete abelian group. The above upper bound (4.5) is finite if

$$\int_{\Gamma_0\backslash\overline{G(\mathbb{R})_T}} |\sum_{\gamma\in\Gamma_0} f(x^{-1}\gamma x) - \int_{\Gamma_\infty} f(x^{-1}nx)dn|dx < \infty. \tag{4.7}$$

We may apply the theory of Fourier transforms, as in [We1], to the quotient $\Gamma_0\backslash\Gamma_\infty$. Choosing Haar measures appropriately, there is a non-trivial character ψ of Γ_∞ such that the Fourier transform

$$\varphi^\wedge(\gamma) = \int_{\Gamma_\infty}\varphi(\gamma')\psi(<\gamma,\gamma'>)d\gamma',\quad \varphi\in L^2(\Gamma_0\backslash\Gamma_\infty),$$

satisfies

$$\varphi^\wedge(1) = \int_{\Gamma_\infty} \varphi(\gamma')d\gamma'.$$

Take $\varphi(\gamma) = \varphi_x(\gamma) = f(x^{-1}\gamma x)$, so by the Poisson summation formula [We1],

$$\sum_{\gamma\in\Gamma_0} f(x^{-1}\gamma x) - \int_{\Gamma_\infty} f(x^{-1}nx)dn = \sum_{\gamma^*\in\Gamma_0^*} \varphi_x^\wedge(\gamma^*),$$

where Γ_0^* denotes the dual lattice of Γ_0. Since f is smooth, $\varphi_x^\wedge$ is rapidly decreasing. For sufficiently regular T, standard arguments now suffice to show that, for any constant $C > 0$, we have

$$\int_{\Gamma_0\backslash\overline{G(\mathbb{R})_T}} |\sum_{\gamma^*\in\Gamma_0^*} \varphi_x^\wedge(\gamma^*)|dx <<_{C,f,n} T^{-C}.$$

This verifies (4.7), which proves the lemma. □

The above proof actually establishes more, namely,

Corollary 4.5. *(of the above proof) For each constant $C > 0$, we have*

$$\int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} k^T(x,f)dx - \int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})_T}} K^G(x,x)dx <<_{C,f,n} T^{-C}.$$

Here

$$G(\mathbb{A})_T = N(\mathbb{A})_u \cdot A(\mathbb{A})^T \cdot K$$

where $A(\mathbb{A})^T = \{a \in A(\mathbb{A}) \mid |\delta(a)|_{\mathbb{A}} \leq T\}$.

In fact, the proof yields more than this. If we let

$$k_{\mathcal{O}}^T(x,f) = \sum_{P\in\{B,G\}} (-1)^{\dim(A_P)} \sum_{\delta\in P(\mathbb{Q})\backslash G(\mathbb{Q})} K_{P,\mathcal{O}}(\delta x,\delta x)\tau_P^\wedge(H(\delta x) - T),$$

for $\mathcal{O} \in \mathcal{C}$, then the above proof establishes

Corollary 4.6. *(of the above proof) For each constant $C > 0$, we have*

$$\sum_{\mathcal{O}\in\mathcal{C}} \int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} |k_{\mathcal{O}}^T(x,f)dx - F^G(x,T,T_0)K_{G,\mathcal{O}}(x,x)|dx <<_{C,f,n} T^{-C}.$$

4.2 Computing the integral of $k^T(x,f)$

For a class $\mathcal{O} \in \mathcal{C}$ and a standard parabolic $P = MN$, let

$$J_{P,\mathcal{O}}(x,y) = \sum_{\mu \in M(\mathbb{Q}) \cap \mathcal{O}} \sum_{\varsigma \in N_{\mu_s}(\mathbb{Q}) \backslash N(\mathbb{Q})} \int_{N_{\mu_s}(\mathbb{A})} f(x^{-1}\varsigma^{-1}\mu n \varsigma y) dn,$$

where $\mu = \mu_s \mu_u$ denotes the Jordan decomposition (over $\overline{\mathbb{Q}}$), and let

$$j^T_{\mathcal{O}}(x,f) = \sum_{P \in \{B,G\}} (-1)^{\dim(A_P)} \sum_{\delta \in P(\mathbb{Q}) \backslash G(\mathbb{Q})} J_{P,\mathcal{O}}(\delta x, \delta x) \tau_P^{\wedge}(H(\delta x) - T).$$

the above proof establishes

Corollary 4.7. *(of the above proof) For each constant $C > 0$, we have*

$$\sum_{\mathcal{O} \in \mathcal{C}} \int_{G(\mathbb{Q}) \backslash \overline{G(\mathbb{A})}} |j^T_{\mathcal{O}}(x,f) dx - F^G(x,T,T_0) J_{G,\mathcal{O}}(x,x)| dx <<_{C,f,n} T^{-C}.$$

Let

$$\begin{aligned} J^T_{\mathcal{O}}(f) &= \int_{G(\mathbb{Q}) \backslash \overline{G(\mathbb{A})}} j^T_{\mathcal{O}}(x,f) dx, \\ K^T_{\mathcal{O}}(f) &= \int_{G(\mathbb{Q}) \backslash \overline{G(\mathbb{A})}} k^T_{\mathcal{O}}(x,f) dx. \end{aligned}$$

Proposition 4.8. *Both $J^T_{\mathcal{O}}(f)$ and $K^T_{\mathcal{O}}(f)$ are polynomials in T.*

Define

$$J_{\mathcal{O}}(f) = J^T_{\mathcal{O}}(f)|_{T=0}. \tag{4.8}$$

We will prove this below. First, we record a

Corollary 4.9. $\sum_{\mathcal{O} \in \mathcal{C}} J^T_{\mathcal{O}}(f) = \sum_{\mathcal{O} \in \mathcal{C}} K^T_{\mathcal{O}}(f)$.

proof: The difference between them is a polynomial, p say, in T by the proposition. Since $J_{G,\mathcal{O}}(x,y) = K_{G,\mathcal{O}}(x,y)$, this polynomial p must be the zero polynomial by Corollaries 4.6 and 4.7. $\square$

Let $\mathcal{O} \in \mathcal{C}$ denote an unramified semi-simple class [A2, p. 949]. If $\mu \in \mathcal{O}$ then its centralizer G_μ is either G or a Cartan subgroup of G defined over $\mathbb{Q}$.

As in [A2], we have

$$j_{\mathcal{O}}^T(x,f) = \sum_{\eta \in G_\mu(\mathbb{Q})\backslash G(\mathbb{Q})} f(x^{-1}\eta^{-1}\mu\eta x) \sum_{P \in \{B,G\}} (-1)^{\dim(A_P)} \tau_P^\wedge(H(\eta x) - T), \tag{4.9}$$

so

$$\begin{aligned} J_{\mathcal{O}}^T(f) &= \int_{G_\mu(\mathbb{Q})\backslash \overline{G(\mathbb{A})}} f(x^{-1}\mu x) \sum_{P \in \{B,G\}} (-1)^{\dim(A_P)} \tau_P^\wedge(H(x) - T) dx \qquad (4.10) \\ &= vol(A_{\mathcal{O}}(\mathbb{R})^0 G_\mu(\mathbb{Q})\backslash \overline{G_\mu(\mathbb{A})} \int_{\overline{G_\mu(\mathbb{A})}\backslash \overline{G(\mathbb{A})}} f(x^{-1}\mu x) v_{\mathcal{O}}(x,T) dx, \end{aligned}$$

where $A_{\mathcal{O}}$ denotes the center of M_μ and

$$v_{\mathcal{O}}(x,T) = \int_{A_{\mathcal{O}}(\mathbb{R})^0} \sum_{P \in \{B,G\}} (-1)^{\dim(A_P)} \tau_P^\wedge(H(ax) - T) da.$$

Note that the global cocycle β splits over $A_{\mathcal{O}}(\mathbb{R})^o \subset A(\mathbb{R})^o$. In the case where $A_{\mathcal{O}} = A$, this may be computed to be

$$v_{\mathcal{O}}(x,T) = 2(T - H(x)).$$

If $\mu \in \mathcal{O}$ is semi-simple but not regular then

$$j_{\mathcal{O}}^T(x,f) = f(\mu) \sum_{P \in \{B,G\}} (-1)^{\dim(A_P)} \tau_P^\wedge(H(x) - T)$$

so

$$J_{\mathcal{O}}^T(f) = f(\mu) \int_{G(\mathbb{Q})\backslash \overline{G(\mathbb{A})}} \sum_{P \in \{B,G\}} (-1)^{\dim(A_P)} \tau_P^\wedge(H(x) - T) dx.$$

4.3 Basic properties

4.3.1 Definitions

Let S be a subset of places of $\mathbb{Q}$ and let $||...|| : G(\mathbb{A}_S) \to \mathbb{R}_+$ be a "height" function as in Section 2. Let $N > 0$ and let $\Gamma \subset \Pi(\overline{K})$ be a finite set of inequivalent genuine irreducible representations of the n-fold cover of $K = K_{0,S} = \prod_{p \in S} K_{0,p}$.

We call a genuine factorizable function $f = \prod_{p\in S} f_p : \overline{G(\mathbb{A}_S)} \to \mathbb{C}$ **smooth** if

(a) $f_\infty \in C^\infty(\overline{G(\mathbb{R})})_{gen}$ if $p = \infty \in S$, and

(b) $f_p \in C_c^\infty(\overline{G(\mathbb{Q}_p)})_{gen}$ if $p \in S - \{\infty\}$.

In general, a genuine function $f : \overline{G(\mathbb{A}_S)} \to \mathbb{C}$ is **smooth** if it is a finite linear combination of factorizable smooth functions. Define $\mathcal{H}_N(\overline{G(\mathbb{A}_S)})_\Gamma$ to be the vector space of all genuine smooth complex valued functions $f : \overline{G(\mathbb{A}_S)} \to \mathbb{C}$ such that

$$\begin{array}{l} \text{supp}(f) \subset \overline{G(\mathbb{A}_S, N)} = \{g \in \overline{G(\mathbb{A}_S)} \mid ||g|| \leq e^N\} \\ \text{The spans of } k \longmapsto f(kg) \text{ generates a repn in } \Gamma, \\ \text{The spans of } k \longmapsto f(gk) \text{ generates a repn in } \Gamma. \end{array} \tag{4.11}$$

The semi-norms

$$||f||_D = \sup_{x\in\overline{G(\mathbb{A}_S)}} |Df(x)|,$$

D a differential operator on $\overline{G(\mathbb{R})}$, induces a topology on $\mathcal{H}_N(\overline{G(\mathbb{A}_S)})_\Gamma$ making it into a topological vector space. By [Ru], ch 1, we may complete this space using the above family of seminorms and obtain a Frechét space.

Define

$$\mathcal{H}(\overline{G(\mathbb{A}_S)}) = \lim_\Gamma \mathcal{H}(\overline{G(\mathbb{A}_S)})_\Gamma$$

where

$$\mathcal{H}(\overline{G(\mathbb{A}_S)})_\Gamma = \lim_N \mathcal{H}_N(\overline{G(\mathbb{A}_S)})_\Gamma.$$

Therefore, each function in $\mathcal{H}(\overline{G(\mathbb{A}_S)})$ is both right and left $\overline{K}$-finite.

Similarly, if A denotes the usual diagonal torus of $G = SL(2)$, then we define $\mathcal{H}(\overline{A(\mathbb{A}_S)})$.

From [A11, section 11], another space of functions of "almost compact" support, $\mathcal{H}_{ac}(\overline{G(\mathbb{A}_S)})$, is introduced, as well as its subspace of "moderate" functions, which we denote by $\mathcal{H}_{mod}(\overline{G(\mathbb{A}_S)})$. However, in the case $G = SL(2)$ it turns out that

$$\begin{array}{l} \mathcal{H}_{mod}(\overline{G(\mathbb{A}_S)}) = \mathcal{H}_{ac}(\overline{G(\mathbb{A}_S)}) = \mathcal{H}(\overline{G(\mathbb{A}_S)}), \\ \mathcal{H}_{mod}(\overline{A(\mathbb{A}_S)}) = \mathcal{H}_{ac}(\overline{A(\mathbb{A}_S)}) = \mathcal{H}(\overline{A(\mathbb{A}_S)}). \end{array}$$

We shall only sketch the definition of the Schwartz space $\mathcal{C}(\overline{G(\mathbb{A}_S)})$: this is the space spanned by factorizable functions f each of whose local components belong to the local Schwartz space, defined for example in [J1]. This space will play a bigger role in Section 6 below.

4.4 An equivalence relation on $G(\mathbb{Q})$

Let $\gamma, \gamma' \in G(\mathbb{Q})$ and let $\gamma = \gamma_u \gamma_s$ denote its Jordan decomposition. We say that γ, γ' are **equivalent**, written $\gamma \sim \gamma'$, if γ_s is $G(\mathbb{Q})$-conjugate to γ'_s. Clearly if γ, γ' are conjugate and semi-simple then they are equivalent. In fact, by the uniqueness of the Jordan decomposition, it can be shown that if γ, γ' are conjugate then they are equivalent. Define $\mathcal{C}$ to be the set of all equivalence classes. This definition holds in general.

At least for $G = SL(n)$, the classical analog of this definition is the following: Let $\gamma, \gamma' \in G(\mathbb{Z})$. We say that γ, γ' are **equivalent**, written $\gamma \sim \gamma'$, if γ_s is $G(\mathbb{C})$-conjugate to γ'_s. In other words, γ, γ' are equivalent if they have the same set of eigenvalues.

We call $(\gamma, \zeta) \in \overline{G(\mathbb{A})}$ semi-simple if $\gamma \in G(\mathbb{A})$ is semi-simple. Let $\mathcal{O} \in \mathcal{C}$ denote a class consisting entirely of semisimple elements (we call such a class **semi-simple**). If $\gamma \in \mathcal{O}$ then the centralizer G_γ is reductive and anisotropic modulo its center:

$$G_\gamma = \begin{cases} G, & \gamma = \pm 1, \\ A^x, & \gamma \neq \pm 1 \text{ and } \gamma \sim a \in A, \\ T, & \text{otherwise}, \end{cases} \tag{4.12}$$

where $A^x = x^{-1}Ax$ and $x^{-1}\gamma x \in A$. In particular, all such classes are unramified in the sense of [A2], p. 949.

Let $\mathcal{O} \in \mathcal{C}$ denote an unramified semi-simple class, let $P = MN \in \{B, G\}$, and let $\mu \in \mathcal{O} \cap M(\mathbb{Q})$. Then

$$\begin{aligned} J_{B,\mathcal{O}}(\delta x, \delta x) &= \sum_{\eta \in M_\mu(\mathbb{Q})\backslash M(\mathbb{Q})} \sum_{\varsigma \in N(\mathbb{Q})} f(x^{-1}\delta^{-1}\varsigma^{-1}\eta^{-1}\mu\eta\varsigma\delta x) \\ &= \sum_{\eta \in M_\mu(\mathbb{Q})\backslash P(\mathbb{Q})} f(x^{-1}\delta^{-1}\eta^{-1}\mu\eta\delta x). \end{aligned} \tag{4.13}$$

If $P = B$ then $M = A$ and $M_\mu = A$ for $\mu \in \mathcal{O}$ as above. If $P = G$ then $M = G$ and $M_\mu = G_\mu$ is one of the above centralizers for $\mu \in \mathcal{O}$ as above. As we saw above, if $P = B$ then

$$\begin{aligned} J_{B,\mathcal{O}}(\delta x, \delta x) &= \sum_{\varsigma \in N(\mathbb{Q})} f(x^{-1}\delta^{-1}\varsigma^{-1}\mu\varsigma\delta x) \\ &= \sum_{\eta \in A(\mathbb{Q})\backslash B(\mathbb{Q})} f(x^{-1}\delta^{-1}\eta^{-1}\mu\eta\delta x). \end{aligned} \tag{4.14}$$

We have then by an equation above

$$\begin{aligned} j_{\mathcal{O}}^T(x,f) &= \sum_{P\in\{B,G\}}(-1)^{\dim A_P}\sum_{\delta\in P(\mathbb{Q})\backslash G(\mathbb{Q})} J_{P,\mathcal{O}}(\delta x,\delta x)\tau_P^\wedge(H_B(\delta x)-T) \\ &= \sum_{P\in\{B,G\}}(-1)^{\dim A_P}\sum_{\eta\in M_\mu(\mathbb{Q})\backslash G(\mathbb{Q})} f(x^{-1}\eta^{-1}\mu\eta x)\tau_P^\wedge(H_B(\eta x)-T) \\ &= \sum_{\eta\in G_\mu(\mathbb{Q})\backslash G(\mathbb{Q})} f(x^{-1}\eta^{-1}\mu\eta x)\sum_{P\in\{B,G\}}(-1)^{\dim A_P}\tau_P^\wedge(H_B(\eta x)-T). \end{aligned} \tag{4.15}$$

Also

$$\begin{aligned} J_{\mathcal{O}}^T(f) &= \int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} j_{\mathcal{O}}^T(x)dx \\ &= \int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}}\sum_{\delta\in G_\mu(\mathbb{Q})\backslash G(\mathbb{Q})} f(x^{-1}\delta^{-1}\mu\delta x)\sum_{P\in\{B,G\}}(-1)^{\dim A_P}\tau_P^\wedge(H_B(\delta x)-T)dx \\ &= \int_{G_\mu(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} f(x^{-1}\mu x)\sum_{P\in\{B,G\}}(-1)^{\dim A_P}\tau_P^\wedge(H_B(x)-T)dx \\ &= vol(G_\mu(\mathbb{Q})\backslash\overline{G(\mathbb{A})}_\mu)\int_{\overline{G(\mathbb{A})}_\mu\backslash\overline{G(\mathbb{A})}} f(x^{-1}\mu x)\sum_{P\in\{B,G\}}(-1)^{\dim A_P}\tau_P^\wedge(H_B(x)-T)dx \\ &= vol(A_{\mathcal{O}}(\mathbb{R})^oG_\mu(\mathbb{Q})\backslash\overline{G(\mathbb{A})}_\mu)\int_{\overline{G(\mathbb{A})_\mu}\backslash\overline{G(\mathbb{A})}} f(x^{-1}\mu x)v(x,T)dx \end{aligned} \tag{4.16}$$

(see also [Sho, (5.5)]), where

$$v(x,T) = \int_{A_{\mathcal{O}}(\mathbb{R})^o}\sum_{P\in\{B,G\}}(-1)^{\dim A_P}\tau_P^\wedge(H_B(ax)-T)da. \tag{4.17}$$

If $A_{\mathcal{O}} = A$, that is to say if there is an $a \in A(\mathbb{Q}) - \{\pm I_2\}$ in the class $\mathcal{O}$ then $v(x,T)$ is the "volume" of the convex hull of

$$\{T - H(x), -(T - H(wx))\}.$$

Thus

$$v(x,T) = 2T - \log\delta(x) - \log\delta(wx). \tag{4.18}$$

This proves the following

Theorem 4.10. *Let $\mathcal{O}\in\mathcal{C}$ and suppose that $\mathcal{O}\cap(A(\mathbb{Q})-\{\pm I_2\})\neq\emptyset$. Then*

$$J_{\mathcal{O}}^T(f) = vol(A_{\mathcal{O}}(\mathbb{R})^oG_\mu(\mathbb{Q})\backslash G_\mu(\mathbb{A}))\int_{G_\mu(\mathbb{A})\backslash G(\mathbb{A})} f(x^{-1}\mu x)v(x,T)dx.$$

These are called **weighted orbital integrals**. The case where $1\in\mathcal{O}$ will be discussed below.

4.5 The definition of $J_M(u, f)$

Let $f \in C_c^\infty(\overline{G(\mathbb{A}_S)})_{gen}$ and let $\overline{\gamma} = (\gamma, 1) \in \overline{M(\mathbb{A}_S)}$ have Jordan decomposition $\gamma = \sigma u$. Suppose that $\overline{M(\mathbb{A}_S)}_{\overline{\gamma}} = \overline{G(\mathbb{A}_S)}_{\overline{\gamma}}$. Assume that S is a "sufficiently large" finite set which contains the archimedian places. In this case, define

$$J_M(\overline{\gamma}, f) = |D(\gamma)|^{1/2} \int_{\overline{M(\mathbb{A}_S)}\backslash\overline{G(\mathbb{A}_S)}} \int_{\mathcal{O}_{\overline{\gamma}}(\overline{M(\mathbb{A}_S)})} f(x^{-1}\mu x) v_M(x) d\mu dx, \quad (4.19)$$

where $\mathcal{O}_{\overline{\gamma}}(\overline{M(\mathbb{A}_S)}) = \{m^{-1}\overline{\gamma}m \; m \in \overline{M(\mathbb{A}_S)}\}$. Here v_M is the "volume" of the convex hull of the points $\{H_B(x), H_{B^-}(x)\}$ in $\mathfrak{a}$. If $x = nk$ then $H_B(x) = 0$, so $v_M(nk) = |H_{B^-}(x)|$. In any case, this integral can be written as

$$J_M(\overline{\gamma}, f) = |D(\gamma)|^{1/2} \int_{\overline{G(\mathbb{A}_S)}_{\overline{\gamma}}\backslash\overline{G(\mathbb{A}_S)}} f(x^{-1}\overline{\gamma}x) v_M(x) dx.$$

This is a function on $\overline{M(\mathbb{A}_S)}$.

If $\gamma \in \overline{M(\mathbb{A}_S)}$ is *arbitrary* and $a \in \overline{A_M(\mathbb{A}_S)}$ is chosen so that $M(\mathbb{A}_S)_{a\gamma} = G(\mathbb{A}_S)_{a\gamma}$ then $J_M(a\gamma, f)$ is well-defined. We may consider $a \to 1$.

As in [A7, §3], let

$$r_\beta(\lambda, u, a) = |a^\beta - a^{-\beta}|^{\rho(\beta,u)\lambda(\beta^\vee)}, \quad \lambda \in \mathfrak{a}_{M,\mathbb{C}}^*,$$

and let

$$r_P(\lambda, u, a) = \prod_{v \in S} \prod_\beta r_\beta(\frac{\lambda}{2}, u_v, a_v),$$

where β runs over the reduced roots of $(P_{\sigma_v}, A_{M_{\sigma_v}})$. By definition,

$$r_M(u, a) = \lim_{\lambda \to 0} \sum_{P \in \mathcal{P}(M)} r_P(\lambda, u, a) \theta_P(\lambda)^{-1},$$

and

$$v_M(x) = \lim_{\lambda \to 0} \sum_{P \in \mathcal{P}(M)} e^{-\lambda(H_P(x))} \theta_P(\lambda)^{-1}.$$

Following [A7, (8.1)], we define $J_M(\gamma, f)$ to be the limit at $a = 1$ of

$$\begin{aligned} |D(a\gamma)|^{1/2} \textstyle\int_{\overline{G(\mathbb{A}_S)}_\sigma\backslash\overline{G(\mathbb{A}_S)}} \int_{\overline{M(\mathbb{A}_S)}_\gamma\backslash\overline{G(\mathbb{A}_S)}_\sigma} f(y^{-1}\sigma x^{-1} a u x y) \times \\ \times \left(\textstyle\sum_{L \in \mathcal{L}(M)} r_M^L(\gamma, a) v_L(xy) \right) dxdy, \end{aligned} \quad (4.20)$$

where $\gamma \in \overline{M(\mathbb{A}_S)}$ is as above.

Example 4.11. *Let* $G = SL(2),\ M = A,\ a = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}$, *where* $t \neq \pm 1$. *In this case,*

$$r_A(\gamma, a) = 1,$$

and

$$v_A(x) = \sum_{p \in S} \log(\sqrt{|1 + n_p^2|_p}),$$

where $x = \prod_{p \in S} a_p \cdot \begin{bmatrix} 1 & n_p \\ 0 & 1 \end{bmatrix} \cdot k_p$, *for* $a_p \in \overline{A(\mathbb{Q}_p)}$, $k_p \in \overline{K_{0,p}}$ *([A6], Intro., [Sho], chap. 6). Therefore,* $J_A(z, f)$, *for* $z \in Z(G(\mathbb{Q}))$, *is the limit as* $a \to 1$ *of*

$$|D(a)|^{1/2} \int_{\overline{A(\mathbb{A}_S)} \backslash \overline{G(\mathbb{A}_S)}} f(y^{-1} a z y) v_A(y) dy \tag{4.21}$$

[Sho, section 6.1], p. 64.

4.6 A more explicit form of the geometric terms

Definition 4.12. *Let*

$$K_{P,unip}(x, y) = \sum_{\gamma \in \mathcal{U}_{M_P}(\mathbb{Q})} \int_{N_P(\mathbb{A})} f(x^{-1} \gamma n y) dn,$$

and for $T_0 = 0$, *let*

$$\begin{aligned} J_{unip}(f) &= J_{unip}^{T_0}(f) \\ &= \int_{G(\mathbb{Q}) \backslash \overline{G(\mathbb{A})}} \sum_P (-1)^{\dim(A_P / A_G)} \sum_{\delta \in P(\mathbb{Q}) \backslash G(\mathbb{Q})} K_{P,unip}(\delta x, \delta x) \tau_P^{\wedge}(H(\delta x) - T_0) dx. \end{aligned}$$

Example 4.13. *Let* $G = SL(2),\ M = A$. *Then*

$$K_{P,unip}(x, y) = \sum_{\gamma \in \mathcal{U}_{M_P}(\mathbb{Q})} \int_{N_P(\mathbb{A})} f(x^{-1} \gamma n y) dn$$

and

$$\begin{aligned} J_{unip}(f) &= \int_{G(\mathbb{Q}) \backslash \overline{G(\mathbb{A})}} \sum_P (-1)^{\dim(A_P / A_G)} \sum_{\delta \in P(\mathbb{Q}) \backslash G(\mathbb{Q})} K_{P,unip}(\delta x, \delta x) \tau_P^{\wedge}(H(\delta x) - T_0) dx \\ &= \int_{G(\mathbb{Q}) \backslash \overline{G(\mathbb{A})}} (K_{G,unip}(x, x) - \sum_{\delta \in B(\mathbb{Q}) \backslash G(\mathbb{Q})} K_{B,unip}(\delta x, \delta x) \tau_B^{\wedge}(H(\delta x) - T_0)) dx \end{aligned}$$

Definition 4.14. *Let $J_M(u,f)$ denote the weighted unipotent orbital integrals defined in the previous subsection, $u \in M(\mathbb{A}_S)$ unipotent. We define $a^M(S,u)$ by the property that*

$$J_{unip}(f) = \sum_P \sum_{u \in \mathcal{U}_{M_P}(\mathbb{Q})} |W_0^M| \cdot |W_0^G|^{-1} a^M(S,u) J_M(u,f).$$

(The verification that this definition makes sense can be found for example in [Sho, section 6.2, pp. 66–67].) For $\gamma \in \overline{M(\mathbb{A}_S)}$, we define $a^M(S,\gamma)$ by

$$a^M(S,\gamma) = \varepsilon^M(\sigma)|\iota^M(\sigma)|^{-1} \sum_{\{u \mid \sigma u \sim \gamma\}} a^{M_\sigma}(S,u),$$

where

$$\varepsilon^M(\sigma) = \begin{cases} 1, & \text{if } \sigma \text{ is } k-\text{elliptic on } M \\ 0, & \text{otherwise.} \end{cases}$$

Furthermore, in the case of $G = SL(2)$, if $\gamma = \sigma u$ is the Jordan decomposition then $a^M(S,\gamma) = \epsilon^M(\sigma) a^M(S,u)$.

The relationship between the $J_{\mathcal{O}}(f)$ and the $J_M(f)$ is given by the following:

Proposition 4.15. *([A12, Theorem 8.1]) If S is a sufficiently large finite set of valuations and $f \in C_c^\infty(\overline{G(\mathbb{A}_S)})_{gen}$ then*

$$J_{\mathcal{O}}(f) = \sum_{M \in \{A,G\}} |W_0^M| \cdot |W_0^G|^{-1} \sum_{\gamma \in M(\mathbb{Q}) \cap \mathcal{O}} a^M(S,\gamma) J_M(\gamma,f)$$

where the constants $a^M(S,\gamma)$ defined above satisfy, if γ is semisimple, the property

$$a^M(S,\gamma) = \begin{cases} vol(M(\mathbb{Q})_\gamma \backslash \overline{M(\mathbb{A}_S)}^1_\gamma), & \text{if } \gamma \text{ is } \mathbb{Q}-\text{elliptic in } M(\mathbb{Q}), \\ 0, & \text{otherwise,} \end{cases}$$

Example 4.16. *For $G = SL(2)$, $1 \in \mathcal{O}$, we have*

$$J_{\mathcal{O}}(f) = \sum_{\gamma \in \mathcal{O}} a^G(S,\gamma) J_G(\gamma,f) + \frac{1}{2} a^A(S,1) J_A(1,f)$$

where $J_A(1,f)$ is in Example 4.13 above and $J_G(\gamma,f)$ is defined in (4.19) above. A similar result holds for the $\mathcal{O}$ with $-1 \in \mathcal{O}$.

5 The spectral side of the trace formula

The spectral side of the trace formula comprises the collection of terms associated with the representations of the group. This side, though analogous to the already discussed geometric side of the trace formula in terms of the overall structure, requires more background to state. Essentially, we will have to leave the entire theory of Eisenstein series as "known" to the reader. Fortunately, there are several excellent expositions of this theory in the case of $SL(2)$ and related groups: [L1], [K2], [KP] for example.

A general remark may be in order here. The spectral theory used in [A2] relies on the general theory of Eisenstein series on a quotient $\Gamma\backslash H$ developed in [L1], where H denotes a Lie group with finitely many connected components and Γ denotes a discrete subgroup satisfying certain assumptions. By passing to the archimedian place, the quotient $G(\mathbb{Q})\backslash\overline{G(\mathbb{A})^1}$ may be regarded as a finite union of quotients $\Gamma_i\backslash H$ as in [L1] (see Lemma 2.7 above). Thus we may apply the results of [L1] here.

We begin with a brief overview of some of the background. Some results we will eventually need are the decomposition (5.4), Proposition 5.7 and the definition following it, and Theorem 5.8. Other than (5.4), much of §5.1.1 can be ignored or skimmed over as we sketch the (slightly different) decompositions used in [A1] and in [A2, A3]. We only need the decomposition (over a set $\mathcal{X}$ introduced below) used in [A2, A3].

5.1 The spectral decomposition

A basic result in representation theory [Ro, ch. 14], [L2, Lemma 3.2] is the following

Lemma 5.1. *Let $\Gamma \subset G$ be a discrete subgroup of a locally compact unimodular group G. The right regular representation of G acting on $L^2(\Gamma\backslash G)$ decomposes as a (not necessarily discrete) Hilbert sum of irreducible representations, each occuring with finite multiplicity.*

For $H < \overline{G(\mathbb{A})}$ a locally compact totally disconnected group of type 1, define $\Pi_u(\overline{H})$ as in §3.1 above.

Another way to look at the spectral decomposition is adelically: The structure of the unitary dual $\Pi_u(\overline{G(\mathbb{A})})$ can be determined from the structure of $\Pi_u(\overline{G(\mathbb{Q}_p)})$. (The proof of this is given in [Ge2] and is a consequence of a theorem of Gelbart mentioned in Example 5.2 below.) The **spectrum** of

$G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$ is the set of elements of $\Pi_u(\overline{G(\mathbb{A})})$ which occur in the representation space $L^2(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})$, on which $\overline{G(\mathbb{A})}$ acts on the right by translation.

5.1.1 The partial decomposition in the adelic case: cuspidals

For each parabolic $P = MN$ of G defined over $\mathbb{Q}$, define

$$\begin{gathered} L^2(M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1})_{cusp} \\ = \{\phi \in L^2(M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1}) \text{ genuine} \mid \text{for any parabolic } P_1 \subset P,\ P_1 \neq P, \\ \text{we have } \int_{N_1(\mathbb{Q})\cap M(\mathbb{Q})\backslash N_1(\mathbb{A})\cap M(\mathbb{A})} \phi(nm)dn = 0, \\ \text{for almost all } m \in \overline{M(\mathbb{A})^1}\}. \end{gathered} \tag{5.1}$$

This is the space of genuine **cusp forms** on $\overline{M(\mathbb{A})^1}$.

Example 5.2. *(representations of adele groups)* Case $P = M = G = SL(2)$*: The genuine unitary representations π of $\overline{G(\mathbb{A})}$ are all "factorizable", i.e., $\pi = \otimes_{p\leq\infty}\pi_p$ on $\overline{G(\mathbb{A})}$, where π_p is a genuine irreducible unitary representation of $\overline{G(\mathbb{Q}_p)}$ and almost all π_p are unramified (i.e., has a non-trivial $\overline{K_p}$-fixed vector). Conversely, given a collection*

$$\{\pi_p \mid \pi_p \in \Pi_u(\overline{G(\mathbb{Q}_p)}),\ \text{almost all } \pi_p \text{ are unramified}\},$$

the restricted tensor product $\pi = \otimes_{p\leq\infty}\pi_p$ defines an element of $\Pi_u(\overline{G(\mathbb{A})})$. For a proof, see for example, [Ge2, §3].

Proposition 5.3. *For each $\rho \in \Pi_u(\overline{M(\mathbb{A})^1})$ there is an integer $m(\rho) \geq 0$ such that*

$$L^2(M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1})_{cusp} \cong \bigoplus_{\rho\in\Pi_u(\overline{M(\mathbb{A})^1})} m(\rho)\rho.$$

If $\rho \in \Pi_u(\overline{M(\mathbb{A})^1})$ and if $m(\rho) \neq 0$ then we call ρ a genuine **cuspidal representation** of $\overline{M(\mathbb{A})^1}$.

Two parabolics $P = MN$ and $P' = M'N'$ are said to be **associate** parabolics if

$$\Omega(\mathfrak{a}, \mathfrak{a}') = \{s \in \Omega \mid s\mathfrak{a}_{\mathbb{C}} = \mathfrak{a}'_{\mathbb{C}}\} \neq \emptyset.$$

Here $\mathfrak{a}_{\mathbb{C}}$ denotes the complexification of the Lie algebra $\mathfrak{a}$ of the center A of M. In particular, a parabolic is always associate to its opposite. We identify the Weyl group Ω with a subgroup of $G(\mathbb{Q})$ (hence as a subgroup of $\overline{G(\mathbb{A})}$) by $s \longmapsto w_s \in G(\mathbb{Q})$. If P, P' are associate and if $\rho \in \Pi_u(\overline{M(\mathbb{A})^1})$, $\rho' \in$

$\Pi_u(\overline{M'(\mathbb{A})^1})$ are cuspidal then we call (M,ρ) **equivalent** to (M',ρ') if there exists an $s \in \Omega(\mathfrak{a},\mathfrak{a}')$ such that $s\rho$ is unitarily equivalent (as a representation of $\overline{M'(\mathbb{A})^1}$) to ρ', where

$$(s\rho)(m') = \rho(w_s^{-1} m' w_s), \quad m' \in \overline{M'(\mathbb{A})^1}.$$

Notation: Let $\mathcal{X} = \mathcal{X}^G$ denote the set of equivalent classes of such pairs (M,ρ). For each $\chi \in \mathcal{X}$, we denote by $\mathcal{P}_\chi$ the corresponding class of associated parabolics. Let $\mathcal{X}(G)$ denote the set of equivalent classes of cuspidal pairs (G,ρ).

Example 5.4. *Two parabolics P, P' of $G = SL(2)$ with Levi A are associate if and only they are opposite. Therefore (M,ρ) is equivalent to (M',ρ') if and only if $M = M' = G$ and $\rho = \rho'$ (as equivalence classes of representations of $\overline{G(\mathbb{A})}$), or $M = M' = A$ and $\rho = \rho'^w$, where $w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ denotes the non-trivial element of the Weyl group of A.*

For each $P = MN$ and each $\chi \in \mathcal{X}$, we let

$$L^2(M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1})_{cusp,\chi} \cong \bigoplus_{\rho \in \Pi_u(\overline{M(\mathbb{A})^1}), (M,\rho)\in\chi} m(\rho)\rho. \tag{5.2}$$

Note that if $P \notin \mathcal{P}_\chi$ then this sum is empty (which, by convention, we define to be 0).

The above theorem has the following

Corollary 5.5. $L^2(M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1})_{cusp} = \bigoplus_{\chi\in\mathcal{X}} L^2(M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1})_{cusp,\chi}$. *In particular,*

$$L^2(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{cusp} = \bigoplus_{\chi\in\mathcal{X}} L^2(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{cusp,\chi}.$$

In other words, the space of cuspidal automorphic functions decomposes into "irreducible" subspaces on which the action of $\overline{G(\mathbb{A})}$ is the more-or-less the same.

5.1.2 The partial decomposition in the adelic case: theta functions

We seek an analogous result for all of $L^2(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{gen}$.

Let $\mathfrak{Z}_M$ denote the center of the universal enveloping algebra $\mathfrak{U}_M$ of M. Each $X \in \mathfrak{Z}_M$ defines a densely defined unbounded operator on $L^2(M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1})_{cusp}$.

Let $\mathcal{H}_P^0$ denote

$$\begin{array}{c} \mathcal{H}_P^0 = \{\varphi : N(\mathbb{A})M(\mathbb{Q})\backslash\overline{G(\mathbb{A})} \to \mathbb{C} \text{ genuine } | \\ (i)\ \forall x \in \overline{G(\mathbb{A})},\ \text{the function } m \longmapsto \varphi(mx),\ m \in M(\mathbb{A}), \\ \text{is } \mathfrak{Z}_{M(\mathbb{R})}-\text{finite}, \\ (ii)\ \text{the span of all functions } \varphi_k : x \longmapsto \varphi(xk),\ k \in \overline{K}, \\ \text{is finite dimensional}, \\ (iii)\ ||\varphi||^2 = \int_{\overline{K}} \int_{M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1}} |\varphi(mk)|^2 dmdk < \infty\}. \end{array} \tag{5.3}$$

Let $\mathcal{H}_P$ denote the closure of $\mathcal{H}_P^0$ with respect to this norm in (iii).

For $\varphi \in \mathcal{H}_P$, $\Lambda \in \mathfrak{a}_{\mathbb{C}}$, $x, y \in \overline{G(\mathbb{A})}$, define $I_P(\Lambda) : \overline{G(\mathbb{A})} \to End(\mathcal{H}_P)$ by

$$(I_P(\Lambda, y)\varphi)(x) = \varphi(xy)\exp(<\Lambda+\rho_P, H_P(xy)>)\exp(-<\Lambda+\rho_P, H_P(x)>).$$

Here $\rho_P \in \mathfrak{a}$ is the element such that $|\det(Ad(m))_{\mathfrak{n}(\mathbb{A})}| = \exp(<2\rho_P, H_P(m)>)$, $m \in \overline{M(\mathbb{A})}$.

For each $P = MN$ and each $\chi \in \mathcal{X}$, we define $L^2(M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1})_\chi$ as follows. If there is no $P_1 \in \mathcal{P}_\chi$ with $P_1 \subset P$ then we define $L^2(M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1})_\chi$ to be 0. Otherwise, let $\Psi(P_1, P, \chi)$ denote the space of all genuine complex-valued functions $\psi : N(\mathbb{A})M_1(\mathbb{Q})\backslash\overline{G(\mathbb{A})} \to \mathbb{C}$, $P_1 = N_1M_1 \in \mathcal{P}_\chi$ with $P_1 \subset P$, such that

1. $\psi(amk)$, as a function of $a \in A_1(\mathbb{Q})\backslash\overline{A_1(\mathbb{A})}$ (fixed $m \in M_1(\mathbb{Q})\backslash\overline{M_1(\mathbb{A})^1},\ k \in K$), is compactly supported,
2. $\psi(amk)$, as a function of $k \in \overline{K_\infty}$, belongs to the vector space defined by the span of the matrix coefficients of some irreducible representation W of $\overline{K_\infty}$,
3. $\psi(amk)$, as a function of $m \in M$, belongs to $L^2(M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1})_{cusp,\chi}$.

Given $\psi \in \Psi(P_1, P, \chi)$, $P_1 \subset P$, let

$$\psi^{\wedge,M}(m) = \sum_{\delta \in P_1(\mathbb{Q})\cap M(\mathbb{Q})\backslash M(\mathbb{Q})} \psi(\delta m), \qquad m \in M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1},$$

so that $\psi^{\wedge,M} \in L^2(M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1})$ [L1, Lemma 1]. Let $L^2(M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1})_\chi$ denote the L^2-closure of the span of all the $\psi^{\wedge,M}$ with $\psi \in \Psi(P_1, P, \chi)$, some $P_1 \subset P$ as above. [L1, Lemma 2] implies

$$L^2(M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1}) = \bigoplus_{\chi \in \mathcal{X}^G} L^2(M(\mathbb{Q})\backslash\overline{M(\mathbb{A})^1})_\chi. \tag{5.4}$$

5.2 Eisenstein series in the adelic case

For $\varphi \in \mathcal{H}_P^0,\ \lambda \in \mathfrak{a}_{\mathbb{C}},\ s \in \Omega(\mathfrak{a}, \mathfrak{a}_1),\ x \in \overline{G(\mathbb{A})}$, let $M(s,\lambda)$ denote the intertwining integral

$$\begin{gathered} M(s,\lambda)\phi(x) \\ = \exp(- < s\lambda + \rho_1, H_P(x) >) \times \\ \times \int_{N_1(\mathbb{A}) \cap w_s N(\mathbb{A}) w_s^{-1} \backslash N_1(\mathbb{A})} \varphi(w_s^{-1} n x) \exp(< \lambda + \rho_P, H_P(w_s^{-1} n x) >) dn, \end{gathered} \tag{5.5}$$

with a suitably normalized measure dn. We define $M(s, \pi_\zeta)$ below (see (5.9)).

Let $\mathfrak{a}^+$ denote the positive Weyl chamber $\mathfrak{a}^+ = \{X \in \mathfrak{a} \mid \alpha(X) > 0,\ \forall \alpha > 0\}$. For $\varphi \in \mathcal{H}_P^0,\ \lambda \in \mathfrak{a}_{\mathbb{C}},\ Re(\lambda) \in \rho_P + \mathfrak{a}^+,\ x \in \overline{G(\mathbb{A})}$, let

$$E(x, \varphi, \Lambda) = \sum_{\delta \in P(\mathbb{Q}) \backslash G(\mathbb{Q})} \varphi(\delta x) \exp(< \lambda + \rho_P, H_P(\delta x) >). \tag{5.6}$$

This is the **Eisenstein series** associated to P and φ and is known to converge absolutely.

Let $P = MN$ denote a parabolic and let $\pi \in \Pi_u(\overline{M(\mathbb{A})^1})$. Following [A2], we define

$$\begin{gathered} \mathcal{H}_P^0(\pi) = \{\varphi : N(\mathbb{A}) M(\mathbb{Q}) \backslash \overline{G(\mathbb{A})} \to \mathbb{C} \text{ genuine} \mid \\ (i)\ \phi \text{ is right } K-\text{finite}, \\ (ii)\ \forall x \in \overline{G(\mathbb{A})},\ \text{the function } m \longmapsto \varphi(mx)\ (m \in \overline{M(\mathbb{A})^1}) \\ \text{is a matrix coefficient of } \pi, \\ (iii)\ ||\varphi||^2 = \int_K \int_{M(\mathbb{Q}) \backslash \overline{M(\mathbb{A})^1}} |\varphi(mk)|^2 dm dk < \infty\}. \end{gathered} \tag{5.7}$$

Let $\mathcal{H}_P(\pi)$ denote the completion of $\mathcal{H}_P^0(\pi)$ with respect to the norm $||...||$ in (iii). For $\phi \in \mathcal{H}_P(\pi)$, $\varsigma \in i\mathfrak{a}^*$, let

$$\begin{aligned} \phi_\varsigma(x) &= \phi(x) e^{\varsigma(H_P(x))}, \quad x \in \overline{G(\mathbb{A})}, \\ \pi_\varsigma &= \pi \otimes \chi_\varsigma, \quad \chi_\varsigma(x) = e^{\varsigma(H_P(x))}, \quad x \in \overline{G(\mathbb{A})}, \\ (I_P(\pi_\varsigma, y)\phi_\varsigma)(x) &= \phi_\varsigma(xy) \delta_P(xy)^{1/2} \delta_P(x)^{-1/2}, \quad x, y \in \overline{G(\mathbb{A})}. \end{aligned}$$

Let $P = MN$ denote a parabolic, let $\pi \in \Pi_u(\overline{M(\mathbb{A})})$, let f be a function on $\overline{M(\mathbb{A})}$, and let $\zeta \in \mathfrak{a}_{\mathbb{C}}^*$. Define π_ζ by

$$\begin{gathered} f_\zeta(m) = f(m) e^{\zeta(H_M(m))}, \\ \pi_\zeta(m) = \pi(m) e^{\zeta(H_M(m))}, \quad m \in \overline{M(\mathbb{A})}. \end{gathered} \tag{5.8}$$

The space of (genuine) matrix coefficients of π, which we denote by $\mathcal{A}(\pi)$, is isomorphic to that of π_ζ, for $\zeta \in i\mathfrak{a}^*$:

$$\mathcal{A}(\pi) \cong \mathcal{A}(\pi_\zeta).$$

(proof: We may let $\pi,\ \pi_\zeta$ have the same representation space V with inner product $(...,...)_V$. If $v_1, v_2 \in V$ and $f(g) = (\pi(g)v_1, v_2)_V$ then $f_\zeta \in \mathcal{A}(\pi_\zeta)$ and every element of $\mathcal{A}(\pi_\zeta)$ arises in this way in a unique fashion. $\square$) We have a natural isomorphism

$$\begin{gathered}\mathcal{H}_P(\pi) \to \mathcal{H}_P(\pi_\zeta),\\ \phi \longmapsto \phi_\zeta,\end{gathered}$$

For $\pi \in \Pi_u(\overline{M(\mathbb{A})})$, $\phi \in \mathcal{H}_P^0(\pi)$, $\varsigma \in \mathfrak{a}_{P,\mathbb{C}}^*$, $s \in \Omega(\mathfrak{a}, \mathfrak{a}')$, $Re(\varsigma) \in \rho_P + (\mathfrak{a}_P^*)^+$, define

$$\begin{gathered}E(x, \phi_\varsigma) = \textstyle\sum_{\delta \in P(\mathbb{Q})\backslash G(\mathbb{Q})} \phi_\varsigma(\delta x)\delta_P(\delta x)^{1/2},\\ (M(s, \pi_\varsigma)\phi_\varsigma)(x) = \delta_P(x)^{-1/2} \textstyle\int_{N'(\mathbb{A}) \cap w_s N(\mathbb{A}) w_s^{-1} \backslash N'(\mathbb{A})} \phi_\varsigma(w_s^{-1}nx)\delta_P(w_s^{-1}nx)^{1/2} dn.\end{gathered} \tag{5.9}$$

Analogously, if $P \subset P_2$, $\varsigma \in \mathfrak{a}_{\mathbb{C}}^*$ is sufficiently large then define

$$E_{P_2}(x, \phi_\varsigma) = \sum_{\delta \in P(\mathbb{Q})\backslash P_2(\mathbb{Q})} \phi_\varsigma(\delta x)\delta_P(\delta x)^{1/2}.$$

Let

$$\begin{gathered}\mathcal{H}_P(\pi)_\chi = \{\phi \in \mathcal{H}_P(\pi) \mid m \longmapsto \varphi(mx)\ (m \in M(\mathbb{Q})\backslash \overline{M(\mathbb{A})}^1)\\ \text{belongs to } L^2(M(\mathbb{Q})\backslash \overline{M(\mathbb{A})^1})_\chi\}.\end{gathered}$$

Proposition 5.6. $\mathcal{H}_P(\pi) = \bigoplus_{\chi \in \mathcal{X}} \mathcal{H}_P(\pi)_\chi$.

Let $K \subset G(\mathbb{A}_f) \cap K_0$ be an open compact subgroup and let

$$\mathcal{H}_P(\pi)_{\chi, \overline{K}} \subset \mathcal{H}_P(\pi)_\chi \tag{5.10}$$

denote the subspace which is right invariant under $\overline{K}$. Let W denote a genuine irreducible representation of $\overline{K_\infty}$ and let

$$\mathcal{H}_P(\pi)_{\chi, \overline{K}, W} \subset \mathcal{H}_P(\pi)_{\chi, \overline{K}} \tag{5.11}$$

denote the subspace of functions ϕ such that $k \longmapsto \phi(xk)$ $(k \in \overline{K_\infty})$ is a matrix coefficient of W.

Proposition 5.7. $\dim \mathcal{H}_P(\pi)_{\chi,\overline{K},W} < \infty$.

We apply this to fixing a choice of a basis: For each connected component $\Pi^0 \subset \Pi_u(\overline{M(\mathbb{A})}^1)$, fix an element $\pi^0 \in \Pi^0$. Every other $\pi \in \Pi^0$ may be uniquely written as $\pi = \pi^0_\zeta$, for some $\zeta \in i\mathfrak{a}_P^*$. Let $\mathcal{B}_P(\pi^0)_\chi$ denote a basis of $\mathcal{H}_P(\pi^0)_\chi$ satisfying the following properties:

1. each $\phi \in \mathcal{B}_P(\pi^0)_\chi$ belongs to $\mathcal{H}_P(\pi^0)_{\chi,\overline{K},W}$ for some K, W (possibly depending on ϕ),

2. the elements of $\mathcal{B}_P(\pi^0)_\chi$ form an orthonormal set with respect to the inner product

$$(\phi_1, \phi_2) = \int_K \int_{M(\mathbb{Q})\backslash \overline{M(\mathbb{A})}^1} \phi_1(mk)\overline{\phi_2(mk)} dm dk,$$

3. if $\varsigma \in i\mathfrak{a}^*$ then

$$\mathcal{B}_P(\pi^0_\varsigma)_\chi = \{\phi_\varsigma \mid \phi \in \mathcal{B}_P(\pi^0)_\chi\}.$$

$\mathcal{B}_P(\pi)_\chi$ is a basis for $\mathcal{H}_P(\pi)_\chi$, if $\pi = \pi^0$, and is a subset of $\mathcal{H}_P(\pi)$ in general.

Let $\mathcal{P}$ denote a class of associated parabolics. Let $L^\wedge_\mathcal{P}$ denote the set of collections $F = \{F_P \mid P \in \mathcal{P}\}$ of measurable functions $F_P : \Pi_u(\overline{M(\mathbb{A})}) \to \mathcal{H}_P$ such that

(i) if $s \in \Omega(\mathfrak{a}, \mathfrak{a}_1)$ then

$$F_{P_1}(s\pi) = M(s, \pi)F_P(\pi),$$

(ii) if $N(A)$ denotes the number of chambers in $\mathfrak{a}$ (which is equal to 1 for $SL(2)$) then

$$||F||^2 = \sum_{P \in \mathcal{P}} \frac{1}{2\pi}^{dim(A)} \int_{\Pi_u(\overline{M(\mathbb{A})})} ||F_P(\pi)||^2 d\pi < \infty.$$

The following result describes some basic properties of Eisenstein series.

Theorem 5.8. *For $s \in \Omega(\mathfrak{a}, \mathfrak{a}_1)$, we have (Langlands [A2, p. 927])*

(a) If $\varphi \in \mathcal{H}^0_P(\pi)$ then $E(x, \varphi_\varsigma)$ and $M(s, \pi_\varsigma)\varphi_\varsigma$ can be analytically continued to a meromorphic function on $\mathfrak{a}^*_\mathbb{C}$. On $i\mathfrak{a}$ the Eisenstein series $E(x, \varphi_\varsigma)$

is regular and the intertwining operator $M(s,\pi_\varsigma)$ is unitary. For $\overline{K_0}$-finite $f \in C_c^\infty(\overline{G(\mathbb{A})})_{gen}$, $t \in \Omega(\mathfrak{a}_1,\mathfrak{a}_2)$, we have

$$\begin{array}{cc} (i) & E(x, I_P(\pi,f)\varphi) = \int_{\overline{G(\mathbb{A})}} f(y)E(xy,\varphi)dy, \\ (ii) & E(x,M(s,\pi)\varphi) = E(x,\varphi), \\ (iii) & M(ts,\pi) = M(t,s\pi)M(s,\pi). \end{array}$$

(b) There is a dense subspace of $L^\wedge_{\mathcal{P}}$ on which the map

$$F \longmapsto \sum_{P\in\mathcal{P}} n(A)^{-1}\left(\frac{1}{2\pi i}\right)^{\dim(A)} \int_{\Pi_u(\overline{M(\mathbb{A})})} E(x,F_P(\pi))d\pi$$

is defined. This extends to all of $L^\wedge_{\mathcal{P}}$ and defines a unitary map of $L^\wedge_{\mathcal{P}}$ onto a closed subspace

$$L^2_{\mathcal{P}}(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}) \subset L^2(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{gen}.$$

Moreover,

$$L^2(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{gen} = \bigoplus_{\mathcal{P}} L^2_{\mathcal{P}}(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}).$$

5.3 The integrability of $k^T_\chi(x,f)$

For K-finite compactly supported f as above, let

$$K_{P,\chi}(x,y) = \sum_{P_1\subset P} n_P(A_1)^{-1}\int_{\Pi_u(\overline{M(\mathbb{A})})} \sum_{\phi\in\mathcal{B}_{P_1}(\pi)_\chi} E_P(x,I_{P_1}(\pi,f)\phi)\overline{E_P(y,\phi)}d\pi \tag{5.12}$$

and

$$k^T_\chi(x,f) = \sum_P (-1)^{\dim(A_P/Z)} \sum_{\delta\in P(\mathbb{Q})\backslash G(\mathbb{Q})} K_{P,\chi}(x,x)\tau^\wedge_P(H(\delta x)-T). \tag{5.13}$$

Lemma 5.9. *$K_{P,\chi}(x,y)$ is the kernel function for the regular representation $R_P(f)$ of $\overline{G(\mathbb{A})}$ on*

$$L^2(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{\mathcal{P},\chi} = L^2(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{\mathcal{P}} \cap L^2(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_\chi,$$

where $\mathcal{P}$ denotes the associate class of P. In particular, $K_{G,\chi}(x,y) = K_\chi(x,y)$ is the kernel function for the regular representation $R_P(f)$ of $\overline{G(\mathbb{A})}$ on $L^2(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_\chi$.

Remark 5.10. *In [A15], different notation is used: $R_P(f)$ denotes the convolution operator associated to right translation on*

$$L^2(N_P(\mathbb{A})M_P(\mathbb{Q})\backslash\overline{G(\mathbb{A})})) = \oplus_\chi L^2(N_P(\mathbb{A})M_P(\mathbb{Q})\backslash\overline{G(\mathbb{A})}))_\chi.$$

Theorem 5.11. $\sum_{\chi\in\mathcal{X}} K_{P,\chi}(x,y) = \sum_{\mathcal{O}\in\mathcal{C}} K_{P,\mathcal{O}}(x,y)$.

proof: Both sides represent the kernel function for the regular representation $R_P(f)$ of $\overline{G(\mathbb{A})}$ on $L^2(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})$. □

The trace formula is obtained by integrating both sides of

$$\sum_{\chi\in\mathcal{X}} \Lambda^T K_{P,\chi}(x,x) = \sum_{\mathcal{O}\in\mathcal{C}} \Lambda^T K_{P,\mathcal{O}}(x,x)$$

over $G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$, where Λ^T is the truncation operator defined below. We know from earlier sections that the right-hand side is a polynomial in T, so the left-hand side is too. The **global trace formula** is the constant term of this expression.

To prove the integrability of $k_\chi^T(x,f)$ a truncation operator is introduced. If $\phi \in C(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{gen}$ then define, as in [A3, p 89], the **truncation operator** by

$$\Lambda^T\phi(x) = \sum_P (-1)^{\dim(A/Z)} \sum_{\delta\in P(\mathbb{Q})\backslash G(\mathbb{Q})} \int_{N(\mathbb{Q})\backslash N(\mathbb{A})} \phi(n\delta x)dn \cdot \tau_P^\wedge(H(\delta x) - T). \tag{5.14}$$

More generally, if $P_1 \subset G$ is a parabolic and if $\phi \in C(P_1(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{gen}$ then define, as in [A3, p 97], the **truncation operator** by

$$\Lambda^{T,P_1}\phi(x) = \sum_{R,P_0\subset R\subset P_1} (-1)^{\dim(A_R/A_1)} \sum_{\delta\in R(\mathbb{Q})\backslash P_1(\mathbb{Q})} \int_{N_R(\mathbb{Q})\backslash N_R(\mathbb{A})} \phi(n\delta x)dn\cdot\tau_R^{\wedge,1}(H(\delta x)-T). \tag{5.15}$$

In the classical case of $SL(2)$, the truncation operator acting on the Eisenstein series has the effect of subtracting off the "y^s term" in its Fourier expansion. This allows one to compute the inner product of two truncated Eisenstein series (see Theorem 5.27 below).

Example 5.12. *For $G = SL(2), P_1 = G$ this formula becomes*

$$\Lambda^T\phi(x) = \phi(x) - \sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})} \int_{N(\mathbb{Q})\backslash N(\mathbb{A})} \phi(n\delta x)dn \cdot \tau_B^\wedge(H(\delta x) - T). \tag{5.16}$$

Lemma 5.13. *The truncation operator is a self-adjoint idempotent on* $C(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{gen} \cap L^2(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})$.

proof: This is an immediate consequence of the non-metaplectic version [A3] since the metaplectic cover splits over N. □

Lemma 5.14. *([A3, Lemma 1.5]) For* $\phi \in C(P(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{gen}$, *we have*

$$\sum_{P_1, P_0 \subset P_1 \subset P} \sum_{\delta \in P_1(\mathbb{Q})\backslash P(\mathbb{Q})} \Lambda^{T,P_1}\phi(\delta x)\tau_1^P(H_0(\delta x) - T) = \int_{N(\mathbb{Q})\backslash N(\mathbb{A})} \phi(nx)dn.$$

proof in the case $G = SL(2)$: Suppose $P = G$. First, from the fact that

$$\sum_{P_1, R \subset P_1 \subset Q} (-1)^{\dim(A_R/A_1)}\tau_R^{\wedge,1}(X)\tau_1^Q(X) = \begin{cases} 1, & Q = R, \\ 0, & Q \neq R, \end{cases} \tag{5.17}$$

(this is Langlands' combinatorial lemma; see [A3]) , we have $\tau_B^{\wedge,B}(X)\tau_B^G(X) = \tau_B^{\wedge,G}(X)\tau_G^G(X) = \tau_B^{\wedge}(X)$. The LHS of the equation in the above lemma is

$$\begin{aligned}
&\sum_{P_1, B \subset P_1 \subset G} \sum_{\delta \in P_1(\mathbb{Q})\backslash G(\mathbb{Q})} \Lambda^{T,P_1}\phi(\delta x)\tau_1^G(H_0(\delta x) - T) \\
= \;& \Lambda^{T,G}\phi(x)\tau_G^G(H_0(x) - T) + \\
&+ \sum_{\delta \in B(\mathbb{Q})\backslash G(\mathbb{Q})} \Lambda^{T,B}\phi(\delta x)\tau_B^G(H_0(\delta x) - T) \\
= \;& \Lambda^{T,G}\phi(x) + \sum_{\delta \in B(\mathbb{Q})\backslash G(\mathbb{Q})} \int_{N(\mathbb{Q})\backslash N(\mathbb{A})} \phi(n\delta x)dn \cdot \tau_B^{\wedge,B}(H_0(\delta x) - T) \cdot \tau_B^G(H_0(\delta x) - T) \\
= \;& \Lambda^{T,G}\phi(x) + \sum_{\delta \in B(\mathbb{Q})\backslash G(\mathbb{Q})} \int_{N(\mathbb{Q})\backslash N(\mathbb{A})} \phi(n\delta x)dn \cdot \tau_B^{\wedge}(H_0(\delta x) - T) \\
= \;& \phi(x),
\end{aligned}$$

by the previous example.

Suppose $P = B$. The LHS of the equation in the above lemma is

$$\Lambda^{T,B}\phi(x)\tau_B^B(H_0(x) - T) = \int_{N(\mathbb{Q})\backslash N(\mathbb{A})} \phi(nx)dn,$$

since $\tau_B^B(X) = 1$.

This verifies the lemma in both cases. □

Lemma 5.15. *([A3, Lemma 2.2])*

$$k_\chi^T(x,f) = \sum_{P_1,P_2 \text{s.t.} P_0\subset P_1\subset P_2} \sum_{\delta\in P_1(\mathbb{Q})\backslash G(\mathbb{Q})} \sigma_1^2(H_0(\delta x)-T)\times$$
$$\times\left\{\sum_{P,P_1\subset P\subset P_2}(-1)^{\dim(A/Z)}\Lambda_2^{T,P}K_{P,\chi}(\delta x,\delta x)\right\}.$$

(The subscript 2 *in* $\Lambda_2^{T,P}$ *means that it is to be applied to the second variable.)*

Example 5.16. *Let* $G = SL(2)$*. The above lemma becomes*

$$k_\chi^T(x,f) = \sum_{P_1,P_2 \text{s.t.}(P_1,P_2)\in\{(B,B),(B,G),(G,G)\}} \sum_{\delta\in P_1(\mathbb{Q})\backslash G(\mathbb{Q})} \sigma_1^2(H_0(\delta x)-T)\times$$
$$\times\left\{\sum_{P,P_1\subset P\subset P_2}(-1)^{\dim(A/Z)}\Lambda_2^{T,P_1}K_{P,\chi}(\delta x,\delta x)\right\}$$
$$= -\sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})}\sigma_B^B(H_0(\delta x)-T)\Lambda_2^{T,B}K_{B,\chi}(\delta x,\delta x) \quad \text{(the } (B,B) \text{ term)}$$
$$+\sigma_G^G(H_0(x)-T)\Lambda_2^{T,G}K_{G,\chi}(x,x) \quad \text{(the } (G,G) \text{ term)}$$
$$+\sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})}\sigma_B^G(H_0(\delta x)-T)\times$$
$$\times\left\{-\Lambda_2^{T,B}K_{B,\chi}(\delta x,\delta x)+\Lambda_2^{T,B}K_{G,\chi}(\delta x,\delta x)\right\} \quad \text{(the } (B,G) \text{ term)}$$
$$= \Lambda_2^{T,G}K_{G,\chi}(x,x)+$$
$$+\sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})}\tau_B(H_0(\delta x)-T)\left\{-\Lambda_2^{T,B}K_{B,\chi}(\delta x,\delta x)+\Lambda_2^{T,B}K_{G,\chi}(\delta x,\delta x)\right\}$$
$$= \Lambda_2^{T,G}K_{G,\chi}(x,x)+\sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})}\tau_B(H_0(\delta x)-T)\Lambda_2^{T,B}K_{G,\chi}(\delta x,\delta x).$$

by (2.5)*. The definition is*

$$k_\chi^T(x,f) = K_{G,\chi}(x,x) - \sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})} K_{B,\chi}(\delta x,\delta x)\tau_B^\wedge(H_0(\delta x)-T). \tag{5.18}$$

We claim now that

$$\Lambda_2^{T,G}K_{G,\chi}(x,x) = K_{G,\chi}(x,x) - \sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})}\tau_B(H(\delta x)-T)\Lambda_2^{T,B}K_{G,\chi}(\delta x,\delta x). \tag{5.19}$$

To see this, note

$$\Lambda_2^{T,G}K_{G,\chi}(x,x) = K_{G,\chi}(x,x) - \sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})}\int_{N(\mathbb{Q})\backslash N(\mathbb{A})}\Lambda_2^{T,B}K_{G,\chi}(x,n\delta x)dn\cdot\tau_B^{\wedge}(H(\delta x)-T).$$

On the other hand,

$$\Lambda_2^{T,B}K_{G,\chi}(x,x) = \int_{N(\mathbb{Q})\backslash N(\mathbb{A})}K_{G,\chi}(x,nx)dn\cdot\tau_B^{\wedge,B}(H(\delta x)-T),$$

so

$$\begin{aligned}\sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})}&\int_{N(\mathbb{Q})\backslash N(\mathbb{A})}\Lambda_2^{T,B}K_{G,\chi}(x,n\delta x)dn\cdot\tau_B^{\wedge}(H(\delta x)-T)\\ &= \sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})}\Lambda_2^{T,B}K_{G,\chi}(x,\delta x)\cdot\tau_B^{\wedge}(H(\delta x)-T)\\ &= \sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})}\Lambda_2^{T,B}K_{G,\chi}(\delta x,\delta x)\cdot\tau_B(H(\delta x)-T), \qquad (5.20)\end{aligned}$$

since the volume of $N(\mathbb{Q})\backslash N(\mathbb{A})$ and $\tau_B(X) = \tau_B^{\wedge}(X)$ in this case. The claimed result follows.

Example 5.17. *Let $G = SL(2),\ P = G$. Then*

$$E_G(x,\phi) = E(x,\phi) = \sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})}\phi(\delta x)\delta_B(\delta x)^{1/2}.$$

If $P = B$ then $E_B(x,\phi) = \phi(x)\delta_B(x)^{1/2}$.

Example 5.18. *Let $G = SL(2),\ P = B$. For $\pi\in\Pi_u(\overline{A(\mathbb{A})})$, $\chi\in\mathcal{X}$, the set $\mathcal{B}_B(\pi)_\chi$ is an orthonormal basis for $\mathcal{H}_B(\pi)_\chi$. We have*

$$\mathcal{H}_B(\pi)_\chi = \left\{\begin{array}{c}\phi : N(\mathbb{A})A(\mathbb{Q})\setminus\overline{G(\mathbb{A})}\to\mathbb{C}\text{ genuine }|\\ (1)\ \ \phi\text{ is right }\overline{K_0}\text{ −finite},\\ (2)\ \ \phi(ag)=\pi(a)\phi(g),\ \ a\in\overline{A(\mathbb{A})},\ g\in\overline{G(\mathbb{A})},\\ (3)\ \ ||\phi||<\infty\end{array}\right\}.$$

Example 5.19. *Let $G = SL(2),\ P = G$. For $\pi\in\Pi_u(\overline{G(\mathbb{A})})$ cuspidal, $\chi\in\mathcal{X}$, the set $\mathcal{B}_G(\pi)_\chi$ is an orthonormal basis for $\mathcal{H}_G(\pi)_\chi$. If $\chi = (G,\pi)$ (in $\mathcal{X}$) then this space is non-zero and consists of (measurable equivalence classes*

of) square-integrable genuine functions on $G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$ which are cuspidal automorphic forms of type π_∞. Since almost every π_p is unramified, the action of almost all $\overline{K_{0,p}}$ is trivial on the functions in this space. Since the functions in this space are all $\overline{K_0}$-finite, we may decompose it by the action of $\overline{K_0}$ on the right:

$$\mathcal{H}_G(\pi)_\chi = \bigoplus_{\sigma\in\Pi(\overline{K_0})} \mathcal{H}_G(\pi)_{\chi,\sigma}$$

where each irreducible representation $\sigma \in \Pi(\overline{K_0})$ occurs at most with finite multiplicity. For the basis above, one must pick a element in each one of these "$\overline{K_0}$-types".

Example 5.20. *Let $G = SL(2)$, $P = B$. Recalling the determination of $\mathcal{B}_B(\pi)_\chi$ above, we have*

$$K_{B,\chi}(x,y) = \int_{\Pi_u(\overline{A(\mathbb{A})})} \sum_{\phi\in\mathcal{B}_B(\pi)_\chi} I_B(\pi,f)\phi(x)d\pi\cdot\delta_B(x)\overline{\phi(y)}$$

Example 5.21. *Let $G = SL(2)$, $P = G$. Recalling the determination of $\mathcal{B}_B(\pi)_\chi$ above, we have*

$$K_{G,\chi}(x,y) = \int_{\Pi_u(\overline{G(\mathbb{A})})} \sum_{\phi\in\mathcal{B}_G(\pi)_\chi} E(x,I_G(\pi,f)\phi)d\pi\cdot\overline{E(y,\phi)} + \int_{\Pi_u(\overline{A(\mathbb{A})})} \sum_{\phi\in\mathcal{B}_B(\pi)_\chi} I_B(\pi,f)\phi(x)d\pi\cdot\delta_B(x)\overline{\phi(y)}.$$

Example 5.22. *Let $G = SL(2)$. We have*

$$\begin{aligned} k_\chi^T(x,f) &= K_{G,\chi}(x,x) - \sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})} K_{B,\chi}(\delta x,\delta x)\tau_B^\wedge(H_0(\delta x)-T) \\ &= \int_{\Pi_u(\overline{G(\mathbb{A})})} \sum_{\phi\in\mathcal{B}_G(\pi)_\chi} E_G(x,I_G(\pi,f)\phi)d\pi\cdot\overline{E_G(x,\phi)} + \\ &+ \int_{\Pi_u(\overline{A(\mathbb{A})})} \sum_{\phi\in\mathcal{B}_B(\pi)_\chi} I_B(\pi,f)\phi(\delta x)d\pi\cdot\delta_B(\delta x)\overline{\phi(\delta x)}(1-\tau_B^\wedge(H(x)-T)) \\ &- \sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q}),\delta\neq 1} \int_{\Pi_u(\overline{A(\mathbb{A})})} \sum_{\phi\in\mathcal{B}_B(\pi)_\chi} I_B(\pi,f)\phi(\delta x)d\pi\cdot\delta_B(\delta x)\overline{\phi(\delta x)}\tau_B^\wedge(H(x)-T) \end{aligned}$$

Let

$$K_\chi^T(f) = \int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} \Lambda_2^T K_\chi(x,x)dx. \tag{5.21}$$

Define

$$J_\chi^T(x,y) = \sum_P n(A)^{-1} \int_{\Pi_u(\overline{M(\mathbb{A})})} \sum_{\phi\in\mathcal{B}_P(\pi)_\chi} E(x, I_P(\pi,f)\phi)\overline{\Lambda^T E(y,\phi)}d\pi \tag{5.22}$$

and $j_\chi^T(x,f) = J_\chi^T(x,x)$. Let

$$J_\chi^T(f) = \int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} J_\chi^T(x,x)dx$$

and let

$$J_\chi(f) = J_\chi^T(f)|_{T=0}. \tag{5.23}$$

Lemma 5.23. *The integrals $K_\chi^T(f), J_\chi^T(f)$ are absolutely convergent and $K_\chi^T(f) = J_\chi^T(f)$ if T is sufficiently regular.*

Remark 5.24. *The equality above is the metaplectic analog of [A3, Lemma 2.4] for $SL(2)$.*

proof in the case $G = SL(2)$: The integral

$$K_\chi^T(f) = \int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} \Lambda_2^T K_\chi(x,x)dx.$$

is absolutely convergent since the integrand is compactly supported on $G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$.

Let $F(P_1,P_2)$ denote the subset of elements of $P_1(\mathbb{Q})\backslash P_2(\mathbb{Q})$ which do not belong to for some parabolic P with $P_1 \subset P \subset P_2,\ P \neq P_2$. Then

$$F(P_1,P_2) = \begin{cases} \{1\}, & P_1 = P_2, \\ B(\mathbb{Q})\backslash G(\mathbb{Q}) - \{1\}, & P_1 = B,\ P_2 = G, \end{cases}$$

and, from [A3, p. 105],

$$\begin{aligned}
&\int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} k_\chi^T(x,f)dx \\
&= \sum_{P_1\subset P_2} (-1)^{\dim(A_2/Z)} \int_{P_1(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} \sum_{\gamma\in F(P_1,P_2)} \sigma_1^2(H(x)-T)\Lambda_2^{T,P_1}K_{P_1,\chi}(\gamma x,x)dx \\
&= \int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} \sigma_G^G(H(x)-T)\Lambda_2^{T,G}K_{G,\chi}(x,x)dx \\
&\quad (the\ P_1=P_2=G\ term) \\
&+ \int_{B(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} \sigma_B^G(H(x)-T) \sum_{\gamma\in B(\mathbb{Q})\backslash G(\mathbb{Q})-\{1\}} \Lambda_2^{T,B}K_{B,\chi}(\gamma x,x)dx \\
&\quad (the\ P_1=B,\ P_2=G\ term) \\
&- \int_{B(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} \sigma_B^B(H(x)-T)\Lambda_2^{T,B}K_{B,\chi}(x,x)dx \qquad (P_1=P_2=B)
\end{aligned}$$

Each of these integrals converge absolutely since the integral is continuous and compactly supported. □

Lemma 5.25. $\sum_{\chi=(M,\rho)\in\mathcal{X},M\neq G}\int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}}|j_\chi^T(x,f)|dx<\infty$.

proof in the case $G=SL(2)$: Since $K_P(x,y)=\sum_{\chi\in\mathcal{X}}K_{P,\chi}(x,y)$ is absolutely and uniformly convergent on compacta in x,y, $K_P(x,y)$ is continuous in x,y. The same reasoning as in the previous lemma therefore applies here as well. □

Lemma 5.26. *For $\chi\in\mathcal{X}$, we have*

$$\int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} j_\chi^T(x,f)dx = \sum_P n(A)^{-1}\int_{\Pi_u(\overline{M(\mathbb{A})})}\sum_{\phi\in\mathcal{B}_P(\pi)_\chi}\Lambda^T E(x,I_P(\pi,f)\phi)\overline{\Lambda^T E(y,\phi)}d\pi.$$

proof: This follows from Lemma 5.23 above on the equality of $J_\chi^T(f)$ and $K_\chi^T(f)$, along with the metaplectic analog of the fact that fact $\Lambda^T\circ\Lambda^T=\Lambda^T$ [A3, Corollary 1.2] and that Λ^T is self-adjoint [A3, Lemma 1.3] (see Lemma 5.13 above). □

Let $L_P\subset\mathfrak{a}_P^G$ denote the integral lattice generated by the co-roots $\{\alpha^\vee\mid\alpha\in\Delta_P\}$. For $\phi\in\mathcal{H}_{P_1}^0(\pi),\ s\in\Omega(\mathfrak{a}_1,\mathfrak{a}_2),\varsigma\in\mathfrak{a}_{1,\mathbb{C}}^*$, let

$$M(s,\pi,\varsigma)\phi=(M(s,\pi_\varsigma)\phi_\varsigma)_{-s\varsigma}.$$

Theorem 5.27. *If $\chi \in \mathcal{X}$ is an unramified class then ([A3, section 4])*

$$J_\chi^T(f) = \sum_{P \in \mathcal{P}_\chi} n(A)^{-1} \int_{\Pi_u(\overline{M(\mathbb{A})})} tr(M_P^T(\pi)_\chi I_P(\pi, f)_\chi) d\pi,$$

where

$$M_P^T(\pi)_\chi = vol(\mathfrak{a}_P^G / L_P) \lim_{\varsigma \to 0} \sum_{P_2 \in \mathcal{P}_\chi} \sum_{s \in \Omega(\mathfrak{a}, \mathfrak{a}_2)} \frac{e^{(s\varsigma)(T)} M(s,\pi)^{-1} M(s,\pi,\varsigma)}{\prod_{\alpha \in \Delta_2}(s\varsigma)(\alpha^\vee)}.$$

For a similar but more general result (in the non-metaplectic case), see [A14, Theorem 8.2], which does not assume that χ is unramified.

Example 5.28. *For $G = SL(2)$, each $(G, \pi) \in \chi \in \mathcal{X}$ is unramified. Moreover, $(A, \mu) \in \chi \in \mathcal{X}$ is unramified if and only if $\mu^w \neq \mu$. ($P = G, \chi = (G, \rho)$) $M_G^T(\pi)_\chi = 1$*
($P = B, \chi = (G, \rho)$) $M_B^T(\pi)_\chi = 1$
($P = B, \chi = (A, \rho)$ unramified)

$$\begin{aligned} & M_B^T(\pi)_\chi \\ &= vol(\mathbb{R}/\mathbb{Z}) \lim_{\varsigma \to 0} \left[\frac{e^{\varsigma(T)} M(1,\pi)^{-1} M(1,\pi,\varsigma)}{\varsigma(\alpha^\vee)} + \frac{e^{-\varsigma(T)} M(s,\pi)^{-1} M(s,\pi,\varsigma)}{-\varsigma(\alpha^\vee)} \right] \\ &= M(s,\pi)^{-1} \lim_{\varsigma \to 0} \frac{M(s,\pi) - e^{-2\varsigma(T)} M(s,\pi,\varsigma)}{\varsigma(\alpha^\vee)}, \end{aligned}$$

where α denotes the unique positive root. We identify $\mathfrak{a}^*$ with $\mathbb{R}$, $\mathfrak{a}$ with $\mathbb{R}$, and $\mathfrak{a}^* \times \mathfrak{a} \to \mathbb{R}$ with the map $(x, y) \longmapsto 2xy$ of real numbers. Thus, at a function ϕ on K, the above limit becomes

$$\begin{aligned} & M(s,\pi)^{-1} \lim_{\varsigma \to 0} \frac{M(s,\pi)\phi(k) - e^{-2\varsigma(T)} e^{\varsigma(H(k))} M(s,\pi_\varsigma)\phi_\varsigma(k)}{\varsigma(\alpha^\vee)} \\ &= M(s,\pi)^{-1} \lim_{\varsigma \to 0} \frac{M(s,\pi)\phi(k) - e^{-2\varsigma(T)} M(s,\pi_\varsigma)\phi_\varsigma}{\varsigma(\alpha^\vee)} \\ &= -M(s,\pi)^{-1} M'(s,\pi)\phi(k) + 4\phi(k)T, \end{aligned}$$

where the prime denotes differentiation with respect to ς. Thus

$$M_B^T(\pi)_\chi = -M(s,\pi)^{-1} M'(s,\pi) + 4 \cdot I \cdot T,$$

where I denotes the identity. (There appears to be a sign mistake here - the minus sign perhaps differs with some calculations of Arthur. However, the sign it gives for the corresponding expression in the trace formula seems to agree with the corresponding sign in [LL].)

For examples and properties of more general limits, see [A6, section 6].

5.4 The trace formula

The geometric side of the trace formula is the sum of the following terms:

(a) the sum over classes $\mathcal{O} \subset G(\mathbb{Q})$ such that $\mathcal{O} \cap Z(G(\mathbb{Q})) \neq \emptyset$ of

$$J_{\mathcal{O}}(f) = a^G(S, z) f(z) + \frac{1}{2} a^A(S, z) J_A(z, f), \tag{5.24}$$

where J_A and a^M are given in §§4.5–4.6 (these include the "nilpotent terms"),

(b) the sum over elliptic regular classes $\mathcal{O} \subset G(\mathbb{Q})$ of

$$J_{\mathcal{O}}(f) = a^G(S, \gamma) J_G(\gamma, f), \tag{5.25}$$

where $\gamma \in \mathcal{O}$ and J_G (an orbital integral) and a^G (a volume) are given in §§4.5–4.6,

(c) the sum over "hyperbolic" regular classes $\mathcal{O} \subset G(\mathbb{Q})$ of

$$J_{\mathcal{O}}(f) = a^A(S, \gamma) J_A(\gamma, f), \tag{5.26}$$

where J_A (a weighted orbital integral) and a^A (a volume) are given in §§4.5–4.6.

The spectral side of the trace formula is the sum of the following terms:

(a) the sum over the cuspidal classes $\chi \in \mathcal{X}(G)$ of

$$J_\chi(f) = \mathrm{tr}(\pi(f)), \quad (\pi, G) \in \chi, \tag{5.27}$$

(b) the sum over classes $\chi \in \mathcal{X} - \mathcal{X}(G)$ of

$$J_\chi(f) = \frac{1}{2} \int_{\Pi_u(\overline{A(\mathbb{A}_S)})} \mathrm{tr}(M_A(\pi)_\chi I_B(\pi, f)_\chi) d\pi, \tag{5.28}$$

where $M_A(\pi)_\chi = M_A^T(\pi)_\chi|_{T=0}$ is (essentially a logarithmic derivative of an intertwining operator) given by Example 5.28.

6 The global trace formula in invariant form

The global trace formula is an equality of the form

$$\sum_{\chi \in \mathcal{X}} J_\chi(f) = \sum_{\mathcal{O} \in \mathcal{C}} J_{\mathcal{O}}(f), \tag{6.1}$$

where $J_\chi(f) = J_\chi^T(f)|_{T=0}$, $J_{\mathcal{O}}(f) = J_{\mathcal{O}}^T(f)|_{T=0}$, and where $J_\chi^T(f)$ is given in Section 5 and $J_{\mathcal{O}}^T(f)$ is given in Section 4.

We can very crudely index the terms $J_{\mathcal{O}}$ by the centralizer of the semi-simple part s of a $\gamma = su \in \mathcal{O}$. As a general principle, the "smaller" the centralizer the easier it is to deal (analytically) with the term.

Let $\mathcal{C}(G)$ denote the subset of elliptic regular equivalence classes in $\mathcal{C}$ and let $\mathcal{X}(G)$ denote the subset of $\chi \in \mathcal{X}$ of the form $\chi = (G, \rho)$ for some cuspidal $\rho \in \Pi_u(\overline{G(\mathbb{A})})$.

Not all the expressions in the above trace formula (6.1) are invariant in the sense that $J(f) = J(f^y)$ for all $f \in \mathcal{H}(\overline{G(\mathbb{A})})$ and all $y \in \overline{G(\mathbb{A})}$. The terms $J_{\mathcal{O}}(f)$ with $\mathcal{O} \in \mathcal{C}(G)$ are invariant and the terms $J_\chi(f)$ with $\chi \in \mathcal{X}(G)$ are invariant. We follow [A6], [A9], [A10], in putting the remaining terms in invariant form.

Fix a finite set of places S of $\mathbb{Q}$ which unless stated otherwise will contain the archimedian place and the primes $p|2n$. Let $\mathbb{A}_S = \Pi_{p\in S}\mathbb{Q}_p$ and embed $\mathcal{H}(\overline{G(\mathbb{A}_S)}) \hookrightarrow \mathcal{H}(G(\mathbb{A}))$ in the natural way. We identify $f \in \mathcal{H}(\overline{G(\mathbb{A}_S)})$ with the $f_S \in \mathcal{H}(G(\mathbb{A}))$ which agrees with f on $\overline{G(\mathbb{A}_S)}$ and with the unit element of $\mathcal{H}(\overline{G(\mathbb{Q}_p)}, K_p)$ on $\overline{G(\mathbb{Q}_p)}$ for $p \notin S$.

Define

$$I_\chi(f) = I_\chi^S(f) = J_\chi(f), \qquad \chi \in \mathcal{X}(G), \tag{6.2}$$

and

$$I_{\mathcal{O}}(f) = I_{\mathcal{O}}^S(f) = J_{\mathcal{O}}(f), \qquad \mathcal{O} \in \mathcal{C}(G), \tag{6.3}$$

for $f \in \mathcal{H}(\overline{G(\mathbb{A}_S)})$. One of the objects to this subsection is to define $I_\chi(f) = I_\chi^S(f)$, $\chi \in \mathcal{X} - \mathcal{X}(G)$, and $I_{\mathcal{O}}(f) = I_{\mathcal{O}}^S(f)$, $\mathcal{O} \in \mathcal{C} - \mathcal{C}(G)$, following [A6]. Once this is done, we will use the formalism of [A6] to deduce the global invariant trace formula in the form

$$\sum_{\chi\in\mathcal{X}} I_\chi(f) = \sum_{\mathcal{O}\in\mathcal{C}} I_{\mathcal{O}}(f). \tag{6.4}$$

To do this requires, in particular, verifying for each $p \in S$ the local assumptions of [A6], of which there are three types:

(1) [A6, Assumption 5.1, p. 30],

(2) [A6, Assumption 5.2, p. 32],

(3) the statements [A6, (7.1)–(7.6)] regarding normalized intertwining operators. (In our case, only the statements [A6, (7.1)–(7.4)] need proving since $SL(2)/k$ is rank one.)

6.1 Local assumptions

We shall now make precise three types of statements alluded to above which must be verified.

Let $Q = M_Q N_Q$ be a parabolic of G and let $y \in G(\mathbb{A}_S)$. For $m \in M(\mathbb{A}_S)$, let

$$f_{Q,y}(m) = \delta_Q(m)^{1/2} \int_K \int_{N_Q(\mathbb{A}_S)} f(k^{-1}mnk) u'_Q(k,y) dn dk, \tag{6.5}$$

where

$$u'_Q(k,y) = \int_{\mathfrak{a}_Q} \sum_{P, Q \subset P} (-1)^{dim(A_Q/A_G)} \tau_Q^P(X) \hat{\tau}_P(X + H_Q(ky)) dX,$$

as in [A6, 3.3] (modulo a typo). (Recall (2.4), (2.5) for $\tau_Q^P, ...$)

Example 6.1. *Let $G = SL(2)$. Then*

$$\begin{gathered} u'_G(k,y) = \tau_G^G(X)\hat{\tau}_G(X + H_G(ky)) = 1, \\ u'_Q(k,y) = \int_{\mathbb{R}} -\tau_B^B(X)\hat{\tau}_B(X + H_B(ky)) + \tau_B^G(X)\hat{\tau}_G(X + H_B(ky)) dX \\ = \int_{\mathbb{R}} -\hat{\tau}_B(X + H_B(ky)) + \tau_B(X) dX \\ = |H_B(ky)|. \end{gathered}$$

6.1.1 Assumption 5.1

First, a

Definition 6.2. *Let $\phi : U \to V$ denote a continuous linear transformation between two complete topological vector spaces. Let U' denote the space of continuous linear functionals on U (similarly for V'). The* **transpose** *$\phi' : V' \to U'$ is the map defined by*

$$\phi'(\lambda) : f \longmapsto \lambda(\phi(f)), \quad \lambda \in V', \ f \in U.$$

Let $C_N^\infty(\mathfrak{a}_{M,S})$ denote the topological vector space of smooth functions on $\mathfrak{a}_{M,S}$ which are supported on

$$\{X \in \mathfrak{a}_{M,S} \mid ||X|| \leq N\},$$

where

$$\mathfrak{a}_{M,S} = \{H_M(m) \mid m \in \overline{M(\mathbb{A}_S)}\}. \tag{6.6}$$

We follow [A11, section 11] (as opposed to [A6, sections 4,5] where the assumptions are stated), restricting ourselves to $G = SL(2)$ and simplifying whenever possible.

Let N, Γ be as in §4.3.1 and let $I_N(\overline{G(\mathbb{A}_S)})_\Gamma$ denote the vector space of complex valued functions ϕ on the tempered dual $\Pi_t(\overline{G(\mathbb{A}_S)})$ such that

(a) if $\pi|_K$ contains no representation in Γ then $\phi(\pi) = 0$,

(b) if $M \subset G$ is an admissible Levi (e.g., $M = A$) and $\sigma \in \Pi_t(\overline{M(\mathbb{A}_S)})$ then

$$\phi(\sigma, X) = \int_{\mathfrak{a}_{M,S}} \phi(\sigma, X) e^{-\Lambda(X)} d\Lambda, \qquad X \in \mathfrak{a}_{M,S}, \tag{6.7}$$

converges to a function of X which belongs to $C_N^\infty(\mathfrak{a}_{M,S})$.

If we fix a family of semi-norms $||...||_{M,\sigma}$ on $C_N^\infty(\mathfrak{a}_{M,S})$ then there are seminorms

$$\phi \longmapsto ||\phi(\sigma, *)||_{M,\sigma}, \qquad \sigma \in \Pi_t(\overline{M(\mathbb{A}_S)}), \tag{6.8}$$

which make $I_N(\overline{G(\mathbb{A}_S)})_\Gamma$ into a topological vector space. By [Ru, ch 1], we may complete $I_N(\overline{G(\mathbb{A}_S)})_\Gamma$ using this family of seminorms to obtain a Frechét space.

Define

$$I(\overline{G(\mathbb{A}_S)}) = \lim_\Gamma I(\overline{G(\mathbb{A}_S)})_\Gamma \tag{6.9}$$

where

$$I(\overline{G(\mathbb{A}_S)})_\Gamma = \lim_N I_N(\overline{G(\mathbb{A}_S)})_\Gamma.$$

In [A11], the spaces $I_{ac}(\overline{G(\mathbb{A}_S)})$ are also introduced, as well as the subspace of moderate functions, which we denote by $I_{mod}(\overline{G(\mathbb{A}_S)})$. Actually only the non-metaplectic versions of these spaces were introduced in [A11], their metaplectic analogs are left to the reader. Note that for $G = SL(2)$ or $G = A$, the diagonal maximal split torus of $SL(2)$, we have

$$I_{ac}(\overline{G(\mathbb{A}_S)}) = I(\overline{G(\mathbb{A}_S)}) = I_{mod}(\overline{G(\mathbb{A}_S)}), \quad I_{ac}(\overline{A(\mathbb{A}_S)}) = I(\overline{A(\mathbb{A}_S)}) = I_{mod}(\overline{A(\mathbb{A}_S)}).$$

Let $M \in \{A, G\}$, let $\mathcal{H}_{ac}$ be as in [A11] (see also §4.3.1 above), and let

$$U_M = \mathcal{H}_{mod}(\overline{M(\mathbb{Q}_p)}) = \mathcal{H}(\overline{M(\mathbb{Q}_p)}).$$

These are smooth, compactly supported (in our case), K-finite functions. Let

$$V_M = I_{mod}(\overline{M(\mathbb{Q}_p)}) = I(\overline{M(\mathbb{Q}_p)})$$

denote the space of functions on $\Pi_u(\overline{M(\mathbb{Q}_p)})$ as in [A11] (and as above). Note that these function spaces are not the same as those in [A6]. I will assume that [A6, Assumption 5.1] is intended for the spaces in [A11, section 11]. Then the assumptions are

Conjecture 6.3. *The map $\phi = \phi_M^M : U_M \to V_M$ given by $\phi(f) : \pi \longmapsto tr(\pi(f))$ is surjective.*

The non-metaplectic analog of this was proven in [A10, Lemma 6.1] (see also [BDK]). For a discussion of the metaplectic conjecture, see [J1].

Conjecture 6.4. *The image of $\phi' : V_M' \to U_M'$ is the space of all invariant distributions on $\overline{M(\mathbb{Q}_p)}$.*

In [A6, Lemma 8.3, Theorem 12.1], it is shown that these imply

Theorem 6.5. *Assume the conjectures above. There is a map $\phi_M^G : U_G \to V_M$ satisfying*

(a) if $G = M$ then ϕ_M^G is as above,

(b) in general, we have

$$\phi_M^G(f^y) = \sum_{Q, M \subset M_Q} \phi_M^{M_Q}(f_{Q,y}), \qquad f \in U_G.$$

In particular, $\phi_G^G(f^y) = \phi_G^G(f)$.

(c) for $f \in U_G$, $\pi \in \Pi_t(\overline{M(\mathbb{A}_S)})$, we have $\phi_M^G(f, \pi) = \mathrm{tr}(I_{P_0}(\pi, f) R_M^G(\pi, P_0))$, where $R_M^G(\pi, P_0))$ is defined in [A6, section 7] via the (G, M)-family

$$R_P(\lambda, \pi, P_0)) = R_{P|P_0}(\pi)^{-1} R_{P|P_0}(\pi_\lambda),$$

where the $R_{P|P_0}(\pi_\lambda)$ are as in §6.1.3 below.

Example 6.6. *Let $G = SL(2)$. In this case,*

$$\phi_A^G(f, \pi) = \mathrm{tr}(I_B(\pi, f) R_A^G(\pi, B))$$

is a global "weighted character" similar to the integrand of the expression in Theorem 5.27 above. The term $R_A^G(\pi, B)$ calculated in a similar fashion to $M_B^T(\pi)_\chi$ in Example 5.28.

Let $\sigma \in \Pi(\overline{A(\mathbb{A}_S)})$ and let

$$R_{\overline{B}|B}(\sigma) = m(\sigma)^{-1} M_{\overline{B}|B}(\sigma)$$

Then we may write

$$R_{\overline{B}|B}(\sigma_\lambda)^{-1} R'_{\overline{B}|B}(\sigma_\lambda)|_{\lambda=0} = -m(\sigma_\lambda)^{-1} m'(\sigma_\lambda)|_{\lambda=0} + M_{\overline{B}|B}(\sigma_\lambda)^{-1} M'_{\overline{B}|B}(\sigma_\lambda)|_{\lambda=0}$$

Eventually we will want to plug this into the term (5.28) (using the last equation in Example 5.28).

6.1.2 Assumption 5.2

Let $M \in \{A, G\}$, $S = \{p\}$.

A complex valued function ϕ on $\Pi(\overline{M(\mathbb{Q}_p)})$ extends to the complex Grothendieck group $R(\Pi(\overline{M(\mathbb{Q}_p)})) \otimes \mathbb{C}$ by linearity. Thus, if L is any Levi of M then ϕ is defined on the image of the induction map $I_L^M : \Pi(L(\mathbb{Q}_p)) \to R(\Pi(\overline{M(\mathbb{Q}_p)}))$. In particular, for $\sigma \in \Pi(\overline{M(\mathbb{Q}_p)})$ and $\zeta \in \oplus_{v \in S} i\mathfrak{a}_v^*$, $\phi(I_L^M(\sigma_\zeta))$ is defined. Let

$$D_\sigma \phi(\sigma^M) = \begin{cases} \frac{d}{d\zeta} \phi(I_L^M(\sigma_\zeta))|_{\zeta=0}, & \text{if it exists,} \\ \infty, & \text{otherwise.} \end{cases} \tag{6.10}$$

If $p < \infty$ is a prime and $\sigma \in \Pi(\overline{M(\mathbb{Q}_p)})$ then define $||\sigma|| = 0$. If $p = \infty$, $\sigma \in \Pi(\overline{M(\mathbb{Q}_p)})$, and Δ is a Casimer operator on $\overline{M(\mathbb{Q}_p)}$ (which is the same as the Casimer operator on $M(\mathbb{Q}_p)$) then define $||\sigma||$ to be the smallest (positive) eigenvalue of $\sigma(\Delta)$. For $\sigma \in \Pi(\overline{M(\mathbb{Q}_p)})$, define

$$||\sigma|| = \sup_{p \in S} ||\sigma_p||.$$

If $U_M = \mathcal{C}(\overline{M(\mathbb{Q}_p)})$, let V_M be defined as follows. Let

$$\begin{gathered} \mathcal{I}(\overline{M(\mathbb{Q}_p)})_{\overline{K}} = \{\phi : \Pi(\overline{M(\mathbb{Q}_p)}) \to \mathbb{C} \mid \\ (a)\ \phi(\pi) = 0 \ \text{ unless } \pi \text{ has a } \overline{K \cap M(\mathbb{Q}_p)} - \text{ fixed vector,} \\ (b) \text{ for each Levi } L \text{ of } M, \\ ||\phi||_{D,n} = \sup_{\sigma \in \Pi_t(\overline{L(\mathbb{Q}_p)})} (1 + ||\sigma||)^n |D_\sigma \phi(\sigma^M)| \\ \text{is finite}\}, \end{gathered}$$

topologized by the semi-norm topology. Let

$$V_M = \mathcal{I}(\overline{M(\mathbb{Q}_p)}) = \lim_K \mathcal{I}(\overline{M(\mathbb{Q}_p)})_{\overline{K}}, \tag{6.11}$$

topologized using the direct limit topology. See also [A16, section 3].

Then the assumptions are

Conjecture 6.7. *The map $\phi = \phi_M^M : U_M \to V_M$ given by $\phi(f) : \pi \longmapsto tr(\pi(f))$ is surjective.*

This is proven in [A16, appendix].

Conjecture 6.8. *The image of $\phi' : V'_M \to U'_M$ is the space of all tempered invariant distributions on $\overline{M(\mathbb{Q}_p)}$.*

When $M = G = SL(2)$ these conjectures are discussed in [J1], where they are shown to be a consequence of other conjectures (!).

In [A6], it was shown that these imply

Theorem 6.9. *Assume the two conjectures above. There is a map $\phi_M^G : U_G \to V_M$ satisfying*

(a) if $M = G$ then ϕ_M^G is as above.

(b) In general, we have

$$\phi_M^G(f^y) = \sum_{Q, M \subset M_Q} \phi_M^{M_Q}(f_{Q,y}), \qquad f \in U_G.$$

In particular, $\phi_G^G(f^y) = \phi_G^G(f)$.

(c) ϕ_M^G is given as in Theorem 6.5(c) above.

6.1.3 Normalized intertwining operators

Let P, P', P'' be parabolics of G defined over $\mathbb{Q}$ with Levi A. Let $\chi \in \Pi_u(\overline{A(\mathbb{Q}_p)})$ have representation space V_χ - a finite dimensional complex vector space with an $A(\mathbb{Q}_p)$-invariant inner product $(..., ...)_\chi$ - and let $\mathcal{K}_P(\chi)$ denote the space of all genuine V_χ-valued functions ϕ whose norms are square-integrable over K_p:

$$\int_{K_p} (\phi, \phi)_\chi dk < \infty.$$

Conjecture 6.10. *There are canonically normalized intertwining operators*

$$R_{P'|P}(\chi) : \mathcal{K}_P(\chi) \to \mathcal{K}_{P'}(\chi)$$

such that

(1) for all $x \in \overline{G(\mathbb{Q}_p)}$, we have

$$R_{P'|P}(\chi) I_P(\chi, x) = I_P(\chi, x) R_{P'|P}(\chi),$$

(2) $R_{P'|P}(\chi)$ *is unitary and*

$$R_{P''|P}(\chi) = R_{P''|P'}(\chi)R_{P'|P}(\chi),$$

(3) if $\psi \in \mathcal{K}_P(\chi)$ *is such that* $\psi(k) = \psi(1)$ *for all* $k \in \overline{K_p}$ *then*

$$R_{P'|P}(\chi)\psi = \psi,$$

(4) (for $p = \infty$*) if* $W \in \Pi(\overline{K_p})$ *and*

$$R_{P'|P}(\chi)_W = R_{P'|P}(\chi)|_{\mathcal{K}_P(\chi)_W}$$

then for all $D \in Diff(i\mathfrak{a}_P^*)$ *there is a* $C > 0,\ N > 0$ *such that*

$$||D_\chi R_{P'|P}(\chi)_W|| \leq C \cdot (1 + ||\chi||)^N \cdot (1 + ||W||)^N,$$

for all $\chi \in \Pi_u(\overline{A(\mathbb{Q}_p)})$*. Here* $||\chi||,\ ||W||$ *are as in [A6, section 5] (see also §6.1.2 above).*

If $p = \infty$, $G = SL(2)$ then these follow either from [Sa]. For $p < \infty$, they are discussed in [J1] (see [A16, section 2] for the case $n = 1$).

6.2 The terms in the invariant trace formula

Recall $\mathcal{C}$ denotes the set of equivalence classes in $G(\mathbb{Q})$ (two elements are equivalent if and only if their semi-simple Jordan components are $G(\mathbb{Q})$-conjugate). Let $\sigma \in G(\mathbb{Q})$ be semi-simple. The (G, S)**-equivalence class of** σ is the set

$$G(\mathbb{Q}) \cap (\sigma\mathcal{U})^{G(\mathbb{Q})} = \{\delta^{-1}\sigma u\delta \mid \delta \in G(\mathbb{Q}),\ u \in \mathcal{U} \cap G(\mathbb{Q})\},$$

where $\mathcal{U}$ is a unipotent conjugacy class in $\overline{G(\mathbb{A}_S)}_\sigma$.

Lemma 6.11. *Each* $\mathcal{O} \in \mathcal{C}$ *breaks up into a finite set of* (G, S)*-classes* $\mathcal{O}_{(G,S)}$.

More generally, we may define this finer notion of equivalence for any Levi M. Let $\gamma \in M(\mathbb{Q})$ have semi-simple Jordan component σ. Let $\gamma' \in M(\mathbb{Q})$. We say that γ' is (M, S)**-equivalent** to γ if there is a $\delta \in M(\mathbb{Q})$ such that
(1) σ is the semi-simple Jordan component of $\delta^{-1}\gamma'\delta$,
(2) $\sigma^{-1}\gamma$ and $\sigma^{-1}\delta^{-1}\gamma'\delta$ are $\overline{M(\mathbb{A}_S)}_\sigma$-conjugate.

Example 6.12. *Let $G = SL(2)$, $\gamma, \gamma' \in G(\mathbb{Q})$ and let S be non-empty.*

Let $M = G$ and $\gamma = 1$. γ is (G, S)-equivalent to γ' if and only if they are equal.

Let $M = A$ and $\gamma = 1$. γ is (A, S)-equivalent to γ' if and only if they are equal.

Let $M = G$ and let γ be regular semi-simple. γ is (G, S)-equivalent to γ' if and only if they are $G(\mathbb{Q})$-conjugate.

Let $M = A$ and $\gamma \in A(\mathbb{Q})$ be regular. γ is (A, S)-equivalent to γ' if and only if they are equal.

Lemma 6.13. *If $G = SL(2)$ and S is non-empty then γ' is (G, S)-equivalent to $\gamma \in G(\mathbb{Q})$ only if γ' is equivalent to γ.*

Example 6.14. *Let $G = SL(2)$ and let $\mathcal{O} \in \mathcal{C} - \mathcal{C}(G)$, so $\mathcal{O}$ is not an elliptic class. If a denotes the Jordan component of some $\gamma \in \mathcal{O}$ and a is regular then we have $\mathcal{O} \cap A(\mathbb{Q}) = \{a, a^w\}$.*

Let $G = SL(2)$. Assume that $\mathcal{O} \in \mathcal{C} - \mathcal{C}(G)$ contains a regular element. We next define a term $J'_{\mathcal{O}}(f)$ such that

$$I_{\mathcal{O}}(f) = J_{\mathcal{O}}(f) - J'_{\mathcal{O}}(f) \tag{6.12}$$

is an invariant distribution on $\mathcal{H}(\overline{G(\mathbb{A}_S)})$. To define $J'_{\mathcal{O}}(f)$, let $\chi \in \Pi_u(\overline{A(\mathbb{A}_S)})$, let $M(\chi) = M(w, \chi)$ denote the intertwining operator

$$M(w, \chi) : I_A(\chi) \to I_A(\chi^w),$$

and let $M(\chi, \eta)$ denote the operator $M(w, \eta) = M(w, I_A(\chi), \eta)$ in [A2, section 4, p. 113], where $\eta \in \mathfrak{a}^*_{\mathbb{C}}$, $P_1 = B$, $P_2 = B^w$. Let $m(\chi, \eta)$ denote the scalar factor such that

$$R(\chi, \eta) = m(\chi, \eta)^{-1} M(\chi, \eta) : \mathcal{K}_B(\chi) \to \mathcal{K}_B(\chi^w) \tag{6.13}$$

satisfies the normalized intertwining operator satisfying the conditions in §6.1.3 above. The relation between the logarithmic derivatives of the two is given by

$$M(\chi, \eta)^{-1} M'(\chi, \eta) = m(\chi, \eta)^{-1} m'(\chi, \eta) + R(\chi, \eta)^{-1} R'(\chi, \eta). \tag{6.14}$$

Let

$$\varphi_A(f, \chi) = \mathrm{tr}(R(\chi, 0)^{-1} R(\chi, 0)' I_A(\chi, f)), \tag{6.15}$$

where the derivative (the prime on R) is with respect to η and then evaluated at 0. This makes sense since the range of $R(\chi, \eta)$ is independent of η. Roughly speaking, the correction term $J'_{\mathcal{O}}(f)$ can be regarded as an average of the inverse Fourier transform of $\varphi_A(f)$ over $\mathcal{O} \cap A(\mathbb{Q})$ (by analogy with the case of $SL(2)$ or $GL(2)$ - see the introduction to [A6] and [LL]). This is essentially the way it works in the formalism here but more work will be required to set things up properly.

We now turn to [A6, section 4] and recall from §3.4 above the definition of the invariant distributions $I^A_{\mathcal{O}}$, I^A_{χ} on $\overline{A(\mathbb{A}_S)}$. These will be used to define $I^G_{\mathcal{O}}$, I^G_{χ} on $\overline{G(\mathbb{A}_S)}$ using the metaplectic analog of [A6, Proposition 4.1 and its proof]. In fact, we have

$$J'_{\mathcal{O}}(f) = |W_G|^{-1}(I^A_{\mathcal{O}})^{\wedge}(\varphi_A(f)), \tag{6.16}$$

where $(I^A_{\mathcal{O}})^{\wedge}$ denotes the distribution on the Schwartz space $\mathcal{S}(\Pi_u(A(\mathbb{A}_S)))$ such that

$$(\varphi^A_G)'((I^A_{\mathcal{O}})^{\wedge}) = I^G_{\mathcal{O}},$$

where

$$\varphi^A_G : \mathcal{H}(\overline{G(\mathbb{A}_S)}) \to I(\Pi_u(\overline{A(\mathbb{A}_S)}))$$

is the Fourier transform map.

Proposition 6.15. *([A6, Introduction and Proposition 4.1]) The distribution* (6.12) *on* $\mathcal{H}(\overline{G(\mathbb{A}_S)})$ *is invariant, where* $J'_{\mathcal{O}}(f)$ *is given by* (6.16).

Assume that $\chi \in \mathcal{X} - \mathcal{X}(G)$. Next we define a term $J'_{\chi}(f)$ such that

$$I_{\chi}(f) = J_{\chi}(f) - J'_{\chi}(f) \tag{6.17}$$

is an invariant distribution on $\mathcal{H}(\overline{G(\mathbb{A}_S)})$. As in (6.16), let

$$J'_{\chi}(f) = |W_G|^{-1}(I^A_{\chi})^{\wedge}(\varphi_A(f)), \tag{6.18}$$

where $(I^A_{\chi})^{\wedge}$ denotes the distribution on the space $I(\Pi_u(\overline{A(\mathbb{A}_S)}))$ such that

$$(\varphi^A_G)'((I^A_{\chi})^{\wedge}) = I^G_{\chi}.$$

Proposition 6.16. *([A6, Introduction and Proposition 4.1]) The distribution* (6.17) *on* $\mathcal{H}(\overline{G(\mathbb{A}_S)})$ *is invariant, where* $J'_{\chi}(f)$ *is given by* (6.18).

The invariant formula is the equality given by (6.4).

6.3 The terms $I_M(\gamma, f)$ and $I_M(\pi, f)$

The term $I_M(\gamma, f)$ is defined inductively by

$$I_M(\gamma, f) = J_M(\gamma, f) - \sum_{L, M \subset L L \neq G} \hat{I}_M^L(\gamma, \phi_L(f)), \tag{6.19}$$

where ϕ_L is as in §6.1. For a general explicit expression, see [A16].

There is an "invariant version" of Proposition 4.15, where J is replaced by I, (substitute (6.12), (6.16), and (6.19) into Proposition 4.15):

Let $(M(\mathbb{Q}) \cap \mathcal{O})_{M,S}$ denote the set of (M, S)-equivalence classes in $M(\mathbb{Q}) \cap \mathcal{O}$.

Proposition 6.17. *([A10, Theorem 3.3])*

$$I_{\mathcal{O}}(f) = \sum_{M \in \mathcal{L}(G)} |W_0^M| \cdot |W_0^G|^{-1} \sum_{\gamma \in (M(\mathbb{Q}) \cap \mathcal{O})_{M,S}} a^M(S, \gamma) I_M(\gamma, f),$$

where S is a sufficiently large set of valuations and the $a^M(S, \gamma)$ are as in §4.6.

In §6.1.3 we introduced normalized intertwining operators $R_{P'|P}(\pi)$, for $P', P \in \mathcal{P}(M)$. We will use these to make the distributions $I_M(\gamma)$ and $I_M(\pi)$ more explicit. For a fixed $P_0 \in \mathcal{P}(M)$, $\nu \in i\mathfrak{a}_M^*$, $\pi \in \Pi_u(\overline{M(\mathbb{A}_S)})$, $\lambda \in \mathfrak{a}_{M,\mathbb{C}}^*$, let

$$R_P(\mu, \pi_\lambda, P_0) = R_{P|P_0}(\pi_\lambda)^{-1} R_{P|P_0}(\pi_{\lambda+\mu}). \tag{6.20}$$

This more than likely (the metaplectic analog of) a (G, M)-family [A11, section 6]. In any case, the quantity

$$R_M(\pi_\lambda, P_0) = \lim_{\mu \to 0} \sum_{P \in \mathcal{P}(M)} R_P(\mu, \pi_\lambda, P_0) \theta_P(\mu)^{-1}$$

exists. Here $R_G(\pi_\lambda, P_0) = 1$ and

$$R_A(\pi_\lambda, P_0) = R_{\overline{P_0}|P_0}(\pi_\lambda)^{-1} \frac{d}{d\lambda} R_{\overline{P_0}|P_0}(\pi_\lambda)$$

by the computation in Example 5.28 (see also [A11] and [A6, (6.5)]).

Example 6.18. *If $G = SL(2)$ and $M = A$ then*

$$I_A(\gamma, f) = J_A(\gamma, f) - \hat{I}_A^A(\gamma, \phi_A(f)),$$

for $\gamma \in A(\mathbb{Q})$. The term $\hat{I}_A^A(\gamma, \phi_A(f))$ is the inverse Fourier transform of $\phi_A(f)(\chi) = tr(I_A(\chi) R_A(\chi, B))(f)$ evaluated at γ.

Let

$$J_M(\pi_\lambda, f) = \mathrm{tr}(R_M(\pi_\lambda, P_0) I_{P_0}(\pi_\lambda, f)), \tag{6.21}$$

where $f \in \mathcal{H}(\overline{G(\mathbb{A}_S)})$ and where $I_{P_0}(\pi)$ denotes the usual induced representation. Define

$$J_M(\pi, X, f) = \int_{i\mathfrak{a}^*_{M,S}} J_M(\pi_\lambda, f) e^{\lambda(X)} d\lambda, \tag{6.22}$$

where f is in the Hecke algebra as above, $X \in \mathfrak{a}_{M,S}$ is as in (6.6), and where

$$\begin{aligned} \mathfrak{a}^*_{M,S} &= \mathfrak{a}^*_M / \mathfrak{a}^\vee_{M,S} \\ \mathfrak{a}^\vee_{M,S} &= \mathrm{Hom}(\mathfrak{a}_{M,S}, 2\pi\mathbb{Z}). \end{aligned}$$

For $\pi \in \Pi_u(\overline{M(\mathbb{A}_S)})$, inductively define $I_M(\pi, X, f) = I^G_M(\pi, X, f)$ by

$$I_M(\pi, X, f) = J_M(\pi, X, f) - \sum_{L, M \subset L L \neq G} \hat{I}^L_M(\pi, X, \phi_L(f)), \tag{6.23}$$

where ϕ_L is as in §6.1 and [A9, section 3]. In general, define $I_A(\pi, f) = I_A(\pi, 0, f)$.

Example 6.19. *If $G = SL(2)$ and $M = A$ then*

$$I_A(\pi, X, f) = J_A(\pi, X, f) - \hat{I}^A_A(\pi, X, \phi_A(f)) = J_A(\pi, X, f) - (\phi_A(f))(\pi),$$

for $\pi \in \Pi_u(\overline{A(\mathbb{A}_S)})$.

We have

Proposition 6.20. *([A9, Lemma 3.1]) If $\pi \in \Pi_t(\overline{M(\mathbb{A}_S)})$ then*

$$I_M(\pi, X, f) = \begin{cases} f_G(\pi, X), & M = G, \\ 0, & M \neq G, \end{cases}$$

where

$$f_G(\pi, X) = \int_{i\mathfrak{a}^*_{G,S}} \mathrm{tr}(\pi_\lambda(f)) e^{\lambda(X)} d\lambda,$$

and $X \in \mathfrak{a}_{G,S}$.

(This vanishing result for I_M, $M \neq G$ is false in general if π is not tempered.

Example 6.21. *If $G = SL(2)$ and $\pi \in \Pi_t(\overline{M(\mathbb{A}_S)})$ then*

$$I_M(\pi, X, f) = \begin{cases} tr(\pi(f)), & M = G, \\ 0, & M = A. \end{cases}$$

We assume (as usual) that S contains the archimedian place. If π is an automorphic representation of $\overline{M(\mathbb{A}_S)}^1$, the infinitesimal character of the archimedian component of $I_M(\pi)$, is a W-orbit in the set of $\mathfrak{a}^*_{M,\mathbb{C}}$-orbits in $\mathfrak{h}^*_{\mathbb{C}}$. In this double orbit there is a unique W-orbit in $\mathfrak{h}^*_{\mathbb{C}}$ of smallest norm, which we denote by ν_π. If $\chi \in \mathcal{X}$ is the class of the orbit of (M, π), π an irreducible cuspidal representation of M then let $\nu_\chi = \nu_\pi$.

Theorem 6.22. *([A10, Corollary 4.5])*

$$I_\chi(f) = \sum_{M \in L(G)} |W_0^M| \cdot |W_0^G|^{-1} \int_{\Pi(M,\chi)} a^M(\pi) I_M(\pi, f) d\pi,$$

where

$$\Pi(M, \chi) = \{\pi \in \Pi(\overline{M(\mathbb{A}_S)^1}) \mid v_\pi = \nu_\chi\},$$

$a^M(\pi)$ satisfies [A10, (4.5)] and I_χ is defined by (6.17).

6.4 The invariant trace formula

The geometric side of the trace formula is the sum of the following terms:

(a) the sum over classes $\mathcal{O} \subset G(\mathbb{Q})$ such that $\mathcal{O} \cap Z(G(\mathbb{Q})) \neq \emptyset$ of

$$I_\mathcal{O}(f) = \sum_{\gamma \in \mathcal{O}_{(G,S)}} a^G(S, \gamma) I_G(\gamma, f) + \frac{1}{2} a^A(S, z) I_A(z, f), \tag{6.24}$$

where I_M and a^M are given in (6.19) and §4.6, resp., and the (G, S)-classes wre described in Example 6.12 above (these include the "nilpotent terms"),

(b) the sum over elliptic regular classes $\mathcal{O} \subset G(\mathbb{Q})$ of

$$I_\mathcal{O}(f) = a^G(S, \gamma) J_G(\gamma, f), \tag{6.25}$$

where $\gamma \in \mathcal{O}$ and J_G (an orbital integral) and a^G (a volume) are given in §4.6,

(c) the sum over "hyperbolic" regular classes $\mathcal{O} \subset G(\mathbb{Q})$ of

$$I_\mathcal{O}(f) = a^A(S, \gamma) I_A(\gamma, f), \tag{6.26}$$

where I_A and a^A (a volume) are given in above and in §4.6.

The spectral side of the trace formula is the sum of the following terms:

(a) the sum over classes $\chi \in \mathcal{X}(G)$ of

$$I_\chi(f) = \mathrm{tr}(\pi(f)), \quad (\pi, G) \in \chi, \tag{6.27}$$

(b) the sum over classes $\chi \in \mathcal{X} - \mathcal{X}(G)$ of

$$I_\chi(f) = \frac{1}{2} \int_{\Pi_u(A,\chi)} a^A(\pi) I_A(\pi, f) d\pi, \tag{6.28}$$

where I_A is given above and $\Pi_u(A, \chi)$ is as in Theorem 6.22 above. (Recall that $I_A(\pi, f) = 0$ for π *tempered*.)

7 The global trace formula in stable(r) form

Our primary references for this section are [LL], [L3].

7.1 Definitions and a general discussion of the results

We define $G_1 = GL(2)$.

Suppose that $\gamma, \gamma' \in G(\mathbb{Q})$. Following the well-known definition, if there is a $g \in G(\overline{\mathbb{Q}})$ such that $\gamma' = g^{-1}\gamma g$ then we call γ, γ' **stably conjugate**. Here $\overline{\mathbb{Q}}$ denotes an algebraic closure of $\mathbb{Q}$. As is well-known, two elements $\gamma, \gamma' \in G(\mathbb{Q})$ are conjugate under $G(\overline{\mathbb{Q}})$ if and only if they are conjugate under $G_1(\mathbb{Q})$. See [L3], [LL].

Here is reformulation of this definition: If there is a $g \in G_1(\mathbb{Q}) \hookrightarrow G_1(\mathbb{A}_S)$ such that $\gamma' = g^{-1}\gamma g$ then γ, γ' are **stably conjugate**. This condition has the following local analog which we make our local definition: Suppose that $\gamma, \gamma' \in \overline{G(F)}$. If there is a $g \in G_1(F)$ such that $\gamma' = g^{-1}\gamma g$ then γ, γ' are **stably conjugate**.

Remark 7.1. *We shall mention briefly that the paper of Springer-Steinberg contains a comment (actually an exercise) which may be worth mentioning in connection with these definitions: [SpSt, ch IV, (2.15)] says that if E is algebraically closed and $G \subset GL(n)$ is either a symplectic group or an orthogonal group then then $g_1, g_2 \in G(E)$ are conjugate in $G(E)$ if and only if they are conjugate in $GL(n, E)$. (I believe it is assumed throughout that E is not of characteristic 2.) In particular, if F is a p-adic field, $\overline{F}$ is*

its algebraic closure and if $g_1, g_2 \in G(F)$ are conjugate in $GL(n,F)$ then they are conjugate in $G(\overline{F})$. Recall the fact that if $g_1, g_2 \in GL(n,F)$ are conjugate in $GL(n,\overline{F})$ then they are conjugate in $GL(n,F)$ ("each conjugacy class in $GL(n)$ is stable", i.e., "$GL(n)$ has no L-indistinguishability"). These in conjunction imply that stable conjugacy in the symplectic or orthogonal groups may be "realized" as $GL(n,F)$-conjugacy.

The global invariant trace formula is an equality of the form (6.4). We will say that a distribution T on $\mathcal{H}(\overline{G(\mathbb{A}_S)})$ is **stably invariant** in case $T(f) = T(f^y)$ for all $f \in \mathcal{H}(\overline{G(\mathbb{A}_S)})$ and all $y \in G_1(\mathbb{A}_S)$. Not all the expressions in the invariant trace formula (6.4) are stably invariant. The object of this section is to write the invariant trace formula as the sum of stable terms which we write in the form (as in [L3])

$$\begin{gathered} \mathrm{tr} R_{cusp}(f) = \textstyle\sum_{\chi \in \mathcal{X}(G)} I'_\chi(f) \\ = -\textstyle\sum_{\chi \in \mathcal{X} - \mathcal{X}(G)} I_\chi(f) + S(f) + \sum_{\mathcal{O} \in \mathcal{C}} SI_{\mathcal{O}}(f) + \text{ "endoscopic terms"} \end{gathered} \tag{7.1}$$

where $S = S_1 + S_2 + S_3$ is defined in (7.22)–(7.24), I'_χ is equal to I_χ plus a term described in §7.8 (g) below ($\chi \in \mathcal{X} - \mathcal{X}(G)$), and the endoscopic terms are a sum of terms of the form

$$\begin{gathered} \text{"endoscopic terms"} = \tfrac{1}{4} \textstyle\sum_{H \neq G, \text{cuspidal}} \left[\sum_{\mathcal{O} \in \mathcal{C}^H} SI^H_{\mathcal{O}}(f^H) - \sum_{\theta \in E^G_H} < \theta, f^H > \right] \\ = \tfrac{1}{4} \textstyle\sum_{H \neq G, \text{cuspidal}} \sum_{\theta \in \Pi_u(H(\mathbb{Q}) \backslash H(\mathbb{A}_S)), \theta \neq 1} < \theta, f^H > \end{gathered} \tag{7.2}$$

which are stable distributions on endoscopic groups H of G (not $\overline{G(\mathbb{A}_S)}$) arising from the stable trace formula of H. Here $\mathcal{C}^H = \{\{\gamma\} \mid \gamma \in H(\mathbb{Q})\}$, E^G_H is the subset of $\Pi_u(H(\mathbb{Q}) \backslash H(\mathbb{A}_S))$, and f^H is a function on $H(\mathbb{A}_S)$ associated to $f \in \mathcal{H}(\overline{G(\mathbb{A}_S)})_{gen}$. In the version presented below we will not stabilize the I_χ in this paper. We shall explain the terms "H a cuspidal endoscopic group of G" in (7.2) in the next subsection.

7.2 Some basics on endoscopic groups

We shall define what an endoscopic group is in the simple case of $G = SL(2)$ and discuss the sets $D(T/F)$, $D(T/\mathbb{Q})$, and $D(T/\mathbb{A})$. The important but

more technical distinction between an endoscopic group and the sharper notion of endoscopic data ([L3], [She]) will be glossed over. Our main reference is [LL].

In the case of $SL(2)$, a group H is an endoscopic group if H is a Cartan subgroup of G. Moreover, up to "equivalence" these are the only ones. In general an endoscopic group H is not a subgroup of G nor are they in general abelian. A cuspidal endoscopic group of $SL(2)$ over F is an elliptic Cartan (i.e., not conjugate to $A(\mathbb{Q})$ by an element of $G(\mathbb{Q})$) of $SL(2)$.

If T is an elliptic Cartan of $SL(2)$ defined over a local or global field F then let

$$D(T/F) = T_1(F)\backslash G_1(F)/G(F), \tag{7.3}$$

where $G_1 = GL(2)$ and $T_1 = Cent(T, G_1)$. We identify without further mention this double coset space with the space $\overline{T_1(F)}\backslash\overline{G_1(F)}/\overline{G(F)}$, where $\overline{G_1(F)}$ denotes the n-fold cover defined using the cocyle β in §2.1. If T is an elliptic Cartan of $SL(2)$ defined over $\mathbb{Q}$ then let

$$D(T/\mathbb{A}) = \oplus_p T_1(\mathbb{Q}_p)\backslash G_1(\mathbb{Q}_p)/G(\mathbb{Q}_p). \tag{7.4}$$

We embed $D(T/\mathbb{Q})$ into $D(T/\mathbb{A})$ or $D(T/\mathbb{A}_S)$ diagonally.

For general definitions, see [L3]. The expressions $SI_{\mathcal{O}_H}$ and f^H are discussed there but will also be mentioned (indirectly) below.

7.3 Stabilizing the elliptic regular terms

We shall call an element $\gamma \in G(\mathbb{Q})$ (resp., $(\gamma, \zeta) \in \overline{G(F)}$) **semisimple** if γ is diagonalizable and **regular** if its eigenvalues γ_1, γ_2 are distinct. We shall call a semisimple element $\gamma \in G(\mathbb{Q})$ (resp., $(\gamma, \zeta) \in \overline{G(F)}$) **elliptic** (or G-**elliptic**) if γ belongs to an elliptic Cartan.

We first shall recall the idea of stabilizing the elliptic regular terms of (7.1) following [LL], [L3], [Ko1].

For $\gamma \in G(\mathbb{Q})$, let $\mathcal{O}_\gamma \in \mathcal{C}$ denote the equivalence class containing γ.

We now turn to the trace formula in the rank 0 case, as discussed in Section 3 above.

Proposition 7.2. *(Poisson summation) Let $T \subset G$ be an elliptic Cartan*

defined over $\mathbb{Q}$. *For any* $\phi \in C_c^\infty(\overline{T(\mathbb{A}_S)})_{gen}$, *we have*

$$\begin{aligned} &\mathrm{vol}(T(\mathbb{Q})\backslash\overline{T(\mathbb{A}_S)}) \sum_{\gamma\in T(\mathbb{Q})} \phi(\gamma) \\ = &\sum_{\theta\in\Pi_u(T(\mathbb{Q})\backslash\overline{T(\mathbb{A}_S)})} < \theta, \phi > . \end{aligned}$$

If $\theta = \otimes_p \theta_p$ *and* $\phi = \otimes_p \phi_p$ *then*

$$< \theta, \phi >= \prod_p \mathrm{tr}\theta_p(\phi_p).$$

Remark 7.3. *(***A preview of coming attractions***) We shall eventually apply this with* $\phi(\gamma)$ *equal to*

$$I_\gamma^\kappa(f) = [D(T/\mathbb{Q}) : D(T/\mathbb{A}_S)]^{-1} \sum_{\delta\in D(T/\mathbb{Q})\backslash D(T/\mathbb{A}_S)} \kappa(\det\delta) I_{\mathcal{O}}(f^\delta), \qquad (7.5)$$

where $\kappa \in \Pi_u(D(T/\mathbb{Q})\backslash D(T/\mathbb{A}_S))$, $\gamma \in T(\mathbb{Q})$ *elliptic, and* $f \in \mathcal{H}(\overline{G(\mathbb{A}_S)})$. *These terms* (7.5) *do not occur in the invariant trace formula. However, after grouping terms as in* (7.6) *below and substituting the expression in Lemma 7.5 below, all the terms* (7.5), $\gamma \in T(\mathbb{Q})$, *will occur in the trace formula except for* $\gamma = \pm 1$. *The terms corresponding to* $\gamma = \pm 1$ *will be discussed in §7.5 below. The terms* (7.5) *with* $\kappa = 1$, $\gamma \neq \pm 1$, *will comprise part of the sum over* $\mathcal{O} \in \mathcal{C}$ *in* (7.1). *The terms* (7.5) *with* $\kappa \neq \pm 1$, *will comprise part of the sum in* (7.2).

Next we record a corollary of the above proposition which we will substitute into the trace formula later.

Corollary 7.4. *For each elliptic torus* T, *we have*

$$vol[T(\mathbb{Q}) \setminus \overline{T(\mathbb{A}_S)}] \cdot \sum_{\gamma\in T(\mathbb{Q})} I_\gamma^\kappa(f) = \sum_\theta < \theta, I_*^\kappa(f) >$$

where I_*^κ *is the unstable orbital integral defined in* (7.5) *above (for its properties, as a function on* T, *in particular on the singular set of* T, *see [LL] or §7.5 below).*

This expression will form part of what is meant by (7.2).

As we will see in Lemma 7.21 below, we may use [LL] to to relate $< \theta, I^\kappa_*(f) >$ to a linear combination of characters of automorphic representations of $\Pi_u(\overline{G(\mathbb{A}_S)})$. If $\mathcal{O} = \mathcal{O}_\gamma$ and $\gamma \in \overline{T(\mathbb{A}_S)}$ is elliptic regular, let

$$\begin{array}{c} I^{st}_{\mathcal{O}}(f) = \sum_{\delta \in D(T/\mathbb{Q})} I_{\mathcal{O}}(f^\delta), \\ SI_{\mathcal{O}}(f) = I^1_\gamma(f)\ . \end{array} \tag{7.6}$$

These distributions $SI_{\mathcal{O}}$ are stable.

Lemma 7.5. *(a) If $\mathcal{O} = \mathcal{O}_\gamma$ and $\gamma \in \overline{T(\mathbb{A}_S)}$ is elliptic regular, we have*

$$SI_{\mathcal{O}}(f) = \sum_{\kappa \in \Pi_u(D(T/\mathbb{Q}) \backslash D(T/\mathbb{A}))} I^\kappa_\gamma(f)$$

(b) If $\mathcal{O} = \mathcal{O}_\gamma$ and $\gamma \in \overline{T(\mathbb{A}_S)}$ is elliptic regular, we have

$$I^\kappa_{h^{-1}\gamma h}(f) = \kappa(\det h) I^\kappa_\gamma(f) = I^\kappa_\gamma(f^h), \qquad h \in \overline{G_1(\mathbb{A}_S)}.$$

Any function g satisfying $g(h^{-1}\gamma h) = \kappa(\det(h))g(\gamma)$, for all $h \in \overline{G_1(\mathbb{A}_S)}$, will be called **$\kappa$-stable**.

proof: It is elementary that if A and B are discrete countable abelian groups, with $A \subset B$ of finite index, and if f is any compactly supported function on B then

$$\sum_{a \in A} f(a) = [B : A]^{-1} \sum_{\kappa \in (B/A)^\wedge} \sum_{b \in B} \kappa(b) f(b), \tag{7.7}$$

whenever the sums converge absolutely. We apply this to our sum over δ:

$$\sum_{\delta \in D(T/\mathbb{Q})} I_{\mathcal{O}}(f^\delta) = \sum_{\kappa \in (D(T/\mathbb{A})/D(T/\mathbb{Q}))^\wedge} [D(T/\mathbb{A}) : D(T/\mathbb{Q})]^{-1} \sum_{\delta \in D(T/\mathbb{A})} \kappa(\delta) I_{\mathcal{O}}(f^\delta),$$

which we may equate with

$$\sum_{\kappa \in \Pi_u(D(T/\mathbb{A})/D(T/\mathbb{Q}))} I^\kappa_\gamma(f). \tag{7.8}$$

□

7.4 Stabilizing the A-elliptic regular terms

We shall call a semisimple element $\gamma \in G(\mathbb{Q})$ (resp., $\gamma \in \overline{G(F)}$) A-**elliptic** regular if γ is conjugate to an element $a \in A(F)$ which is elliptic regular in $A(F)$ (i.e., $a \neq \pm 1$).

If $\mathcal{O} = \mathcal{O}_\gamma$ and $\gamma \in G(\mathbb{Q})$ is A-elliptic regular, let

$$\begin{array}{c} I^{st}_{\mathcal{O}}(f) = \sum_{\delta \in D(A/\mathbb{Q})} I_{\mathcal{O}}(f^\delta), \\ I^{\kappa}_{\gamma}(f) = [D(A/\mathbb{Q}) : D(A/\mathbb{A}_S)]^{-1} \sum_{\delta \in D(A/\mathbb{A}_S)} \kappa(\delta) I_{\mathcal{O}}(f^\delta), \end{array} \tag{7.9}$$

where

$$\begin{array}{c} D(A/\mathbb{Q}) = A(\mathbb{Q})Z(G_1(\mathbb{Q}))\backslash G_1(\mathbb{Q})/G(\mathbb{Q}) \\ D(A/\mathbb{A}_S) = A(\mathbb{A}_S)Z(\overline{G_1(\mathbb{A}_S)})\backslash \overline{G_1(\mathbb{A}_S)}/G(\mathbb{A}_S) \\ \kappa \in \Pi_u(D(A/\mathbb{Q})\backslash D(A/\mathbb{A}_S)). \end{array}$$

Let

$$SI_{\mathcal{O}}(f) = I^1_\gamma(f),$$

From (7.7), we have

$$I^{st}_{\mathcal{O}}(f) = \sum_{\kappa \in \Pi_u(D(A/\mathbb{Q})\backslash D(A/\mathbb{A}_S))} I^{\kappa}_{\gamma}(f) \tag{7.10}$$

Lemma 7.6. *If $\mathcal{O} = \mathcal{O}_\gamma$ and $\gamma \in G(\mathbb{Q})$ is A-elliptic regular, then $SI_{\mathcal{O}}(f)$ is stable.*

Recall that this term $SI_{\mathcal{O}}$ is defined using

$$I_{\mathcal{O}}(f) = \sum_{\gamma \in \mathcal{O}} a^G(S,\gamma) I_G(\gamma, f) + \frac{1}{2} \sum_{\gamma \in \mathcal{O} \cap A(\mathbb{Q})} a^A(S,\gamma) I_A(\gamma, f). \tag{7.11}$$

The $I_M(\gamma, f)$ were defined by

$$I_G(\gamma, f) = J_G(\gamma, f),$$

which is an "ordinary" orbital integral over $\overline{G(\mathbb{A}_S)}_\gamma \backslash \overline{G(\mathbb{A}_S)}$, and

$$I_A(\gamma, f) = J_A(\gamma, f) - \hat{I}^A_A(\gamma, \phi_A(f)).$$

(It turns out that $\overline{G(\mathbb{A}_S)}_\gamma \subset \overline{A(\mathbb{A}_S)}$, with equality if γ is an N-th power, and $J_G(\gamma, f) = 0$ if γ is not an N-th power - provided f is genuine as we have

assumed.) To see that the sum $\sum_{\gamma\in\mathcal{O}} a^G(S,\gamma)I_G(\gamma,f)$ is stably invariant, we claim that $I_G(\gamma,f)$ is stably invariant. This is proven in [J3].

The term $J_A(\gamma,f)$ is not invariant. We examine it more closely in Lemma 7.9 and (7.14) below. (See also [A16, Corollary 4.2] for a general formula for it in the case $n=1$.)

The term $\hat{I}_A^A(\gamma,\phi_A(f))$ is not invariant. It is the Fourier transform of the map

$$\chi \to \mathrm{tr}(I_A(\chi,f)R_A(\chi,B)),$$

see (7.18). The instability of $f \longmapsto \phi_A(f)$ is examined in Lemma 7.23 below.

Let $F=\mathbb{Q}_p$, $p<\infty$, $K=G(\mathbb{Z}_p)$ which we identify with a *subset* $(K,1)$ of $\overline{G(F)}$, $\sigma\in\Pi(\overline{A(F)})$. Let $K_{I_A(\sigma)}(k,k')$ denote the kernel function of the induced representation $I_A(\sigma)$ on the space of functions on K, so

$$\mathrm{tr}(I_A(\sigma,f)) = \int_K K_{I_A(\sigma)}(k,k)dk. \tag{7.12}$$

Lemma 7.7. *If* $F(\sigma)=\mathrm{tr}(I_A(\sigma,f))$ *then*

$$F^\vee(\gamma) = |a_1-a_2|J_G(\gamma,f),$$

for $\gamma=(a,\zeta)\in\overline{A(F)}$, $a\neq\pm1$.

proof: This will be proven in [J3].

As a corollary, it follows from this lemma without to much work that $f\longmapsto I_G(\gamma,f)$ is stable, for $\gamma=(a,\zeta)\in\overline{A(F)}$, $a\neq\pm1$.

Let $\delta(g)=|a_1/a_2|$, if $g=(nak,\zeta)\in\overline{G_1(F)}$. Let $N(g)$ denote the operator on the space of functions on K given by

$$f(k)\longmapsto f(k)\log\delta(kg).$$

This operator also occurs in connection with the terms in the invariant trace formula associated to the normalized intertwining operators.

Lemma 7.8. *For* $g\in G_1(F)$, *we have*

$$\mathrm{tr}(I_A(\sigma,f)N(g)) = \int_K K_{I_A(\sigma)}(k,k)\log\delta(kg)dk.$$

Lemma 7.9. *For* $g\in G_1(F)$, *if* $F(\sigma)=\mathrm{tr}(I_A(\sigma,f)N(g))$ *then*

$$J_A(a,f^{g^{-1}}) - J_A(a,f) = F^\vee(a)+F^\vee(a^w)$$

proof: This will be proven in [J3].
If $v > 2$ then we may represent $D(A/F)$ by

$$\{\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} \mid a = 1, \epsilon, \pi, \epsilon\pi\},$$

where π is a uniformizer and ϵ is a unit which is not a quadratic residue. If $v = 2$ the situation is similar... Therefore, if κ is non-trivial we may apply Lemma 7.9 above to compute

$$\sum_{\delta \in D(A/F)} \kappa(\delta) J_A(\gamma, f^\delta).$$

If $x = a \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} k$, $n \in \mathbb{A}_S$, then

$$v_A(x) = v(x, 0) = \sum_v v(0, x_v) = \sum_v \log \sqrt{|1 + n_v^2|_v} \tag{7.13}$$

and

$$\begin{aligned} J_A(\gamma, f) &= |D(\gamma)|^{1/2} \int_{\overline{G(\mathbb{A}_S)}_\gamma \backslash \overline{G(\mathbb{A}_S)}} f(x^{-1}\gamma x) v_A(x) dx \\ &= \sum_w \prod_v |D(\gamma_v)|_v^{1/2} \int_{\overline{G(\mathbb{Q}_p)}_\gamma \backslash \overline{G(\mathbb{Q}_p)}} f_v(x_v^{-1}\gamma_v x_v) v_A(x_w) dx_v \\ &= \sum_v |D(\gamma_v)|_v^{1/2} J_A(\gamma_v, f_v) \prod_{w \neq v} |D(\gamma_w)|_w^{1/2} J_G(\gamma_w, f_w). \end{aligned} \tag{7.14}$$

Remark 7.10. *This comprises metaplectic analog of the $a \neq \pm 1$ part of the distribution denoted S_2 in [LL]. It will be combined with other distribution described in the next section to yield a stably invariant distribution.*

7.5 Stabilizing the singular terms

A term $I_{\mathcal{O}}(f)$ is called **singular** if $\mathcal{O}$ does not contain any M-elliptic elements, for any $M \in \{A, G\}$. An element $\gamma \in G(\mathbb{Q})$ is called **singular** if its eigenvalues γ_1, γ_2 are equal (i.e., γ is not regular).

In this subsection, our goal is to show that part of the singular terms of the geometric side of the invariant trace formula (6.4) "combine" with the sum of the terms $I_\gamma^\kappa(f)$, with $\kappa \neq 1$ and with $\gamma \in T(\mathbb{Q})$ elliptic *singular* (i.e., $\gamma = \pm 1$) to contribute to (7.2). If $\mathcal{O} = \{z\}$, $z = \pm 1$, then define

$$SI_{\mathcal{O}}(f) = c \cdot f(z), \tag{7.15}$$

where $c = vol(G(\mathbb{Q})\backslash\overline{G(\mathbb{A}_S)})$ is a volume.

Lemma 7.11. *If $z \in \mathcal{O}$, with $z = \pm 1$, then*

$$I_{\mathcal{O}}(f) - SI_{\mathcal{O}}(f) = \sum_{\gamma \in \mathcal{O}, \gamma \neq z} a^G(S,\gamma) I_G(\gamma, f) + \frac{1}{2} a^A(S,z) I_A(z,f).$$

proof: This equation follows from Proposition 6.17 (also Proposition 4.15, Example 4.16). $\square$

Definition 7.12. *Let κ be a character of $D(T/F)$ of order 2. Define $\Phi^{T/\kappa}$ on the regular set of $\gamma \in T(F)$ by*

$$\begin{aligned} \Phi^{T/\kappa}(\gamma, f) &= \Delta(\gamma, T, \kappa) \textstyle\int_{\overline{G_1(F)}_\gamma \backslash \overline{G_1(F)}} \kappa(\det g) f(g^{-1}\gamma g) dg \\ &= \lambda(E/F, \psi) \kappa(\tfrac{\gamma_{1,v} - \gamma_{2,v}}{\gamma^0_{1,v} - \gamma^0_{2,v}}) \textstyle\int_{\mathcal{O}} \epsilon(x) f(x) |\omega_\gamma|, \end{aligned}$$

where $\epsilon(g^{-1}\gamma g) = \kappa(\det(g))$, $\gamma^0 \in \overline{T(F)}$ is fixed, and $\mathcal{O}$ denotes the conjugacy class of γ in $\overline{G(F)}$.

We leave the relation between the global expressions $\Phi^{T/\kappa}(\gamma, f)$ and $I^\kappa_\gamma(f)$ to the reader (use the Hilbert product formula).

Let $\mathcal{O}(\gamma)$ denote the orbit of γ in $\overline{G(F)}$ and, for $a \in F^\times$, which we identify with the center of $G_1(F)$, let

$$n(a) = a \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Definition 7.13. *If $z \in Z(\overline{G(F)})$, then we define*

$$\Phi^{T/\kappa}(z, f) = \begin{cases} \textstyle\int_{\mathcal{O}(n(z))} f(x) dx, & T = A,\ \kappa = 1 \\ \lambda(E/F, \psi) \kappa(\frac{\pi_E - \overline{\pi}_E}{\gamma^0_{1,v} - \gamma^0_{2,v}}) \textstyle\int_{\mathcal{O}(n(z))} \epsilon(x) f(x) dx, & T_1(F) = E^\times, \kappa \in \Pi_u(F^\times), \end{cases}$$

where π_E is a uniformizer for E/F, $T_1(F) = E^{\times 1}$, and

$$\epsilon(g^{-1} n(z) g) = \begin{cases} \kappa(-b), & b \neq 0, \\ \kappa(c), & b = 0, \end{cases}$$

if

$$g^{-1} n(z) g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Lemma 7.14. *If $\kappa \in \Pi_u(F^\times)$ is the character of order 2 associated to E/F be local class field theory ($\kappa = 1$ if $E = F \times F$) and $T_1(F) = E^\times$, then $\gamma \longmapsto \Phi^{T/\kappa}(\gamma, f)$ is a smooth function on $T(F)$.*

proof: This follows from the results on Shalika germs. It is proven in [J3].

Lemma 7.15. *For $z \in Z(\overline{G(F)})$, we have*

$$\lim_{\gamma \to z} \int_{\overline{G_1(F)}_\gamma \backslash \overline{G_1(F)}} \kappa(\det g) f(g^{-1}\gamma g) dg$$
$$c \frac{1}{L(1,\kappa)} \int_{\overline{G_1(F)}_{n(1)} \backslash \overline{G_1(F)}} \kappa(\det g) f(g^{-1} n(z) g) dg$$

where $c = |\gamma_{1,v} - \gamma_{2,v}|$.

proof: This will be proven in [J3].

Lemma 7.16. *Let κ be a character of $\mathbb{A}_S^\times$ of order 2 which we extend to $\overline{G_1(\mathbb{A}_S)}$ via the* det *map, and T be a Cartan of G associated to κ by class field theory then*

$$\sum_{\delta \in D(A/\mathbb{A}_S)} \kappa(\delta) I_G(n(z), f^\delta) = c \lim_{\gamma \to z, \gamma \in T(\mathbb{Q})} \sum_{\delta \in D(T/\mathbb{A}_S)} \kappa(\delta) I_G(\gamma, f^\delta), \qquad (7.16)$$

for $z = \pm 1$, $\kappa \in \Pi_u(G_1(\mathbb{A}_S))$ of order 2, where the constant $c = c(T, \kappa, S)$ is given by

$$c = \prod_v |\gamma_{1,v}^0 - \gamma_{2,v}^0|_v^{-1} L(1, \kappa_v).$$

Remark 7.17. *The terms on the right hand side of the above lemma go into (7.20) below. To see this, just relate them to (7.5) by using (7.11) and the fact that $\mathcal{O} \cap A(\mathbb{Q}) = \emptyset$.*

The terms on the left hand side will examined following the proof below.

proof: This is is a consequence of the previous lemma and the definitions. □

If $z \in \mathcal{O}$ is an equivalence class of $G(\mathbb{Q})$ as above then part of $I_{\mathcal{O}}(f)$ may be written as

$$\sum_{\gamma \in \mathcal{O}} J_G(\gamma, f) = J_G(z, f) + \sum_{a \in \mathbb{Q}^\times / \mathbb{Q}^{\times 2}} \int_{\overline{G(\mathbb{A}_S)}_{n(1)} \backslash \overline{G(\mathbb{A}_S)}} f(g^{-1} z \cdot \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} g) dg$$
$$= J_G(z, f) + \sum_{\kappa \in \Pi_u(\mathbb{A}_S^\times / \mathbb{A}_S^{\times 2})} \sum_{a \in \mathbb{A}_S^\times / \mathbb{A}_S^{\times 2}} \kappa(a) \int_{\overline{G(\mathbb{A}_S)}_{n(1)} \backslash \overline{G(\mathbb{A}_S)}} f(g^{-1} z \cdot \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} g) dg$$
$$= J_G(z, f) + \sum_{\kappa \in \Pi_u(\mathbb{A}_S^\times / \mathbb{A}_S^{\times 2})} \int_{\overline{G_1(\mathbb{A}_S)}_{n(1)} \backslash \overline{G_1(\mathbb{A}_S)}} \kappa(\det g) f(g^{-1} n(z) g) dg, \qquad (7.17)$$

which we write as $J_G(z,f)$ plus the sum over $\kappa \in \Pi_u(\mathbb{A}_S^\times/\mathbb{A}_S^{\times 2})$ of the left hand side of the equation in Lemma 7.16 above.

Regarding the expression

$$\frac{a^A(S,z)}{2}\sum_\delta \kappa(\delta)(J_A(z,f^\delta)-\hat{I}_A^A(z,\phi_A(f^\delta)))),$$

we must also compute $a^A(S,z)$ and discuss the terms $\sum_\delta \kappa(\delta)J_A(z,\phi_A(f^\delta))$ and $\sum_\delta \kappa(\delta)\hat{I}_A^A(z,\phi_A(f^\delta))$.

Computation of $a^A(S,z)$: By Proposition 4.15,

$$a^A(S,z) = vol(A(\mathbb{Q})\backslash\overline{A(\mathbb{A}_S)^1}).$$

Computation of $\sum_\delta \kappa(\delta)J_A(z,\phi_A(f^\delta))$: See Example 4.13 above and (7.14). These terms comprise the $a=\pm 1$ part of the distribution denoted by $S_2(f)$ "missing" from (7.14).

Computation of $\sum_\delta \kappa(\delta)\hat{I}_A^A(z,\phi_A(f^\delta))$: First, we note that the map $\phi_A : \mathcal{H}(\overline{G(\mathbb{A}_S)}) \to I(A(\mathbb{A}_S))$ of §6.1.1 is a "weighted character" of an induced representation:

$$\begin{aligned}\phi_A(\sigma,f) &= \mathrm{tr}(R_{\overline{B}|B}(\sigma_\lambda)^{-1}R'_{\overline{B}|B}(\sigma_\lambda)I_A(\sigma_\lambda,f))|_{\lambda=0}\\ &= \textstyle\sum_v \mathrm{tr}(R_{\overline{B}|B}(\sigma_v)^{-1}R'_{\overline{B}|B}(\sigma_v)I_A(\sigma_v,f_v))\prod_{w\neq v}\mathrm{tr}(I_A(\sigma_w,f_w)).\end{aligned} \tag{7.18}$$

for $\sigma \in \Pi_u(\overline{A(\mathbb{A}_S)})$. The inverse Fourier transform of the function $\sigma \longmapsto \phi_A(\sigma,f)$ evaluated at $a=1$ is essentially the distribution denoted by $S_1(f)$ in [LL].

7.6 The geometric side of the stable trace formula

The stable elliptic regular terms are

$$\sum_{\mathcal{O}\in\mathcal{C}_G,\mathcal{O}\text{ regular elliptic}} SI_{\mathcal{O}}(f), \tag{7.19}$$

where the SI's are defined by (7.6), (7.9),

Definition 7.18. *If π is an irreducible admissible genuine representation of $\overline{G(\mathbb{Q}_p)}$ then let $L(\pi)$ denote the set of inequivalent representations π^g where $g \in \overline{G_1(\mathbb{Q}_p)}$.*

Remark 7.19. *For π as above and $n > 1$ then*

$$\#L(\pi) = \prod_{v|p} \#(k_v^\times / k_v^{\times 2}).$$

Lemma 7.20. *If θ_1 is an irreducible genuine representation of $T_1(\mathbb{Q}_p)$ and $\pi_1(\theta_1)$ is the irreducible admissible representation of $G_1(\mathbb{Q}_p)$ associated to it in [J2] (at least if n is odd), let $\theta = \theta_1|_{T(\mathbb{Q}_p)}$ and let $\pi(\theta)$ denote one of the irreducible constituents of $\pi(\theta) = \pi_1(\theta_1)|_{G(\mathbb{Q}_p)}$.*

This is proven in [J3].

The following lemma will not be used here. Note that it related the distribution

$$\begin{array}{c} C_c^\infty(\overline{G(\mathbb{A}_S)})_{gen} \to C_c^\infty(\overline{T(\mathbb{A}_S)})_{gen} \\ f \longmapsto I_*^\kappa(f) \end{array}$$

to a map

$$\begin{array}{c} C_c^\infty(T(\mathbb{A}_S))' \to C_c^\infty(\overline{G(\mathbb{A}_S)})'_{gen} \\ \theta \longmapsto \prod_{p\in S} \sum_{\pi_p \in \Pi(\theta_p)} \kappa_*(\pi_p)\chi_{\pi_p}(f_p), \end{array}$$

after relating $C_c^\infty(\overline{T(\mathbb{A}_S)})_{gen}$ to $C_c^\infty(T(\mathbb{A}_S))$.

Lemma 7.21. *For each elliptic torus T and each factorizable $f \in \mathcal{H}(\overline{G(\mathbb{A}_S)})_{gen}$, we have*

$$\begin{array}{c} < \theta, I_*^\kappa(f) > \\ = [D(T/\mathbb{Q}) : D(T/\mathbb{A}_S)]^{-1} \prod_{p\in S} \sum_{\delta \in D(T/\mathbb{Q}_p)} \sum_{\pi_p \in \Pi(\theta_p^\delta)} \kappa_*(\pi_p)\chi_{\pi_p}(f_p), \end{array}$$

where $\kappa_*(\pi_p) := \kappa(\delta)$ if $\pi_p \in \Pi(\theta_p^\delta)$, θ_p^h denotes the conjugate of θ_p by h, θ_p is a unitary genuine irreducible representation of $\overline{T(\mathbb{A}_S)}$, and $f_p(g) = f_p(g, 1)$ (with apologies for the abuse of notation).

This is proven in [J3].

Remark 7.22. *This result is essentially the metaplectic analog of [LL, (2.4-5)], [LL, p. 762]. This is, after a suitable choice of global measures, related to a special case of [J2, (3.42)] which was only proven under in the case n odd.*

By this lemma and the trace formula for rank 0 (in Proposition 7.2) the terms

$$\sum_{T\ ,G-\text{elliptic}} \text{vol}(T(\mathbb{Q})\backslash\overline{T(\mathbb{A}_S)}) \sum_{\gamma\in T(\mathbb{Q})} \sum_{\kappa\in\Pi_u(D(T/\mathbb{Q})\backslash D(T/\mathbb{A}_S)),\kappa\neq 1} I_\gamma^\kappa(f) \qquad (7.20)$$

may be expressed as a linear combination of characters of representations

$$\sum_T \sum_\theta c_\theta tr(\pi(\theta)(f)), \tag{7.21}$$

for some constants c_θ, where θ runs over $\Pi_u(T(\mathbb{Q})\backslash\overline{T(\mathbb{A}_S)})$. Note that we may rewrite the terms in (7.20) as

$$\sum_{H\ cuspidal} \sum_{\mathcal{O}_H \neq \pm 1, a\ conj, class\ in\ H} SI^H_{\mathcal{O}_H}(f^H)$$

in the notation of (7.2).

In (7.20) the terms $\gamma = \pm 1$ occuring in the middle sum on the right come from the singular elliptic terms in Lemma 7.16 above.

7.7 Stabilizing the remaining terms

Let $F = \mathbb{Q}_p$.

Lemma 7.23. *For all $g \in \overline{G_1(F)}$,*

$$\begin{aligned}\mathrm{tr}(R_{\overline{B}|B}(\sigma)^{-1}R'_{\overline{B}|B}(\sigma)I_A(\sigma, f^{g^{-1}})) - \mathrm{tr}(R_{\overline{B}|B}(\sigma)^{-1}R'_{\overline{B}|B}(\sigma)I_A(\sigma, f)) \\ = \mathrm{tr}(I_A(\sigma, f)N(g)) + \mathrm{tr}(I_A(\sigma^w, f)N(g))\end{aligned}$$

proof: Proven in [J3].

From Example 5.28, (5.28), and Theorem 6.22, let

$$S_1(f) = \sum_{\chi \in \mathcal{X} - \mathcal{X}(G)} \sum_v \frac{1}{2} \int_{\Pi(A,\chi)} \mathrm{tr}(R_{\overline{B}|B}(\sigma_v)^{-1}R'_{\overline{B}|B}(\sigma_v)I_A(\sigma_v, f_v)) \prod_{w \neq v} \mathrm{tr}(I_A(\sigma_w, f_w))|d\pi|. \tag{7.22}$$

Let

$$S_2(f) = -\frac{1}{2} \sum_v \sum_{\{a, a^w\} \in A(\mathbb{Q})} \frac{|D(a)|_v^{1/2} J_A(a, f_v)}{L(1, \mathbb{Q}_v)} \prod_{w \neq v} \frac{|D(a)|_w^{1/2} J_G(a, f_w)}{L(1, \mathbb{Q}_w)}. \tag{7.23}$$

Let

$$\begin{aligned} S_3(f) = \\ = Res_{s=0} \tfrac{1}{s} \textstyle\sum_{z \in Z(G(\mathbb{Q}))} \sum_{x \in \mathbb{Q}^\times / \mathbb{Q}^{\times 2} x \neq 1} vol(Z(G(\mathbb{Q}))\backslash Z(\overline{G(\mathbb{A}_S)})) \times \\ \times \textstyle\int_{\overline{G(\mathbb{A}_S)}_{n(1)}\backslash \overline{G(\mathbb{A}_S)}} f(g^{-1} z \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} g) \lambda(g)^{-s} dg, \end{aligned} \tag{7.24}$$

where $\lambda(g) = \delta(g)\delta(wg)$.

The following will not be needed. It is the analog of a result in [LL], the proof of which used the analogs of Lemmas 7.9 and 7.23 above.

Conjecture 7.24. *Let $S = S_1 + S_2 + S_3$. Then S is stable $S(f^g) = S(f)$ for all $g \in \overline{G_1(\mathbb{A}_S)}$.*

Perhaps the proof of this (assuming it i correct) is similar to the proof of its analog in [LL].

7.8 The stable trace formula

This is the equality

$$\sum_{\mathcal{O}\in\mathcal{C}} SI_{\mathcal{O}}(f) = \sum_{\chi\in\mathcal{X}} I_{\chi}(f) - \sum_{H\ cuspidal} \sum_{\mathcal{O}_H} SI^H_{\mathcal{O}_H}(f^H), \tag{7.25}$$

where the $SI_{\mathcal{O}}$ are as in sections 7.2-7.3.

The geometric side of the stable(r) trace formula is the sum of the following terms:

(a) the sum over (G,S)-classes $\mathcal{O} \subset G(\mathbb{Q})$ such that $\mathcal{O} \cap Z(G(\mathbb{Q})) \neq \emptyset$ of

$$SI_{\mathcal{O}}(f) = a^G(S,z)f(z), \tag{7.26}$$

where a^G is (a volume) is given in §4.6 (these do not include the "nilpotent terms", which got combined with some other terms ...),

(b) the sum over elliptic regular classes $\mathcal{O} \subset G(\mathbb{Q})$ of

$$SI_{\mathcal{O}}(f) = \sum_{\delta\in D(T/\mathbb{A}_S)} a^G(S,\gamma)J_G(\gamma, f^\delta), \tag{7.27}$$

where $\gamma \in \mathcal{O}$, $T = G_\gamma$, and J_G (an orbital integral) and a^G (a volume) are given in §4.6,

(c) the sum over A-elliptic regular classes $\mathcal{O} \subset G(\mathbb{Q})$ of

$$SI_{\mathcal{O}}(f) = \sum_{\delta\in D(A/\mathbb{A}_S)} I_{\mathcal{O}}(f^\delta), \tag{7.28}$$

(d) plus terms associated to the geometric side of the trace formula for each cuspidal endoscopic of G,

(e) the "geometric part" of the endoscopic terms (7.2).

The spectral side of the stable(r) metaplectic trace formula is the sum of the following terms:

(f) the sum over classes $\chi \in \mathcal{X}(G)$ of

$$I_\chi(f) = \mathrm{tr}(\pi(f)), \quad (\pi, G) \in \chi. \tag{7.29}$$

(g) The sum over classes $\chi \in \mathcal{X} - \mathcal{X}(G)$ of

$$I_\chi(f) + \sum_{\sigma,(\sigma,A)\in\chi} \frac{m'(\sigma)}{m(\sigma)} \mathrm{tr}(I_A(\sigma, f)), \tag{7.30}$$

(h) the sum of the distributions S_1, S_2, and S_3

(i) the "spectral part" of the endoscopic terms (7.2) (which we have omitted ...).

8 Some food for thought

It would be nice if we had

(1) provided a natural interpretation of the stabilization of the terms in (h),

(2) uncovered endoscopic analogs of Arthur's inductive formulas (6.19), (6.23) of the invariant distributions $I_\mathcal{O}$ and I_χ (if one exists),

(3) a stabilization of the terms in (f).

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